3 AIR QUALITY IMPACT

3.1 Introduction

3.1.1 This section presents the assessment on potential air quality impacts arising from construction and operation of the Project, which has been conducted in accordance with the criteria and guidelines as stated in section 1 of Annex 4 and Annex 12 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) as well as the requirements given in Clause 3.4.4 of the EIA Study Brief (ESB-291/2015).

3.1.2 An application for an Environmental Permit (EP) would be submitted for the following Schedule 2 Designated Projects (DPs) (refer to **Figure 1.3**) and these DPs during construction and operational phases are included in this air quality impact assessment:

- Construction of new primary distributor road (Road P1) (DP1)
- Construction of eight new distributor roads (Road D1 to D8) (DP2)
- Construction of slip roads: between Road D8 Junction and existing Castle Peak Road; Junction of D8/P1 and Junction of D7/P1; and Kong Sham Western Highway (KSWH) connection to Road D3 (DP5)
- Construction of partly depressed and partly decked-over roads located at Road D2, Road D4 and Road D6 (DP6)
- Construction of four new SPSs at Sites 2-34, 3-41, 3-48 and 4-35, respectively (DP9).
- Construction of Road P1 and slip-road from KSWH to Road D3 located within the "Conservation Area" of Yuen Tau Shan (DP12)

3.1.3 The following DPs would apply the EP through separated EIA studies but their potential impacts during construction phase and/or operational phase are also included in this air quality impact assessment. The locations of these DPs are shown in **Figure 1.2** and the details of the DP refer to **Section 1.5**.

- Construction of new West Rail Hung Shui Kiu (HSK) Station (Site 4-34) (DP3)
- Potential Construction of Environmentally Friendly Transport Services (EFTS) (DP4) – subject to further review
- Potential Construction of a new container back-up and storage area (Sites 3-1, 3-4, 3-5, 3-13 and 3-14) (DP7) – subject to further review
- Construction of New HSK sewage treatment works (STW) (Site 3-26 and part of existing San Wai STW) (DP8)
- Construction of flushing water service reservoirs for reuse of reclaimed water at Tan Kwai Tsuen and Fung Kong Tsuen (Sites 3-3 and 5-40) (DP10)
- Construction of one refuse transfer station (RTS) (Site 3-12) (DP11)

3.2 Environmental Legislation, Policies, Plans, Standards and Criteria

3.2.1 The criteria and guidelines for air quality assessment are laid down in Annex 4 and Annex 12 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM).

3.2.2 The Air Pollution Control Ordinance (APCO) provides the statutory authority for controlling air pollutants from a variety of sources. The Hong Kong Air Quality Objectives (AQOs), which stipulate the maximum allowable concentrations over specific periods for typical pollutants, should be met. The relevant AQOs are listed in **Table 3.1** which would be adopted in the air quality impact assessment.

3.2.3 The EIAO-TM stipulates that the hourly Total Suspended Particulates (TSP) level should not exceed 500 μg/m^3 (measured at 25°C and one atmosphere) for construction dust impact assessment.

3.2.4 In accordance with the EIAO-TM, odour level at an air sensitive receiver should meet 5 odour units based on an averaging time of 5 seconds for odour prediction assessment.

Table 3.1 Hong Kong Air Quality Objectives

Pollutant	Averaging Time	Maximum Average Concentration (μg/m^3) (1)	No. of Exceedances Allowed
Fine Suspended Particulates (FSP) (PM 2.5) (3)	24-hr	75	9
	Annual (2)	35	NA
Respirable Suspended Particulates (RSP) (PM10) (4)	24-hr	100	9
	Annual	50	NA
Sulphur Dioxide (SO_2)	10-min	500	3
	24-hr	125	3
Nitrogen Dioxide (NO_2)	1-hr	200	18
	Annual	40	NA
Carbon Monoxide (CO)	1-hr	30,000	0
	8-hr	10,000	0
Photochemical Oxidants (as ozone)	8-hr	160	9

Note:
(1) Gaseous air pollutants are measured at 293 K and 101.325 kPa.
(2) Arithmetic mean.
(3) Suspended particulates in air with a nominal aerodynamic diameter of 2.5 μm or smaller.
(4) Suspended particulates in air with a nominal aerodynamic diameter of 10 μm or smaller.

Air Pollution Control (Construction Dust) Regulation

3.2.5 Notifiable and regulatory works are under the control of Air Pollution Control (Construction Dust) Regulation. Notifiable works include site formation, reclamation, demolition, foundation and superstructure construction for buildings and road construction. Regulatory works include building renovation, road opening and resurfacing, slope stabilisation, and other activities including stockpiling, dusty material handling, excavation, concrete production, etc. This Project is expected to include notifiable works (site formation, demolition, foundation and superstructure construction for buildings as well as road construction) and regulatory works (road resurfacing, temporary stockpile, dusty material handling, and excavation). Contractors and site agents are required to inform EPD and adopt dust reduction measures to minimise dust emissions, while carrying out construction works, to the acceptable level.

Air Pollution Control Ordinance – Control of Emission from Specified Processes

3.2.6 A licence is required for the operation of specified processes (SPs), such as Cement Work (Concrete Batching Plant), under Part IV of APCO. The list of specified processes which required licence for operation is stated in Annex I of APCO. Application for a licence should be made to EPD. EPD may either grant or refuse or grant a licence subject to whether the applicant can fulfill the environmental standards to avoid causing air pollution. If EPD decides to grant the licence, a set of conditions would be imposed to prevent adverse impacts from the discharge of air pollutant emissions.

Air Pollution Control (Non-road Mobile Machinery) (Emission) Regulation

3.2.7 The Air Pollution Control (Non-road Mobile Machinery) (Emission) Regulation comes into effect on 1 June 2015. Under the Regulation, Non-road mobile machinery (NRMMs), except those exempted, are required to comply with the prescribed emission standards. From 1 September 2015, all regulated machines sold or leased for use in Hong Kong must be approved or exempted with a proper label in a prescribed format issued by EPD. Starting from 1 December 2015, only approved or exempted NRMMs with a proper label are allowed to be used in specified activities and locations including construction sites. The Contractor is required to ensure the adopted machines or non-road vehicle under the Project could meet the prescribed emission standards and requirement.

3.3 Baseline Conditions

Local Air Quality and Environ

3.3.1 The proposed Project is predominantly rural with low population density and located next to the existing Tin Shui Wai and Tuen Mun New Towns. The local air quality is dominated by vehicular exhaust emissions from the Yuen Long Highway, KSWH and Castle Peak Road as well as the scattered industrial emissions within the area.

3.3.2 As the proposed Project is within the vicinity of Shenzhen as well as Guangdong Province, the existing air quality in the assessment area is seasonally affected by the industrial emissions from the Guangdong Province as well.

3.3.3 In accordance with the Revised RODP, it is proposed that the current chicken farm at the Fung Kong Tsuen shall be stayed and some new sites for “Port Backup, Storage and Workshop Uses” (“PBU+SWU”) would be built in the vicinity of the chicken farm.

3.3.4 There is no EPD air quality data available for the assessment area. The nearest EPD air quality monitoring station in the vicinity of the assessment area is Yuen Long air quality monitoring station (AQMS). The annual average monitoring data recorded at this station has shown the trend of pollutants’ concentration in the past five years. The recent five years (Year 2011 to 2015) annual average concentrations are summarised in **Table 3.2**.

Table 3.2 Air Pollutant Concentrations (Year 2011 – 2015) measured at EPD's Yuen Long AQMS

Pollutant	Averaging Time	AQO [1]	Data Summary	Year [2][7]				
				2011	2012	2013	2014	2015
Fine Suspended Particulates ($PM_{2.5}$)	24-hr	75 (9)	Maximum	**109**	**87**	**140**	**119**	**127**
			10th Highest	**76**	65	**106**	**86**	**78**
			No. of Exceedance(s)	13	1	35	18	11
	Annual	35	-	**36**	29	**37**	35	30
Respirable Suspended Particulates (PM_{10})	24-hr	100 (9)	Maximum	**173**	**165**	**184**	**156**	**176**
			10th Highest	**111**	100	**142**	**124**	**102**
			No. of Exceedance(s)	25	10	47	21	12
	Annual	50	-	**54**	44	**56**	50	44
Total Suspended Particulates (TSP)	Daily[6]	500	Maximum	196	161	218	N/A	N/A
Sulphur Dioxide (SO_2)	10-min[5][6]	500 (3)	Maximum	N/A	N/A	N/A	112	67
			4th Highest	N/A	N/A	N/A	92	51
			No. of Exceedance(s)	N/A	N/A	N/A	0	0
	24-hr	125 (3)	Maximum	36	42	53	34	20
			4th Highest	33	29	33	27	17
			No. of Exceedance(s)	0	0	0	0	0
Nitrogen Dioxide (NO_2)	1-hr	200 (18)	Maximum	**257**	180	**230**	**227**	**339**
			19th Highest	188	147	183	165	162
			No. of Exceedance(s)	8	0	7	4	5
	Annual	40	-	**54**	**49**	**54**	**52**	**45**
Carbon Monoxide (CO)	1-hr	30,000	-	3210	2200	2690	2560	2460
	8-hr	10,000	-	2360	1829	1911	2319	2143
Ozone (O_3)	8-hr	160 (9)	Maximum	**195**	**217**	159	**304**	**261**
			10th Highest	138	**163**	130	**177**	**161**
			No. of Exceedance(s)	3	14	0	13	10

Note:
[1] Values in ( ) mean the number of exceedances allowed per year. TSP is not a criteria pollutant in the AQOs.
[2] Bolded and underlined values mean exceedance of relevant AQOs.
[3] Reference conditions of gaseous pollutants concentration data: 298 K and 101.325 kPa (before 2014) and 293K and 101.325 kPa (since 2014).
[4] N/M – Not Measured. N/A – Not Available (Relevant report is not available yet or data is not made available by EPD).
[5] Monitoring data for the AQO of 10-minute SO_2 on or before Year 2013 is currently not publicly available.
[6] Extracted from EPD's Air Quality in Hong Kong of relevant years.
[7] The hourly data of SO_2, NO_2, RSP, FSP, O_3 and CO are provided in EPD's website (http://epic.epd.gov.hk/EPICDI/air/station/?lang=en) and its 8-hour, daily and annual average concentrations are calculated based on the hourly data.

3.3.5 **Table 3.2** indicates that the annual average concentration of NO_2 and RSP/FSP at some years fail the AQO criteria. The 8-hour average O_3 concentrations in Year 2011 to 2015 also fail the criterion. Except the daily FSP level recorded in Year 2012, the 10th highest daily RSP/FSP concentrations fail the criterion in Year 2011 to 2015. The high RSP/FSP level at the Yuen Long Station was probably due to regional air quality problems within the Pearl River Delta Region. In particular, the industrial activities both at Yuen Long Industrial Estate and the Fu Tin Industrial Area in Shenzhen remain active. Heavy traffic in the area (major highways) also contributes to the regional particulates concentrations because NO_2 and RSP are typical pollutants of road emissions.

3.3.6 Apart from EPD's monitoring stations, there are also monitoring stations set up by the CLP Power Hong Kong Ltd. (CLP) in Yuen Long Areas. The latest monitoring results for NO_2 and SO_2 (from Year 2011 to 2015) are listed in **Table 3.3**.

Table 3.3 Air Pollutant Concentrations Measured at CLP's Tin Shui Wai and Lau Fau Shan AQMS [1]

Year	2011	2012 (3)	2013	2014	2015
Pollutant/ Locations (5, 6)	**Annual Average NO_2 Concentration (µg/m³) (4)**				
Tin Shui Wai	39	32	**45**	35	37
Lau Fau Shan	36	30	30	32	24
	Hourly Average NO_2 Concentration (µg/m³), 19th Highest (4)				
Tin Shui Wai	153	127	179	151	149
Lau Fau Shan	174	138	158	147	159
	Daily Average SO_2 Concentration, 4th Highest (µg/m³) (4)				
Tin Shui Wai	24	30	33	35	18
Lau Fau Shan	34	35	22	22	21

Note:
(1) Not use.
(2) Monitoring results exceeded AQOs are shown as bold and underlined characters.
(3) "N/A" means data is not available
(4) Reference conditions of gaseous pollutants concentration data: 298 K and 101.325 kPa.
(5) Annual data are extracted from "Air Quality in Hong Kong" published by EPD.
(6) Hourly and daily data are calculated from the hourly data provided in CLP's website (https://www.clpgroup.com/poweru/eng/air_quality/airQuality_monitoring_detail.aspx).

3.3.7 It is observed that the monitoring results of hourly average NO_2 and daily average SO_2 concentration recorded in Tin Shui Wai and Lau Fau Shan in Year 2011 - 2015 comply with the AQOs. Similar to the EPD's data, the annual average NO_2 concentrations fail the AQO criteria at some years.

Future Background Air Quality

3.3.8 As a general reference, the future prevailing background concentrations can be made reference to the EPD's PATH modelling results for Year 2020. The PATH model is a regional air quality model developed by EPD to simulate air quality over Hong Kong against the Pearl River Delta (PRD) as background. PATH is set up on a three-dimensional grid system with horizontal nesting. Details of the PATH results are presented in **Table 3.4**.

[1] CLP data were extracted from EPD's publication "Air Quality in Hong Kong" reports, and relevant CLP's website (https://www.clpgroup.com/poweru/eng/air_quality/airQuality_monitoring_detail.aspx).

Table 3.4 Future Air Pollutant Concentrations at Year 2020 from PATH-2016 Model

Pollutant	Averaging Time	AQO [1]	Data Summary	PATH Grid Cell [2] [3]					
				(20, 47)	(20, 48)	(20, 49)	(21, 44)	(21, 45)	(21, 46)
Fine Suspended Particulates ($PM_{2.5}$) [4][5]	24-hr	75 (9)	Max.	111.9	110.6	108.7	106.2	104.4	106.3
			10th Max.	72.3	71.3	69.6	69.6	69.5	70.3
			No. of Exceedance(s)	6	5	5	5	5	5
	Annual	35	-	30.1	29.9	29.3	29.9	29.6	29.7
Respirable Suspended Particulates (PM_{10}) [5]	24-hr	100 (9)	Max.	149.2	147.5	144.9	141.6	139.2	141.8
			10th Max.	96.4	95.1	92.9	92.8	92.7	93.7
			No. of Exceedance(s)	6	5	5	5	5	5
	Annual	50	-	42.4	42.1	41.3	42.0	41.7	41.9
Sulphur Dioxide (SO_2)	24-hr	125 (3)	Max.	36.8	37.9	44.7	36.6	36.5	36.4
			4th Max.	32.7	32.2	35.8	26.3	28.3	29.5
			No. of Exceedance(s)	0	0	0	0	0	0
Nitrogen Dioxide (NO_2)	1-hr	200 (18)	Max	131.2	131.9	130.0	118.5	120.0	125.2
			19th Max.	94.6	95.8	99.7	100.0	99.2	97.6
			No. of Exceedance(s)	0	0	0	0	0	0
	Annual	40	-	19.6	19.3	19.2	22.6	22.6	21.0
Carbon Monoxide (CO)	1-hr	30,000	-	996.0	1005.6	1064.4	993.4	989.1	995.6
	8-hr	10,000	-	837.6	851.2	873.2	846.7	835.5	838.1
Photochemical Oxidants (as ozone)	8-hr	160 (9)	Max.	221.2	218.6	224.0	222.2	221.7	218.0
			10th Max.	156.3	158.3	151.2	160.0	159.4	156.6
			No. of Exceedance(s)	8	7	8	10	9	9

Note:
[1] Values in ( ) mean the number of exceedances allowed per year.
[2] Bolded values mean exceedance of the AQOs.
[3] N/A – Not Available.
[4] Since $PM_{2.5}$ is not available from PATH model outputs, 24-hour average $PM_{2.5}$ and annual average $PM_{2.5}$ are estimated by 0.75 x PM_{10} and 0.71 x PM_{10} respectively with reference to Guidelines on the Estimation of PM 2.5 for Air Quality Assessment in Hong Kong.
[5] 10th highest daily and annual RSP concentration is adjusted by adding 37.6 μg/m³ and 21.9 μg/m³ respectively with reference to Guidelines on Choice of Models and Model Parameters. Same adjustment is applied to estimation of $PM_{2.5}$ as well.

Table 3.4 Future Air Pollutant Concentrations at Year 2020 from PATH-2016 Model (Cont'd)

Pollutant	Averaging Time	AQO [1]	Data Summary	PATH Grid Cell [2] [3]					
				(21, 47)	(21, 48)	(21, 49)	(21, 50)	(22, 44)	(22, 45)
Fine Suspended Particulates ($PM_{2.5}$) [4][5]	24-hr	75 (9)	Max.	107.0	108.4	106.7	105.0	113.0	110.4
			10th Max.	70.3	70.5	68.7	69.5	71.9	69.5
			No. of Exceedance(s)	5	5	5	5	7	5
	Annual	35	-	29.3	29.5	28.9	29.1	30.8	29.8
Respirable Suspended Particulates (PM_{10}) [5]	24-hr	100 (9)	Max.	142.7	144.6	142.3	140.0	150.7	147.2
			10th Max.	93.8	94.0	91.6	92.6	95.9	92.7
			No. of Exceedance(s)	5	5	5	5	7	5
	Annual	50	-	41.3	41.5	40.7	40.9	43.4	42.0
Sulphur Dioxide (SO_2)	24-hr	125 (3)	Max.	36.6	36.8	36.8	38.8	36.4	36.5
			4th Max.	30.0	30.2	30.0	33.0	25.8	26.1
			No. of Exceedance(s)	0	0	0	0	0	0
Nitrogen Dioxide (NO_2)	1-hr	200 (18)	Max	126.3	126.1	115.6	124.8	118.7	121.6
			19th Max.	95.2	93.3	92.4	96.6	93.0	99.7
			No. of Exceedance(s)	0	0	0	0	0	0
	Annual	40	-	19.6	18.9	18.6	18.6	20.0	23.0
Carbon Monoxide (CO)	1-hr	30,000	-	1024.6	1049.8	1068.2	1109.5	992.6	993.4
	8-hr	10,000	-	854.0	868.3	891.6	910.5	868.8	865.2
Photochemical Oxidants (as ozone)	8-hr	160 (9)	Max.	219.8	220.7	224.6	227.8	218.6	217.2
			10th Max.	158.7	159.4	158.9	154.2	159.1	156.0
			No. of Exceedance(s)	9	8	9	7	9	9

Note:
[1] Values in ( ) mean the number of exceedances allowed per year.
[2] Bolded values mean exceedance of the AQOs.
[3] N/A – Not Available.
[4] Since $PM_{2.5}$ is not available from PATH model outputs, 24-hour average $PM_{2.5}$ and annual average $PM_{2.5}$ are estimated by 0.75 x PM_{10} and 0.71 x PM_{10} respectively with reference to Guidelines on the Estimation of PM 2.5 for Air Quality Assessment in Hong Kong.
[5] 10th highest daily and annual RSP concentration is adjusted by adding 37.6 μg/m³ and 21.9 μg/m³ respectively with reference to Guidelines on Choice of Models and Model Parameters. Same adjustment is applied to estimation of $PM_{2.5}$ as well.

Table 3.4 Future Air Pollutant Concentrations at Year 2020 from PATH-2016 Model (Cont'd)

Pollutant	Averaging Time	AQO [1]	Data Summary	PATH Grid Cell [2] [3]					
				(22, 46)	(22, 47)	(22, 48)	(22, 49)	(23, 45)	(23, 46)
Fine Suspended Particulates ($PM_{2.5}$) [4][5]	24-hr	75 (9)	Max.	106.5	104.8	105.4	104.4	112.7	109.4
			10th Max.	68.8	68.7	68.8	68.5	70.2	69.1
			No. of Exceedance(s)	4	4	4	4	6	5
	Annual	35	-	29.1	28.9	28.8	28.7	29.8	29.2
Respirable Suspended Particulates (PM_{10}) [5]	24-hr	100 (9)	Max.	142.0	139.7	140.5	139.2	150.3	145.8
			10th Max.	91.8	91.6	91.7	91.3	93.6	92.1
			No. of Exceedance(s)	4	4	4	4	6	5
	Annual	50	-	41.0	40.7	40.6	40.4	41.9	41.1
Sulphur Dioxide (SO_2)	24-hr	125 (3)	Max.	36.5	36.5	36.6	36.6	36.4	36.4
			4th Max.	26.4	28.5	28.8	28.6	25.8	26.1
			No. of Exceedance(s)	0	0	0	0	0	0
Nitrogen Dioxide (NO_2)	1-hr	200 (18)	Max	118.4	118.1	118.1	115.7	118.5	117.7
			19th Max.	98.7	97.5	96.9	93.7	93.8	97.4
			No. of Exceedance(s)	0	0	0	0	0	0
	Annual	40	-	21.6	20.8	20.1	19.2	19.8	21.2
Carbon Monoxide (CO)	1-hr	30,000	-	1007.4	1060.9	1102.1	1136.1	1002.3	997.5
	8-hr	10,000	-	860.9	882.7	899.3	914.9	875.5	880.1
Photochemical Oxidants (as ozone)	8-hr	160 (9)	Max.	216.7	218.3	220.4	222.2	215.3	215.6
			10th Max.	157.1	158.1	158.1	158.0	155.9	158.5
			No. of Exceedance(s)	9	9	9	9	9	9

Note:
[1] Values in ( ) mean the number of exceedances allowed per year.
[2] Bolded values mean exceedance of the AQOs.
[3] N/A – Not Available.
[4] Since $PM_{2.5}$ is not available from PATH model outputs, 24-hour average $PM_{2.5}$ and annual average $PM_{2.5}$ are estimated by 0.75 x PM_{10} and 0.71 x PM_{10} respectively with reference to Guidelines on the Estimation of PM 2.5 for Air Quality Assessment in Hong Kong.
[5] 10th highest daily and annual RSP concentration is adjusted by adding 37.6 μg/m³ and 21.9 μg/m³ respectively with reference to Guidelines on Choice of Models and Model Parameters. Same adjustment is applied to estimation of $PM_{2.5}$ as well.

Table 3.4 Future Air Pollutant Concentrations at Year 2020 from PATH-2016 Model (Cont'd)

Pollutant	Averaging Time	AQO [1]	Data Summary	PATH Grid Cell [2] [3]	
				(23, 47)	(23, 48)
Fine Suspended Particulates ($PM_{2.5}$) [4][5]	24-hr	75 (9)	Max.	106.6	105.1
			10th Max.	68.8	68.6
			No. of Exceedance(s)	4	4
	Annual	35	-	28.9	28.7
Respirable Suspended Particulates (PM_{10}) [5]	24-hr	100 (9)	Max.	142.1	140.2
			10th Max.	91.8	91.5
			No. of Exceedance(s)	4	4
	Annual	50	-	40.8	40.5
Sulphur Dioxide (SO_2)	24-hr	125 (3)	Max.	36.4	36.5
			4th Max.	26.7	27.4
			No. of Exceedance(s)	0	0
Nitrogen Dioxide (NO_2)	1-hr	200 (18)	Max	121.1	123.4
			19th Max.	98.7	100.5
			No. of Exceedance(s)	0	0
	Annual	40	-	21.5	20.6
Carbon Monoxide (CO)	1-hr	30,000	-	1018.1	1067.6
	8-hr	10,000	-	897.4	910.8
Photochemical Oxidants (as ozone)	8-hr	160 (9)	Max.	216.3	217.6
			10th Max.	160.3	161.0
			No. of Exceedance(s)	10	10

Note:
[1] Values in ( ) mean the number of exceedances allowed per year.
[2] Bolded values mean exceedance of the AQOs.
[3] N/A – Not Available.
[4] Since $PM_{2.5}$ is not available from PATH model outputs, 24-hour average $PM_{2.5}$ and annual average $PM_{2.5}$ are estimated by 0.75 x PM_{10} and 0.71 x PM_{10} respectively with reference to Guidelines on the Estimation of PM 2.5 for Air Quality Assessment in Hong Kong.
[5] 10th highest daily and annual RSP concentration is adjusted by adding 37.6 μg/m³ and 21.9 μg/m³ respectively with reference to Guidelines on Choice of Models and Model Parameters. Same adjustment is applied to estimation of $PM_{2.5}$ as well.

3.4 Identification of Air Sensitive Receivers

3.4.1 In accordance with Annex 12 of the EIAO-TM, any domestic premises, hotel, hostel, hospital, clinic, nursery, temporary housing accommodation, school, educational institution, office, factory, shop, shopping centre, place of public worship, library, court of law, sports stadium or performing arts centre are considered as air sensitive receivers (ASRs). Any other premises or place with which, in terms of duration or number of people affected, has a similar sensitivity to the air pollutants as the aforelisted premises and places is also considered to be a sensitive receiver. As stated in the EIA Study Brief, the boundary of the assessment area for air quality assessment should be 500 m from the Project alignment and boundaries of all associated areas under the Project.

3.4.2 The representative planned ASRs within the Project area are listed in **Table 3.5**. The representative existing ASRs within and in the vicinity of the Project area are summarised in **Table 3.6**. The locations of representative ASRs are illustrated in **Figure 3.2** and **Figure 3.3**.

3.4.3 The representative ASRs (existing / planned) that could be affected by the Project within the 500 m assessment area have been identified based on the Revised RODP and the latest following Outline Zoning Plans, Layout Plan, Outline Development Plans, Layout Plans and other relevant published land use plans.

- Ha Tsuen OZP No. S/YL-HT/10
- Lam Tei and Yick Yuen OZP No. S/TM-LTYY/8
- Tong Yan San Tsuen OZP No. S/YL-TYST/12
- Ping Shan OZP No. S/YL-PS/16
- Lau Fau Shan and Tsim Bei Tsui OZP No. S/YL-LFS/7
- Tin Shui Wai OZP No. S/TSW/12

Table 3.5 Representative Planned Air Sensitive Receivers within the Project Area (for Revised RODP)

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P201	Planned Governmental Building	G	4-31	2031	112
P202	Planned Governmental Building	G	4-31	2031	112
P203	Planned Governmental Building	G	4-31	2031	112
P204	Planned Governmental Building	G	4-31	2031	112
P205	Planned District Police Station and Married Quarters	G	4-32	2031	116
P206	Planned District Police Station and Married Quarters	G	4-32	2031	116
P207	Planned District Police Station and Married Quarters	G	4-32	2031	116
P208	Planned District Police Station and Married Quarters	G	4-32	2031	116
P209	Planned Secondary School	E	4-33	2031	36
P210	Planned Secondary School	E	4-33	2031	36
P211	Planned Secondary School	E	4-33	2031	36
P212	Planned Primary School	E	4-36	2031	36
P213	Planned Primary School	E	4-36	2031	36
P214	Planned Primary School	E	4-36	2031	36
P215	Planned Industrial Uses	I	3-50	2030	40
P216	Planned Industrial Uses	I	3-51	2030	40
P217	Planned Industrial Uses	I	3-51	2030	40
P218	Planned Industrial Uses	I	3-51	2030	40
P219	Planned Industrial Uses	I	3-51	2030	40
P220	Planned Industrial Uses	I	3-52	2031	40
P221	Planned Industrial Uses	I	3-52	2031	40
P222	Planned Industrial Uses	I	3-52	2031	40
P223	Planned Industrial Uses	I	3-52	2031	40
P224	Planned Industrial Uses	I	3-52	2031	40
P225	Planned Regional Plaza	OU	4-14	2031	4
P226	Planned Regional Plaza	OU	4-14	2031	4
P227	Planned Regional Plaza	OU	4-14	2031	4
P228	Planned Youth Facility	G	4-15	2031	68
P229	Planned Youth Facility	G	4-15	2031	68
P230	Planned Commercial Building	C	4-16	2031	115
P231	Planned Commercial Building	C	4-16	2031	115
P232	Planned Commercial Building	C	4-16	2031	115
P233	Planned Educational and Related Uses	OU	4-17	2031	36
P234	Planned Educational and Related Uses	OU	4-17	2031	36
P235	Planned Educational and Related Uses	OU	4-17	2031	36
P236	Planned Educational and Related Uses	OU	4-17	2031	36
P237	Planned Educational and Related Uses	OU	4-17	2031	36
P238	Planned Educational and Related Uses	OU	4-17	2031	36
P239	Planned Village Resite	RR4	4-20	2026	8
P240	Planned Village Resite	RR4	4-20	2026	8
P241	Planned Village Resite	RR4	4-20	2026	8
P242	Planned Vehicle Depot	G	4-21	2031	48
P243	Planned Vehicle Depot	G	4-21	2031	48

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P244	Planned Vehicle Depot	G	4-21	2031	48
P245	Planned Vehicle Depot	G	4-21	2031	48
P246	Planned Public Rental Housing	RSc	4-22	2030	116
P247	Planned Public Rental Housing	RSc	4-22	2030	116
P248	Planned Public Rental Housing	RSc	4-22	2030	116
P249	Planned Public Rental Housing	RSc	4-22	2030	116
P250	Planned Public Rental Housing	RSc	4-28	2030	131
P251	Planned Commercial and Residential Development	OU (C&R)	4-29	2031	158
P252	Planned Commercial and Residential Development	OU (C&R)	4-29	2031	158
P253	Planned Commercial and Residential Development	OU (C&R)	4-29	2031	158
P254	Planned Commercial and Residential Development	OU (C&R)	4-29	2031	158
P255	Planned Commercial Uses	C	4-12a	2031	182
P256	Planned Commercial Uses	C	4-12a	2031	182
P257	Planned Commercial Uses	C	4-12b	2031	182
P258	Planned Commercial Uses	C	4-12b	2031	182
P259	Planned Commercial Uses	C	4-12d	2031	182
P260	Planned Commercial Uses	C	4-12d	2031	182
P261	Planned Commercial Uses	C	4-12d	2031	182
P262	Planned Commercial and Big Box Retail	C	4-13a	2031	180
P263	Planned Commercial and Big Box Retail	C	4-13a	2031	180
P264	Planned Commercial and Big Box Retail	C	4-13a	2031	180
P265	Planned Commercial and Big Box Retail	C	4-13a	2031	180
P266	Planned Commercial Uses	C	4-13b	2031	187
P267	Planned Commercial and Residential Development	OU (C&R)	4-25a	2031	162
P268	Planned Commercial and Residential Development	OU (C&R)	4-25a	2031	162
P269	Planned Commercial and Residential Development	OU (C&R)	4-25c	2031	166
P270	Planned Commercial and Residential Development	OU (C&R)	4-25c	2031	166
P301	Planned Subsidised Sales Flat	R2 (SSF)c	4-24	2031	104
P302	Planned Subsidised Sales Flat	R2 (SSF)c	4-24	2031	104
P303	Planned Subsidised Sales Flat	R2 (SSF)c	4-24	2031	104
P304	Planned Subsidised Sales Flat	R2 (SSF)c	4-24	2031	104
P305	Planned Subsidised Sales Flat	R2 (SSF)c	4-24	2031	104
P306	Planned Private Residential Area	R1c	4-26	2031	140
P307	Planned Private Residential Area	R1c	4-26	2031	140
P308	Planned Private Residential Area	R1c	4-26	2031	140
P309	Planned Private Residential Area	R1c	4-26	2031	140
P310	Planned Public Rental Housing	RSc	4-28	2030	131
P311	Planned Public Rental Housing	RSc	4-28	2030	131

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P312	Planned Public Rental Housing	RSc	4-28	2030	131
P313	Planned Commercial and Residential Development	OU (C&R)	4-25a	2031	162
P314	Planned Commercial and Residential Development	OU (C&R)	4-25a	2031	162
P315	Planned Commercial and Residential Development	OU (C&R)	4-25b	2031	162
P316	Planned Commercial and Residential Development	OU (C&R)	4-25b	2031	162
P401	Planned Sports Centre	G	1-30	2036	22
P402	Planned Sports Centre	G	1-30	2036	22
P403	Planned Commercial and Residential Development	OU (C&R)	5-32	2030	120
P404	Planned Commercial and Residential Development	OU (C&R)	5-32	2030	120
P405	Planned Commercial and Residential Development	OU (C&R)	5-32	2030	120
P406	Planned Commercial and Residential Development	OU (C&R)	5-32	2030	120
P407	Committed Residential Area	R2	5-33	N/A	43
P408	Committed Residential Area	R2	5-33	N/A	43
P409	Committed Residential Area	R2	5-33	N/A	43
P410	Committed Residential Area	R2	5-33	N/A	43
P411	Planned Reserved Area	G	3-34	2036	14
P412	Planned Schools	E	5-34	2036	36
P413	Planned Schools	E	5-34	2036	36
P414	Planned Schools	E	5-34	2036	36
P415	Planned Primary School	E	5-37	2031	36
P416	Planned Primary School	E	5-37	2031	36
P417	Planned Primary School	E	5-37	2031	36
P418	Planned Reserved Area	G	5-38	2031	14
P419	Planned Reserved Area	G	5-38	2031	14

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P420	Planned Reserved Area	G	5-38	2031	14
P421	Planned Enterprise and Technology Park	OU (E+TP)	3-42	2036	66
P422	Planned Reserved Area	G	5-42	2036	14
P423	Planned Reserved Area	G	5-42	2036	14
P424	Planned Hospital, Specialist Clinic/ Polyclinic	G	5-13	2036	54
P425	Planned Hospital, Specialist Clinic/ Polyclinic	G	5-13	2036	54
P426	Committed Residential Area	R3	5-22	N/A	50
P427	Committed Residential Area	R3	5-22	N/A	50
P428	Committed Residential Area	R3	5-22	N/A	50
P429	Committed Residential Area	R3	5-22	N/A	50
P432	Planned Subsidised Sales Flat	R2 (SSF)c	5-24	2026	122
P433	Planned Subsidised Sales Flat	R2 (SSF)c	5-24	2026	122
P434	Planned Subsidised Sales Flat	R2 (SSF)c	5-24	2026	122
P435	Planned Subsidised Sales Flat	R2 (SSF)c	5-24	2026	122
P436	Planned Subsidised Sales Flat	R2 (SSF)c	5-24	2026	122
P437	Planned Subsidised Sales Flat	R2 (SSF)	5-26	2024	83
P438	Planned Subsidised Sales Flat	R2 (SSF)	5-26	2024	83
P439	Planned Clinic	G	5-27	2030	22
P440	Planned Clinic	G	5-27	2030	22
P441	Planned Residential Care Home for the Elderly	G	5-28	2030	27
P442	Planned Residential Care Home for the Elderly	G	5-28	2030	27

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P443	Planned Sports Ground	G	1-29	2036	18
P444	Planned Sports Ground	G	1-29	2036	18
P445	Planned Sports Ground	G	1-29	2036	18
P446	Planned Sports Ground	G	1-29	2036	18
P447	Planned Commercial and Residential Development	OU (C&R)	4-25b	2031	162
P501	Planned Hospital, Specialist Clinic/ Polyclinic	G	5-13	2036	54
P502	Planned Schools	E	5-14	2036	36
P503	Planned Schools	E	5-14	2036	36
P504	Planned Schools	E	5-14	2036	36
P505	Planned Schools	E	5-14	2036	36
P601	Planned Enterprise and Technology Park	OU (E+TP)	3-32	2036	66
P602	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-13	2031	80
P603	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-13	2031	80
P604	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-14	2031	80
P605	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-14	2031	80
P606	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-14	2031	80
P607	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-14	2031	80
P608	Planned Logistics Facility	OU(LF)	3-16	2036	88
P609	Planned Logistics Facility	OU(LF)	3-16	2036	88
P610	Planned Logistics Facility	OU(LF)	3-16	2036	88
P611	Planned Logistics Facility	OU(LF)	3-16	2036	88
P612	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-11	2031	80
P613	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-11	2031	80
P614	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-11	2031	80
P615	Planned Logistics Facility cum EFTS Depot	OU(LF)	3-18	2031	90
P616	Planned Logistics Facility cum EFTS Depot	OU(LF)	3-18	2031	90
P617	Planned Logistics Facility cum EFTS Depot	OU(LF)	3-18	2031	90
P618	Planned Logistics Facility cum EFTS Depot	OU(LF)	3-18	2031	90
P619	Planned Logistics Facility	OU(LF)	3-20	2036	88
P620	Planned Logistics Facility	OU(LF)	3-20	2036	88
P621	Planned Logistics Facility	OU(LF)	3-20	2036	88
P622	Planned Logistics Facility	OU(LF)	3-20	2036	88
P623	Planned Logistics Facility	OU(LF)	3-24	2036	88
P624	Planned Logistics Facility	OU(LF)	3-24	2036	88
P625	Planned Logistics Facility	OU(LF)	3-24	2036	88
P626	Planned Logistics Facility	OU(LF)	3-24	2036	88
P627	Planned Logistics Facility	OU(LF)	3-25	2036	88
P628	Planned Logistics Facility	OU(LF)	3-25	2036	88
P629	Planned Logistics Facility	OU(LF)	3-27	2036	88
P630	Planned Logistics Facility	OU(LF)	3-27	2036	88
P631	Planned Logistics Facility	OU(LF)	3-27	2036	88

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P632	Planned Logistics Facility	OU(LF)	3-27	2036	88
P633	Planned Logistics Facility	OU(LF)	3-28	2036	88
P634	Planned Logistics Facility	OU(LF)	3-28	2036	88
P635	Planned Logistics Facility	OU(LF)	3-28	2036	88
P636	Planned Logistics Facility	OU(LF)	3-28	2036	88
P637	Planned Logistics Facility	OU(LF)	3-29	2036	88
P638	Planned Logistics Facility	OU(LF)	3-29	2036	88
P639	Planned Logistics Facility	OU(LF)	3-29	2036	88
P701	Planned Private Residential Area	R2c	1-13	2037	118
P702	Planned Hospital, Specialist Clinic/ Polyclinic	G	5-13	2036	54
P703	Planned Schools	E	1-15	2036	36
P704	Planned Schools	E	1-15	2036	36
P705	Planned Schools	E	1-15	2036	36
P706	Planned Schools	E	1-15	2036	36
P707	Planned Private Residential Area	R2c	1-16	2037	127
P708	Planned Private Residential Area	R2c	1-16	2037	127
P709	Planned Private Residential Area	R2c	1-16	2037	127
P710	Planned Private Residential Area	R2c	1-16	2037	127
P711	Planned Subsidised Sales Flat	R2 (SSF)c	5-16	2030	101
P712	Planned Subsidised Sales Flat	R2 (SSF)c	5-16	2030	101
P713	Planned Subsidised Sales Flat	R2 (SSF)c	5-16	2030	101
P714	Planned Private Residential Area	R2	1-17	2036	103
P715	Planned Private Residential Area	R2	1-17	2036	103
P716	Planned Private Residential Area	R2	1-17	2036	103
P717	Planned Private Residential Area	R2	1-17	2036	103
P718	Planned Public Rental Housing	RSc	5-17	2030	104
P719	Planned Public Rental Housing	RSc	5-17	2030	104
P720	Planned Public Rental Housing	RSc	5-17	2030	104
P721	Planned Public Rental Housing	RSc	5-17	2030	104
P722	Planned Primary Schools	E	1-18	2036	36
P723	Planned Primary Schools	E	1-18	2036	36
P724	Planned Primary Schools	E	1-18	2036	36
P725	Planned Primary Schools	E	1-18	2036	36
P726	Planned Private Residential Area	R2	1-19	2036	87
P727	Planned Private Residential Area	R2	1-19	2036	87
P728	Planned Private Residential Area	R2	1-19	2036	87
P729	Planned Private Residential Area	R2	1-19	2036	87
P730	Planned Private Residential Area	R2c	1-21	2036	118
P731	Planned Private Residential Area	R2c	1-21	2036	118
P732	Planned Private Residential Area	R2c	1-21	2036	118

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P733	Planned Private Residential Area	R2c	1-21	2036	118
P734	Planned Secondary School	E	5-21	2030	36
P735	Planned Secondary School	E	5-21	2030	36
P736	Planned Secondary School	E	5-21	2030	36
P737	Planned Schools	E	1-22	2036	36
P738	Planned Schools	E	1-22	2036	36
P739	Planned Schools	E	1-22	2036	36
P740	Planned Schools	E	1-22	2036	36
P741	Planned Logistics Facility	OU(LF)	3-25	2036	88
P742	Planned Logistics Facility	OU(LF)	3-25	2036	88
P743	Planned Commercial Area	C	5-18a	2030	118
P744	Planned Commercial Area	C	5-18a	2030	118
P745	Planned Commercial Area	C	5-18a	2030	118
P746	Planned Commercial Area	C	5-18b	2030	118
P747	Planned Commercial Area	C	5-18b	2030	118
P748	Planned Commercial Area	C	5-18b	2030	118
P749	Planned Commercial Area	C	5-3a	2030	151
P801	Planned Hospital, Specialist Clinic/ Polyclinic	G	5-13	2036	54
P802	Planned Commercial Area	C	5-1	2031	151
P803	Planned Commercial Area	C	5-1	2031	151
P804	Planned Commercial Area	C	5-1	2031	151
P805	Planned Commercial Area	C	5-1	2031	151
P806	Planned Committed Commercial Area	C	5-2	N/A	50
P807	Planned Committed Commercial Area	C	5-2	N/A	50
P808	Planned Committed Commercial Area	C	5-2	N/A	50
P809	Planned Commercial Area	C	5-4	2036	152
P810	Planned Commercial Area	C	5-4	2036	152
P811	Planned Commercial Area	C	5-4	2036	152
P812	Planned Subsidised Sales Flat	R2 (SSF)c	5-6	N/A	10
P813	Planned Subsidised Sales Flat	R2 (SSF)c	5-6	N/A	10
P814	Planned Subsidised Sales Flat	R2 (SSF)c	5-6	N/A	10
P815	Planned Subsidised Sales Flat	R2 (SSF)c	5-6	N/A	10
P816	Planned Commercial and Residential Development	OU (C&R)	5-8	2036	109
P817	Planned Commercial and Residential Development	OU (C&R)	5-8	2036	109
P818	Planned Commercial and Residential Development	OU (C&R)	5-8	2036	109
P819	Planned Commercial and Residential Development	OU (C&R)	5-8	2036	109
P820	Planned Commercial and Residential Development	OU (C&R)	5-9	2036	147
P821	Planned Commercial and Residential Development	OU (C&R)	5-9	2036	147
P822	Planned Commercial and Residential Development	OU (C&R)	5-9	2036	147
P823	Planned Commercial and Residential Development	OU (C&R)	5-9	2036	147
P824	Planned Commercial Area	C	5-3a	2030	151
P825	Planned Commercial Area	C	5-3a	2030	151
P826	Planned Commercial Area	C	5-3a	2030	151
P827	Planned Commercial Area	C	5-3b	2030	151
P828	Planned Commercial Area	C	5-3b	2030	151
P829	Planned Commercial Area	C	5-3b	2030	151
P830	Planned Commercial Area	C	5-3b	2030	151

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P831	Planned Commercial and Residential Development	OU (C&R)	5-7a	2036	137
P832	Planned Commercial and Residential Development	OU (C&R)	5-7a	2036	137
P833	Planned Commercial and Residential Development	OU (C&R)	5-7a	2036	137
P834	Planned Commercial and Residential Development	OU (C&R)	5-7b	2036	137
P835	Planned Commercial and Residential Development	OU (C&R)	5-7b	2036	137
P836	Planned Commercial and Residential Development	OU (C&R)	5-7b	2036	137
P901	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-7	2026	80
P902	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-8	2026	80
P1001	Planned Public Rental Housing	RSc	2-30	2036	116
P1002	Planned Public Rental Housing	RSc	2-30	2036	116
P1003	Planned Public Rental Housing	RSc	2-30	2036	116
P1004	Planned Public Rental Housing	RSc	2-30	2036	116
P1005	Planned Public Rental Housing	RSc	2-30	2036	116
P1006	Planned Public Rental Housing	RSc	2-30	2036	116
P1007	Planned Public Rental Housing	R2 (SSF)c	2-31	2036	113
P1008	Planned Reserved Area	G	2-33	2036	36
P1009	Planned Reserved Area	G	2-33	2036	36
P1010	Planned Reserved Area	G	2-33	2036	36
P1011	Planned Reserved Area	G	2-33	2036	36
P1012	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-13	2031	80
P1013	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-13	2031	80
P1014	Planned Logistics Facility	OU(LF)	3-15	2036	88
P1015	Planned Logistics Facility	OU(LF)	3-15	2036	88
P1016	Planned Logistics Facility	OU(LF)	3-15	2036	88
P1017	Planned Logistics Facility	OU(LF)	3-15	2036	88
P1018	Planned Port Back-up, Storage and Workshop	OU (PBU+ SWU)	3-1	2031	80
P1019	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-1	2031	80
P1020	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-1	2031	80
P1021	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-1	2031	80
P1022	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-4	2031	80
P1023	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-4	2031	80

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P1024	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-4	2031	80
P1025	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-5	2031	80
P1026	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-5	2031	80
P1027	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-5	2031	80
P1028	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-5	2031	80
P1029	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-6	2026	80
P1030	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-6	2026	80
P1031	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-6	2026	80
P1032	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-6	2026	80
P1033	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-7	2026	80
P1034	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-7	2026	80
P1035	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-7	2026	80
P1036	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-8	2026	80
P1037	Planned Private Residential Area	RR4	2-18	2026	8
P1038	Planned Private Residential Area	RR4	2-18	2026	8
P1039	Planned Private Residential Area	RR4	2-19	2026	8
P1040	Planned Private Residential Area	RR4	2-19	2026	8
P1041	Planned Private Residential Area	RR4	2-19	2026	8
P1042	Planned Reserved Area	G	2-20	2036	14
P1043	Planned Reserved Area	G	2-20	2036	14
P1044	Planned Reserved Area	G	2-20	2036	14
P1045	Planned Reserved Area	G	2-20	2036	14
P1046	Planned Reserved Area	G	2-21	2036	36
P1047	Planned Reserved Area	G	2-21	2036	36
P1048	Planned Reserved Area	G	2-21	2036	36
P1049	Planned Sports Centre	G	2-22	2036	22
P1050	Planned Sports Centre	G	2-22	2036	22
P1051	Planned Sports Centre	G	2-22	2036	22
P1052	Planned Sports Centre	G	2-22	2036	22
P1053	Planned Schools	E	2-24	2036	36
P1054	Planned Schools	E	2-24	2036	36
P1055	Planned Residential Care Home for the Elderly	G	2-25	2036	27
P1056	Planned Residential Care Home for the Elderly	G	2-25	2036	27

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P1057	Planned Residential Care Home for the Elderly	G	2-25	2036	27
P1058	Planned Residential Care Home for the Elderly	G	2-25	2036	27
P1059	Planned Schools	E	2-26	2036	36
P1060	Planned Schools	E	2-26	2036	36
P1061	Planned Schools	E	2-26	2036	36
P1062	Planned Schools	E	2-26	2036	36
P1063	Planned Primary School	E	2-28	2036	36
P1064	Planned Primary School	E	2-28	2036	36
P1065	Planned Primary School	E	2-28	2036	36
P1066	Planned Parking and Operational Facilities for EFTS	OU	2-29	2036	48
P1067	Planned Parking and Operational Facilities for EFTS	OU	2-29	2036	48
P1068	Planned Parking and Operational Facilities for EFTS	OU	2-29	2036	48
P1069	Planned Parking and Operational Facilities for EFTS	OU	2-29	2036	48
P1101	Planned Subsidised Sales Flat	R2 (SSF)c	2-31	2036	113
P1102	Planned Subsidised Sales Flat	R2 (SSF)c	2-31	2036	113
P1103	Planned Subsidised Sales Flat	R2 (SSF)c	2-31	2036	113
P1104	Planned Private Residential Area	R2c	1-13	2037	118
P1105	Planned Private Residential Area	R2c	1-13	2037	118
P1106	Planned Private Residential Area	R2c	1-13	2037	118
P1107	Planned Private Residential Area	R2c	1-13	2037	118
P1108	Planned Government Complex	G	1-1	2036	36
P1109	Planned Government Complex	G	1-1	2036	36
P1110	Planned Schools	E	1-3	2036	36
P1111	Planned Schools	E	1-3	2036	36
P1112	Planned Schools	E	1-3	2036	36
P1113	Planned Schools	E	1-3	2036	36
P1114	Planned Private Residential Area	R2c	1-4	2037	114
P1115	Planned Private Residential Area	R2c	1-4	2037	114
P1116	Planned Private Residential Area	R2c	1-4	2037	114
P1117	Planned Private Residential Area	R2c	1-4	2037	114
P1118	Planned Private Residential Area	R2c	1-5	2037	118
P1119	Planned Private Residential Area	R2c	1-5	2037	118
P1120	Planned Private Residential Area	R2c	1-5	2037	118
P1121	Planned Private Residential Area	R2c	1-5	2037	118
P1122	Planned Private Residential Area	R2c	1-6	2037	127
P1123	Planned Private Residential Area	R2c	1-6	2037	127

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P1124	Planned Private Residential Area	R2c	1-6	2037	127
P1125	Planned Private Residential Area	R2c	1-6	2037	127
P1126	Planned Private Residential Area	R2c	1-8	2037	127
P1127	Planned Private Residential Area	R2c	1-8	2037	127
P1128	Planned Private Residential Area	R2c	1-8	2037	127
P1129	Planned Private Residential Area	R2c	1-8	2037	127
P1130	Planned Private Residential Area	R2c	1-9	2037	124
P1131	Planned Private Residential Area	R2c	1-9	2037	124
P1132	Planned Private Residential Area	R2c	1-9	2037	124
P1133	Planned Private Residential Area	R2c	1-9	2037	124
P1134	Planned Private Residential Area	R2	1-12	2036	110
P1135	Planned Private Residential Area	R2	1-12	2036	110
P1136	Planned Private Residential Area	R2	1-12	2036	110
P1137	Planned Private Residential Area	R2	1-12	2036	110
P1138	Planned Schools	E	2-24	2036	36
P1139	Planned Schools	E	2-24	2036	36
P1301	Planned Private Residential Area	R3	2-14	2037	53
P1302	Planned Private Residential Area	R3	2-14	2037	53
P1303	Planned Private Residential Area	R3	2-14	2037	53
P1304	Planned Private Residential Area	R3	2-14	2037	53
P1305	Planned Primary School	E	2-15	2037	36
P1306	Planned Primary School	E	2-15	2037	36
P1307	Planned Primary School	E	2-15	2037	36
P1308	Planned Primary School	E	2-15	2037	36
P1309	Planned Commercial Area	C	2-1	2037	28
P1310	Planned Commercial Area	C	2-1	2037	28
P1311	Planned Commercial Area	C	2-1	2037	28
P1312	Planned Schools	E	2-16	2037	36
P1313	Planned Schools	E	2-16	2037	36
P1314	Planned Schools	E	2-16	2037	36
P1315	Planned Schools	E	2-16	2037	36
P1316	Planned Schools	E	2-16	2037	36
P1317	Planned Private Residential Area	R3	2-2	2037	46
P1318	Planned Private Residential Area	R3	2-2	2037	46
P1319	Planned Private Residential Area	R3	2-2	2037	46
P1320	Planned Private Residential Area	R3	2-2	2037	46
P1321	Planned Private Residential Area	R3	2-3	2037	53
P1322	Planned Private Residential Area	R3	2-3	2037	53
P1323	Planned Private Residential Area	R3	2-3	2037	53

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P1324	Planned Private Residential Area	R33	2-3	2037	53
P1325	Planned Private Residential Area	R3	2-7	2037	50
P1326	Planned Private Residential Area	R3	2-7	2037	50
P1327	Planned Private Residential Area	R3	2-7	2037	50
P1328	Planned Private Residential Area	R3	2-7	2037	50
P1329	Planned Primary School	E	2-9	2037	36
P1330	Planned Primary School	E	2-9	2037	36
P1331	Planned Primary School	E	2-9	2037	36
P1332	Planned Primary School	E	2-9	2037	36
P1333	Planned Private Residential Area	R3	2-10	2037	56
P1334	Planned Private Residential Area	R3	2-10	2037	56
P1335	Planned Private Residential Area	R3	2-10	2037	56
P1336	Planned Private Residential Area	R3	2-10	2037	56
P1337	Planned Private Residential Area	R3c	2-11	2037	83
P1338	Planned Private Residential Area	R3c	2-11	2037	83
P1339	Planned Private Residential Area	R3c	2-11	2037	83
P1340	Planned Private Residential Area	R3c	2-11	2037	83
P1341	Planned Reserved Area	G	2-17	2037	36
P1342	Planned Reserved Area	G	2-17	2037	36
P1343	Planned Reserved Area	G	2-17	2037	36
P1344	Planned Reserved Area	G	2-17	2037	36
P1345	Planned Private Residential Area	RR4	2-18	2026	8
P1346	Planned Private Residential Area	RR4	2-18	2026	8
P1401	Planned Government Complex	G	1-1	2036	36
P1402	Planned Government Complex	G	1-1	2036	36
P1403	Planned Subsidised Sales Flat	R2 (SSF)c	1-2	2037	98
P1404	Planned Subsidised Sales Flat	R2 (SSF)c	1-2	2037	98
P1405	Planned Subsidised Sales Flat	R2 (SSF)c	1-2	2037	98
P1406	Planned Subsidised Sales Flat	R2 (SSF)c	1-2	2037	98
P1501	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-8	2026	80
P1502	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-8	2026	80
P1503	Planned Port Back-up, Storage and Workshop	OU (PBU +SWU)	3-11	2031	80
P1601	Planned Enterprise and Technology Park	OU (E+TP)	3-32	2036	66
P1602	Planned Enterprise and Technology Park	OU (E+TP)	3-32	2036	66
P1603	Planned Enterprise and Technology Park	OU (E+TP)	3-33	2036	66

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P1604	Planned Enterprise and Technology Park	OU (E+TP)	3-33	2036	66
P1605	Planned Enterprise and Technology Park	OU (E+TP)	3-33	2036	66
P1606	Planned Reserved Area	G	3-34	2036	14
P1607	Planned Enterprise and Technology Park	OU (E+TP)	3-39	2036	66
P1608	Planned Enterprise and Technology Park	OU (E+TP)	3-39	2036	66
P1609	Planned Enterprise and Technology Park	OU (E+TP)	3-39	2036	66
P1610	Planned Enterprise and Technology Park	OU (E+TP)	3-40	2036	66
P1611	Planned Enterprise and Technology Park	OU (E+TP)	3-40	2036	66
P1612	Planned Enterprise and Technology Park	OU (E+TP)	3-40	2036	66
P1613	Planned Enterprise and Technology Park	OU (E+TP)	3-42	2036	66
P1614	Planned Enterprise and Technology Park	OU (E+TP)	3-42	2036	66
P1615	Planned Industrial Uses	I	3-43	2030	40
P1616	Planned Industrial Uses	I	3-43	2030	40
P1617	Planned Industrial Uses	I	3-44	2030	40
P1618	Planned Industrial Uses	I	3-44	2030	40
P1619	Planned Industrial Uses	I	3-44	2030	40
P1620	Planned Industrial Uses	I	3-44	2030	40
P1621	Planned Industrial Uses	I	3-45	2030	40
P1622	Planned Industrial Uses	I	3-45	2030	40
P1623	Planned Industrial Uses	I	3-45	2030	40
P1628	Planned Industrial Uses	I	3-50	2030	40
P1629	Planned Industrial Uses	I	3-50	2030	40
P1630	Planned Industrial Uses	I	3-50	2030	40
P1631	Planned Industrial Uses	I	3-50	2030	40
P1632	Planned Regional Plaza	OU	4-14	2031	4
P1633	Planned Fire Station, Ambulance Depot and Staff Quarters	G	4-1	2031	120
P1634	Planned Fire Station, Ambulance Depot and Staff Quarters	G	4-1	2031	120
P1635	Planned Fire Station, Ambulance Depot and Staff Quarters	G	4-1	2031	120
P1636	Planned Fire Station, Ambulance Depot and Staff Quarters	G	4-1	2031	120
P1637	Planned Community Hall and Sports Centre	G	4-2	2031	27
P1638	Planned Community Hall and Sports Centre	G	4-2	2031	27
P1639	Planned Community Hall and Sports Centre	G	4-2	2031	27
P1640	Planned Community Hall and Sports Centre	G	4-2	2031	27
P1641	Planned Subsidised Sales Flat	R2 (SSF)c	4-3	2031	92
P1642	Planned Subsidised Sales Flat	R2 (SSF)c	4-3	2031	92
P1643	Planned Subsidised Sales Flat	R2 (SSF)c	4-3	2031	92
P1644	Planned Public Rental Housing	RSc	4-4	2031	119
P1645	Planned Public Rental Housing	RSc	4-4	2031	119

ASR ID	Description	Use	Site ID	Intake Year #	Proposed Maximum Height (mAG)
P1646	Planned Public Rental Housing	RSc	4-4	2031	119
P1647	Planned Public Rental Housing	RSc	4-4	2031	119
P1648	Planned Public Rental Housing	RSc	4-4	2031	119
P1649	Planned Public Rental Housing	RSc	4-5	2031	143
P1650	Planned Public Rental Housing	RSc	4-5	2031	143
P1651	Planned Public Rental Housing	RSc	4-5	2031	143
P1652	Planned Public Rental Housing	RSc	4-5	2031	143
P1653	Planned Subsidised Sales Flat	R2 (SSF)c	4-6	2031	131
P1654	Planned Subsidised Sales Flat	R2 (SSF)c	4-6	2031	131
P1655	Planned Schools	E	4-8	2031	36
P1656	Planned Schools	E	4-8	2031	36
P1657	Planned Schools	E	4-8	2031	36
P1658	Planned Schools	E	4-8	2031	36
P1659	Planned Private Residential Area	R1c	4-9	2031	152
P1660	Planned Private Residential Area	R1c	4-9	2031	152
P1661	Planned Private Residential Area	R1c	4-9	2031	152
P1662	Planned Private Residential Area	R1c	4-9	2031	152
P1663	Planned Private Residential Area	R1c	4-10	2031	155
P1664	Planned Private Residential Area	R1c	4-10	2031	155
P1665	Planned Logistics Facility	OU(LF)	3-29	2036	88
P1666	Planned Commercial Uses	C	4-12c	2031	182
P1667	Planned Commercial Uses	C	4-12c	2031	182
P1668	Planned Commercial Uses	C	4-13b	2031	187
P1669	Planned Commercial Uses	C	4-13b	2031	187
P1670	Planned Commercial Uses	C	4-13b	2031	187
P1671	Planned Commercial and Residential Development	OU (C&R)	4-25c	2031	166

Note: Full names of the abbreviations of uses should be referred to the List of Abbreviations.
- Intake Year marks as "N/A" is the planned ASRs with unknown schedule. They are assumed to exist since the commencement of the Project, thus they are considered in all construction phase assessment scenarios.

Table 3.6 Representative Existing Air Sensitive Receiver within and in the vicinity of the Project Area (for Revised RODP)

ASR ID	Description	Use	Height (mAG)
A102	The Sherwood	Residential	55
A103	Botania Villa	Residential	35
A104	Tuen Mun San Tsuen	Village	10
A105	E Lu	Village	13
A106	Lingrade Garden	Residential	10
A107	Fortress Garden	Village	10
A108	Tsing Chuen Wai	Village	10
A109	Handsome Villa	Village	10
A110	Tsoi Yuen Tsuen	Village	10
A111	Kam Fung Garden	Village	10
A112	Ling Sun Kindergarten	Education	10
A201	Nai Wai	Village	10
A202	Full Park	Village	10
A203	Good Rich Garden	Village	10
A204	Kam Cheong Garden	Village	10
A205	Sun Fung Wai	Village	10
A206	Yonking Garden	Village	10
A207	Sun Fai Garden	Village	10
A208	Oaklands Court	Village	10
A209	Ling Liang Church Primary School	Education	8
A301	Mountain View	Village	10
A302	The Richtree Villas	Village	10
A303	The Matas Lodge	Village	10
A304	Sun Fung Wai Sun Tsuen	Village	10
A305	Tan Kwai Tsuen	Village	10
A306	Chung Uk Tsuen	Village	10
A307	Osmanthus Garden	Residential	13
A308	Meadowlands	Residential	10
A309	Jubilee Garden	Residential	10
A310	Tin Ha Road Playground	Recreational	1.5
A311	Bauhinia Garden	Residential	20
A312	Tan Kwai Tsuen	Village	10
A313	Parkview Garden	Residential	15
A314	The Church of Jesus Christ of Latter-day Saints	IC	18
A401	Casa De Oro	Residential	20
A402	Tin Sam Villa	Village	10
A403	Beauty Court	Residential	32
A404	Treasure Court	Residential	20
A405	Shung Tak College	Education	36
A406	Tin Sum Tsuen	Village	10
A407	Bellevue Court	Residential	10
A408	San Lee Uk Tsuen	Village	10
A409	Hung Fuk Estate	Residential	32
A410	Galore Garden	Village	10
A411	Choza Rico	Residential	10
A412	San Sang Tsuen	Village	10
A413	Greenville Park	Residential	10
A414	Shek Po Tusen	Village	10
A415	Tin Sum Tsuen	Village	10
A416	Hung Fuk Estate	Residential	32
A502	Uptown	Residential	60
A503	Le Regent	Residential	15
A504	Fui Sha Wai	Village	10
A505	San Hei Tsuen	Village	10
A506	Flowery Garden	Village	10
A507	Imperial Villas	Residential	15
A508	Ping Shan San Tsuen	Village	10
A601	Tseung Kong Wai	Village	10
A602	Farm House	Agricultural	1.5
A603	Farm House	Agricultural	1.5
A701	Kau Lee Uk Tsuen	Village	10
A702	San Uk Tsuen	Village	10
A703	Sha Chau Lei Tsuen	Village	10
A704	Ha Tsuen Shi	Village	10
A705	Yan Wu Garden	Residential	10

ASR ID	Description	Use	Height (mAG)
A706	Sik Kong Tsuen	Village	10
A707	Pui Shing Catholic Secondary School	Education	36
A708	Sik Kong Wai	Village	10
A801	Hung Uk Tsuen	Village	10
A802	Kiu Tau Wai	Village	10
A803	Tak Ying Garden	Village	15
A804	Tong Fong Tsuen	Village	10
A805	Hang Mei Tsuen	Village	10
A806	Hang Tau Tsuen	Village	10
A807	Sheung Cheung Wai	Village	10
A808	Tang Siu Tong Secondary School	Education	36
A809	Ping Shan Tin Shui Wai Public Library	Recreational	36
A810	Kwok Yat Wai College	Education	36
A811	QE School Old Students' Association Primary School	Education	36
A812	Block R, Tin Shing Court	Residential	128
A813	Block H, Tin Shing Court	Residential	128
A901	Warehouse	Others	10
A902	Farm House	Agricultural	10
A903	Farm House	Agricultural	10
A1001	San Wai	Village	10
A1002	Fung Kong Tsuen	Village	10
A1003	Mountain Royal	Residential	10
A1004	Warehouse	Others	10
A1005	Warehouse	Others	10
A1101	Lo Uk Tsuen	Village	10
A1102	Tung Tau Tsuen	Village	10
A1103	Block 8, Locwood Court	Residential	112
A1104	Oi Chiu House, Tin Shui Estate	Residential	117
A1105	Tin Shui Estate	Residential	117
A1106	Shui Lung House, Tin Shui Estate	Residential	117
A1107	Shui Sing House, Tin Shui Estate	Residential	117
A1108	Caritas Lok Kan School	Education	36
A1109	Chestwood Court	Residential	102
A1201	Locwood Court	Residential	112
A1202	Tin Yiu Estate	Residential	111
A1203	Sherwood Court	Residential	112
A1300	Sha Kong Tsuen	Village	10
A1301	Hang Hau Tsuen	Village	10
A1302	Wing Jan School	Education	10
A1303	Sha Kong Wai Tsai	Village	10
A1304	Ngau Hom Tsuen	Village	10
A1305	Ngau Hom Tsuen	Village	10
A1306	San Hing Tsuen	Village	10
A1307	Deep Bay Grove	Residential	10
A1308	Ngan Hom Tsuen San Hing Wai Tsuen Village Office	G/IC	10
A1309	Lau Fhu Shan Police Station	G/IC	15
A1401	Sha Kong Wai	Village	10
A1402	Tin Wah Estate	Residential	66
A1403	Tin Yan Estate	Residential	87
A1404	Tin Yuet Estate	Residential	128
A1405	Tin Chung Court	Residential	128
A1501	Ha Tsuen Weigh Station	G/IC	1.5
P430	Existing Residential Development (Sheffield Villas) at Site 5-23	Residential	50
P431	Existing Residential Development (Sheffield Villas) at Site 5-23	Residential	50

Note: Height of 1.5 metres above ground is the average respiratory level of human beings.

3.4.4 The assessment heights at the representative ASRs in both construction dust impact assessment and operation phase impact assessment are up to the roof level of the existing buildings or planning parameter under the Revised RODP, including 1.5 metres Above Ground (mAG), 5 mAG, 10 mAG, 20 mAG, 40 mAG and/or 80 mAG, whichever are lower than the roof level of respective ASRs.

3.5 Identification and Description of Pollutant Sources

Construction Phase

3.5.1 Major construction works for the Project including all the 12 DPs that would include site clearance, site formation works, site development works, spoil handing activity at the work areas of the park, open space, community farm and amenity area, carriageways construction including utility installation, construction of sewage treatment works and sewage pumping station and associated facilities, drainage channel re-vitalisation, and stormwater drainage facilities. The construction activities dominantly contribute to construction dust impacts would include excavation, loading/unloading of spoils, material handling and wind erosion of exposed site area. The tentative working hours would be 07:00 to 19:00 from Monday to Saturday for all works sites. Based on the construction programme, rock crushing plants with capacity greater than 5,000 tonnes per annum and concrete batching plants with total silo capacity exceeds 50 tonnes would not be expected within the site. Blasting is not anticipated based on the current preliminary information. Based on the available information of the concurrent projects as presented in **Section 2.12**, the work sites of the concurrent projects with confirmed programme would not be located within 500 m assessment area during construction phase of the Project, therefore no cumulative dust impact from these concurrent projects as mentioned in **Section 2.12** would be anticipated. Regarding the projects in the vicinity of the Project area but without the confirmed construction programme such as Tuen Mun Western Bypass (TMWB), its construction dust impact would not be considered in the assessment.

3.5.2 Apart from the above construction works, there would be some minor supporting works, including demolition works, pile removal, stormwater drain, and other improvement works for the Project. The potential dust impacts from these activities are localised and the extent of excavated areas would be limited.

3.5.3 Vehicular emission from the road traffic, emission from port back-up and logistics activities, industrial emission from chimney and nearby SPs are also the contributors to the cumulative construction dust impact.

Identification of Key Air Pollutants of Emission from Construction Activities

3.5.4 As mentioned above, major construction activities with significant particulate emission include excavation work, loading/unloading of spoils, material handling and wind erosion of exposed site area. On-site use of diesel-powered engines is the potential source for other gaseous pollutants such as NO_2 and SO_2 and smoke. The Air Pollution Control (Non-road Mobile Machinery) (Emission) Regulation came into effect in June 2015 to control emissions from diesel-powered engines. In addition, the number of such equipment required on-site will be limited and the equipment would be under normal operation with proper maintenance which is unlikely to cause significant smoke and gaseous emissions. Thus, particulates from construction activities would be the major air pollutant during construction phase. According to Annex 4 of the EIAO-TM, Total Suspended Particulates (TSP) is the criteria pollutant for construction dust impact assessment. Quantitative assessment of TSP emission impact as well as other particulates, Respirable Suspended Particulates (RSP) and Fine Suspended Particulates (FSP) stipulated in AQOs, is conducted for assessing construction dust impact due to the Project.

Operation Phase

3.5.5 Four major types of air pollution sources have been identified within the assessment area. The first source is vehicular exhaust from the road traffic within 500 m assessment area of the Project including DP1, DP2, DP5, DP6, and DP12, and existing roads including Yuen Long Highway, KSWH, and Castle Peak Road. The locations of all DP roads and existing roads are shown in **Figure 2.1**.

3.5.6 The second source is emission from port back-up and logistic activities at the New Container Back-up and Storage Area (DP7) planned at the north-west part of the Project area adjoining KSWH that can facilitate readily available cross boundary connectivity. As mentioned in **Section 2.5**, the existing brownfield operations in HSK area has created considerable environmental problems. The provision of a dedicated area for port back-up, storage and workshop uses would accommodate some of the existing brownfield operation so as to alleviate existing industrial/residential interface issues resulting from existing container storage and port back-up areas. Since new roads will be provided to directly connect this area to KSWH, it would eliminate the current movement of heavy good vehicles through built-up areas so as to reduce the vehicular emission impact on the air sensitive uses within the Project area.

3.5.7 The third source is emission from industrial chimneys, including the factories, concrete batching plants, crushing plant and asphalt plant operating under the Specified Processes (SP) licenses. Based on the latest chimney survey information collected in end 2015, these chimneys are located within existing villages, while some others are located at the existing factory buildings within Project area. Some major SP, such as crushing plant and asphalt plant are located at the Lam Tei Quarry, which would be terminated before the 1st population intake before 2024. Some existing concrete batching plants within the Project area are required to be removed due to the Project. For air quality impact assessment purpose, an industrial zone planned at the western fringe of the Project near KSWH has assumed for locating three new concrete batching plants. Since new roads will be provided to directly connect this site to KSWH and Road P1, it would avoid the movement of the trip generation from the concrete batching plants through built-up areas so as to reduce the potential vehicular emission impact on the ASRs within the Project area.

3.5.8 The fourth source is odour from the chicken farm, planned refuse transfer station (RTS) (DP11), new HSK sewage treatment works (STW) (DP8), 4 new sewage pumping stations (SPS) (DP9) and existing San Wai STW. The planned RTS and HSK STW would be located next to the existing San Wai STW which is planned to be expanded and upgraded. The new HSK STW and the planned RTS are planned at the north-west fringe of the Project which are located far away from the residential developments and educational institutes to avoid the potential odour impact. Cumulative odour impact would be expected on the nearby planned port back-up and logistic facilities. The chicken farm is an existing facility located at about 200 m southwest of Fung Kong Tsuen (historic village) which may have potential odour impact on the nearby New Container Back-up and Storage Area. The four new SPSs which would be located at different areas within the Project area may have potential odour nuisances to the nearby ASRs.

Identification of Key/Representative Air Pollutants of Vehicle Emissions from Open Road

3.5.9 Vehicular emission comprises a number of pollutants, including Nitrogen Oxides (NOx), RSP, FSP, Sulphur Dioxides (SO_2), Carbon Monoxide (CO), Lead (Pb), Toxic Air Pollutants (TAPs), etc. According to "An Overview on Air Quality and Air Pollution Control in Hong Kong" [2] published by EPD, one of the major air pollution issues is the local street-level pollution. Diesel vehicles are the main source of street-level pollution.

[2] http://www.epd.gov.hk/epd/english/environmentinhk/air/air_maincontent.html

The levels of suspended particulates (RSP/FSP) and nitrogen dioxide (NO_2) at the roadside in Hong Kong have been exceeded the Air Quality Objectives over the years. Motor vehicles, especially diesel vehicles, are the main sources of these pollutants at street level in Hong Kong. For other pollutants such as Diesel Particulate Matters (DPM), Polycyclic Aromatic Hydrocarbons (PAHs) and Volatile Organic Compounds (VOCs), due to the low concentration in vehicular emission, they are not considered as key pollutants for the purpose of this study.

3.5.10 **Nitrogen oxides (NOx)** is a major pollutant from fossil fuel combustion. According to the 2014 Hong Kong Emission Inventory Report published by EPD, marine vessels and public electricity generation are the largest NOx emission sources and accounted for 33% and 33% of the total emission in 2014, respectively. Vehicles were also a major NOx emission source, accounting for 19% of the total.

3.5.11 In the presence of O_3 and VOC, NOx would be converted to NO_2. Increasing traffic flow would inevitably increase the NOx emission and subsequently the roadside NO_2 concentration. Hence, NO_2 is one of the key pollutants for the operational air quality assessment of the Project. 1-hour and annual average NO_2 concentrations at each identified ASRs would be assessed and compared with the relevant prevailing AQOs to determine the compliance.

3.5.12 **Respirable Suspended Particulates (RSP)** refers to suspended particulates with a nominal aerodynamic diameter of 10μm or less. According to the 2014 Hong Kong Emission Inventory Report published by EPD, marine vessels became the largest RSP emission source and accounted for 36% of the total emissions in 2014. Public electricity generation and non-combustion sectors are the 2nd and 3rd major RSP emission sources, accounting for 17% and 15% of the total emission in 2014.

3.5.13 Increasing traffic flow would inevitably increase the roadside RSP concentrations. Hence, RSP is a key pollutant for the operational air quality assessment of the Project. The 24-hour and annual average RSP concentrations at each identified ASR would be assessed and compared with the relevant prevailing AQO to determine the compliance.

3.5.14 **Fine Suspended Particulates (FSP)** refers to suspended particulates with a nominal aerodynamic diameter of 2.5μm or less. According to the 2014 Hong Kong Emission Inventory Report published by EPD, navigation became the largest FSP emission source and accounted for 42% of the total emissions in 2014. Road transport is also a major FSP emission source, accounting for 17% of the total emission in 2014.

3.5.15 Similar to the RSP, increasing traffic flow would increase the roadside FSP. Hence, FSP is also a key pollutant for the operational air quality assessment of the Project. The 24-hour and annual averaged FSP concentrations at each identified ASR would be assessed and compared with the relevant prevailing AQO to determine the compliance.

3.5.16 **Sulphur dioxide (SO_2)** is formed primarily from the combustion of sulphur-containing fossil fuels. In Hong Kong, power stations and marine vessels are the major sources of SO_2, followed by fuel combustion equipment and motor vehicles.[3] SO_2 emission from vehicular exhaust is due to the sulphur content in diesel oil. According to the 2014 Hong Kong Emission Inventory Report released by EPD, SO_2 emissions from vehicles had been substantially reduced (<1% of total) after the introduction of Euro V diesel from December 2007, whose sulphur content are capped at 0.001%. Similarly, industrial sector and construction had been using Euro V diesel since 2009. Starting from 1 October 2008, sulphur content of liquid fuel shall not exceed 0.005% for commercial and industrial process, according to the Air Pollution Control (Fuel Restriction) (Amendment) Regulation (the "Amendment Regulation").

[3] Air Quality in Hong Kong 2014

3.5.17 In view that road transport only contributes a very small amount of SO_2 emission, relatively low measured concentrations and the adoption of low-sulphur and ultra-low-sulphur fuel under the existing government policy, SO_2 would not be a critical air pollutant of concern.

3.5.18 **Carbon Monoxide (CO)** is a typical pollutant emitted from fossil fuel combustion and comes mainly from vehicular emissions. With reference to the "Air Quality in Hong Kong 2014"[4], the highest 1-hour average (3,590 µg/m^3) and the highest 8-hour average (2,609 µg/m^3) CO concentrations were both recorded at the Central and Mong Kok roadside stations; these values were around one eighth and one fourth of the respective AQO limits. In view that there is still a large margin to the AQOs, CO would not be a critical air pollutant of concern.

3.5.19 **Ozone (O_3)** is produced from photochemical reaction between NOx and VOCs in the presence of sunlight, which will not be generated by this Project. Concentration of O_3 is governed by both precursors and atmospheric transport from other areas. When precursors transport under favorable meteorological conditions and sunlight, ozone will be produced. This explains why higher ozone levels are generally not produced in the urban core or industrial area but rather at some distance downwind after photochemical reactions have taken place. In the presence of large amounts of NOx in the roadside environment, O_3 reacts with NO to give NO_2 and thus results in O_3 removal. O_3 is therefore not considered as a key air pollutant for the operational air quality assessment of a road project.

3.5.20 **Lead (Pb)** is not considered as a critical air pollutant of concern. The sale of leaded petrol has been banned in Hong Kong since April 1999. According to the "Air Quality in Hong Kong 2014", the overall 3-month averages were ranging from 29 ng/m^3 (at Central/Western, Kwai Chung and Mong Kok) to 35 ng/m^3 (Yuen Long). The measured concentrations were well below the AQO limits of 500 ng/m^3.

3.5.21 **Toxic Air Pollutants (TAPs)** is a type of the pollutants found in vehicular exhaust, which are known or suspected to cause cancer or other serious health and environmental effects. With reference to EPD's Assessment of Toxic Air Pollutant Measurements in Hong Kong Final Report [5], monitored TAPs in Hong Kong include diesel particulate matters (DPM), toxic elemental species, dioxins, polychlorinated biphenyls (PCBs), polycyclic aromatic hydrocarbons (PAHs), carbonyls, and volatile organic compounds (VOCs). According to the results of Assessment of Toxic Air Pollutant Measurements in Hong Kong Final Report and Sources of PCB emissions [6], vehicular emission is not considered as primary source of dioxins, PCBs, carbonyls and most toxic elemental species in Hong Kong. Therefore, these pollutants are not considered as key pollutants for quantitative assessment for the operation phase of a road project.

3.5.22 **Diesel Particulate Matters (DPM)**, as part of the overall Respirable Suspended Particulates (RSP), is one of the most important parameter contributing to the overall health risk of the population. Local vehicular emission is one of the major sources of DPM.

3.5.23 EPD has embarked on the following three key programmes to reduce the diesel particulate level at the roadside [7]: (a) the LPG taxi and light bus program; (b) the introduction of an advanced test to check diesel vehicle smoke emission; and (c) the retrofit of pre-Euro diesel commercial vehicles with diesel oxidation Catalysts (DOCs). According to EPD's website [8], franchised bus companies have also retrofitted their Euro I buses with diesel oxidation catalysts (DOCs) and Euro II and III buses with diesel

4 http://www.aqhi.gov.hk/api_history/english/report/files/AQR2014e_Update0616.pdf
5 http://www.epd.gov.hk/epd/english/environmentinhk/air/studyrpts/assessment_of_tap_measurements.html
6 http://www.eea.europa.eu/publications/EMEPCORINAIR5/Sources_of_PCB_emissions.pdf/view
7 http://www.epd.gov.hk/epd/english/news_events/legco/files/EA_Panel_110526a_eng.pdf
8 http://www.epd.gov.hk/epd/english/environmentinhk/air/prob_solutions/cleaning_air_atroad.html

particulate filters (DPFs). A DPF can reduce particulate emissions from diesel vehicles by over 80%.

3.5.24 As recommended by EPD's Assessment of Toxic Air Pollutant Measurements in Hong Kong Final Report, elemental carbon (EC) is used as a surrogate for DPM, and with reference to Measurements and Validation for the 2008/2009 Particulate Matter Study in Hong Kong [9], EC showed a significant decrease in concentration from 2001 to 2009 in Hong Kong, i.e. -47.5%, -30.0% and -28.3% at Mong Kok, Tsuen Wan and Hok Tsui Monitoring Sites, respectively. With the continual efforts by EPD to reduce particulate emission from the vehicular fleet, a discernible decreasing trend is noted in the level of particulate matter. Therefore, DPM is not selected as representative pollutant for quantitative assessment for this project.

3.5.25 **Polycyclic Aromatic Hydrocarbons (PAHs)** are organic compounds of two or more fused benzene rings, in linear, angular or cluster conformations. Local vehicular traffic is also an important source of PAHs. For this group, the most important PAH is Benzo[a]pyrene, and it is often selected as a marker for the PAHs[10]. The EU Air Quality Standards for PAHs (expressed as concentration of Benzo[a]pyrene) is 1 ng/m^3 for annual average[11]. With reference to "Air Quality in Hong Kong 2014", annual average concentrations of PAHs (Benzo[a]pyrene) measured at EPD's TAP monitoring stations (Tsuen Wan and Central/Western) were 0.16 ng/m^3 and 0.13 ng/m^3, which is far below the EU Standards. Thus, PAHs are not considered as key pollutants for quantitative assessment for this project.

3.5.26 **Volatile Organic Compounds (VOCs)** are of great concern due to the important role played by them in a range of health and environmental problems. The US EPA has designated many VOC, including those typically found in vehicular emission, as air toxics. According to Assessment of Toxic Air Pollutant Measurements in Hong Kong Final Report, 2003, among the VOC compounds, benzene and 1,3-butadiene are the most significant ones for Hong Kong. The UK Air Quality Standards for benzene and 1,3-butadiene are 5.0 µg/m^3 and 2.25 µg/m^3 respectively[12]. Accordingly to "Air Quality in Hong Kong 2014", annual average concentrations of benzene at EPD's TAP monitoring stations were 1.31 µg/m^3 (Central/Western) – 1.47 µg/m^3 (Tsuen Wan), respectively. The levels of 1,3-butadiene were 0.05 µg/m^3 and 0.08 µg/m^3 for Central/Western and Tsuen Wan districts respectively. They are far below the UK Standards. Thus, VOCs are not considered as key pollutants for quantitative assessment for this project.

Identification of Key/Representative Air Pollutants of Ports Back-up and Logistics Activities

3.5.27 The operations for port back-up and logistics activities would generally include container storage, goods dispatching and logistic activities, and vehicle movement. Such operations would mainly involve on-road and off-road vehicles. It is anticipated that the key pollutants would be the same as that for the open road vehicular emission. Thus, NO_2, RSP and FSP would be the key concerns. The locations of the emission sources of port back-up and logistics activities are shown in **Appendix 3.2**.

[9] http://www.epd.gov.hk/epd/english/environmentinhk/air/studyrpts/files/HKEPDFinalReportRev_11-29-10_v2.pdf
[10] Assessment of Toxic Air Pollutant Measurements in Hong Kong Final Report
[11] http://ec.europa.eu/environment/air/quality/standards.htm
[12] http://www.medway.gov.uk/environmentandplanning/environmentalhealth/airquality/airqualityfordevelopers.aspx

Identification of Key/Representative Air Pollutants of Chimneys and Specified Processes

3.5.28 There are some chimneys located within or in the vicinity of the Project. All installation or alteration of fuel-burning equipment and chimney is controlled under the Air Pollution Control (Furnaces, Oven & Chimneys) (Installation & Alteration) Regulations. The key pollutants generated from combustion activities would be NO_2 and SO_2. The locations of these existing chimneys are shown in **Appendix 3.2**.

3.5.29 Similarly, any industrial activities which involve certain polluting industrial processes known as "Specified Processes" (SP) would be controlled under the Air Pollution Control Ordinance. Through the terms and conditions of the SP licence, the owners of the premises used for the conduct of specified processes are required to use the Best Practicable Means (BPM) to prevent air pollution. The key pollutants induced from the SP within the assessment area have been extracted from the EPD licensing records.

3.5.30 Four existing concrete batching plants (also controlled under SP) within the Project area are required to be removed to make way for the necessary developments under the Project. For air impact assessment purposes, Site 3-47, which is at the western portion next to the northern portal of the TMWB, has assumed for locating three new concrete batching plants in future. Under the latest BPM for concrete batching plant, the new batching plant will be under a tighter emission control with the point source emission concentration limit of 10 mg/m^3 for particulates (rather than the previous 50 mg/m^3) starting from 1st January 2018. The tentative locations of the three concrete batching plants at Site 3-47 are shown in **Appendix 3.2**.

Identification of Key/Representative Air Pollutants of Existing Chicken Farm, Planned RTS, Upgraded Sai Wai STW, New HSK STW and New Sewage Pumping Stations

3.5.31 The existing chicken farm, planned RTS, upgraded Sai Wai STW and new HSK STW (DP8) would cause potential odour impact to the Project sites. As the planned RTS and new STWs are newly planned facilities, their odour emission can be easily abated by design, i.e. to be equipped with odour control technology such as deodorising units. The potential odour sources of the upgraded Sai Wai STW would also be enclosed and the vented air would be treated in the deodorising unit according to the approved EIA Report of “Upgrading and expansion of San Wai Sewage Treatment Works and expansion of Ha Tsuen Pumping Station”. However, the existing chicken farm may not have adequate odour abatement by design and would pose potential odour impact on the planned sites in the Project area. The odour impacts due to existing chicken farm, RTS and the sewage treatment facilities are assessed in this EIA study. The location of the existing chicken farm is shown in **Appendix 3.6** while the locations of the Sai Wai STW, new HSK STW and planned RTS are indicated a figure attached in **Appendix 3.8**.

3.5.32 Four new SPS (DP9) would be constructed at Sites 2-34, 3-41, 3-48 and 4-35 to convey the existing sewage flow to San Wai STW and new sewage flow to the new HSK STW. The locations of these SPSs are indicated in **Figure 1.3**. The design capacity of the SPS at Site 2-34 (SPS1), Site 3-41 (SPS2), Site 3-48 (SPS3) and Site 4-35 (SPS4) would be 27,000 m^3 per day, 39,500 m^3 per day, 11,000 m^3 per day and 68,000 m^3 per day, respectively. It may have potential odour nuisance during operation of SPSs. However, the odorous facilities of these four SPSs would be enclosed and designed with the provision of deodourisers to treat the exhausted gas, insignificant odour impact from the operation of the SPSs would be anticipated.

3.5.33 The refuse collection vehicles should be of fully-enclosed type to avoid odour spread. In addition, the designed route for the refuse collection vehicle to/from the RTS would be via Road P1 and then Road D1 so as to minimise the potential odour impact from these vehicles to the ASRs within the Project area. The drivers of refuse collection vehicles (RCV) should follow the procedures for cleaning of the RCV in accordance with Code of Practice on the Operation of Refuse Collection Vehicles issued by EPD and Transport Department so as to keep the RCV in a clean and hygienic condition. Therefore, no adverse odour nuisance would be expected from the operation of RCVs.

3.6 Assessment Approach and Methodology (Construction Phase Dust Impact)

General Approach

3.6.1 According to "Guidelines on Assessing the 'TOTAL' Air Quality Impacts", three components of contribution should be considered in evaluating the air quality impact due to the Project upon ASRs, namely primary, secondary and background contributions.

Primary Contributions

3.6.2 Primary contributions are the project-induced emission which contributes to the local air quality impact. Thus, construction dust emission associated with the construction works of the Project is the primary contributions during the construction phase.

3.6.3 Construction activities with significant particulate emission are identified from the construction method suggested by engineering design. Construction dust impact is predicted based on emission factors from US Environmental Protection Agency (USEPA) Compilation of Air Pollution Emission Factors (AP-42), 5th edition and activity data from the engineer design. The major dusty construction activities for the Project to be concerned and considered in the modelling assessment include:

- Site Formation Works for all planned developments within the Project area including all DPs
  - Excavation and material handlings within the construction site modelled as heavy construction activities
  - Spoil handling activity at the work areas of the park, open space, community farm and amenity area (Spoil handling activity is considered in Tier 2 and Tier 3 assessment to reflect the work areas with minor excavation works and spoil loading/unloading.)
  - Wind erosion of open active site during non-working hours
- Carriageway construction including Road P1 (DP1), Road D1 - D8 (DP2), DP5, DP6 and DP12
  - Excavation, material handlings and backfilling within the construction site modelled as heavy construction activities
  - Wind erosion of open active site during non-working hours

3.6.4 Works activities and plant would not be concentrated in certain areas of the site close to ASRs for an extended period of time during the construction period. However, notwithstanding that such a scenario would not be expected to occur, a hypothetical Tier 1 screening test assuming 100% active area of construction site of the Project with mitigation measures in place has been undertaken for predicting hourly average TSP, daily average RSP and FSP levels. It aims to highlight the hot spot locations where construction dust may potentially become an issue. However, it should be emphasised that Tier 1 screening test is a hypothetical one which is very conservative and does not occurred in reality.

3.6.5 A tier results have allowed a more focused next tier assessment to be undertaken at the specific hot spot locations where TSP/RSP/FSP non-compliance is predicted under the Tier 1 screening test. A focused next Tier assessment, Tier 2, is undertaken whereby monthly average works areas of the concurrent work contracts according to the construction work programme are considered as active work areas in the assessment. The monthly average works areas per works contract is depending on the total work area and its construction period of its works contract. The calculation of the monthly average works areas of each works contract is presented in **Appendix 3.1**. The locations of the works contracts are shown in the Development Phasing Plan as attached in **Appendix 3.13**. In Tier 2 assessment, the active work areas of concurrent work contracts are positioned closest to the potentially worst affected ASRs, and the individual affected ASR would be influenced by one active work area of each concurrent work contract. Hoarding is erected along the construction site boundary adjacent to the non-construction areas such as residential, educational institutes or recreation area in use so as to minimise the dust impact. When the active works areas are positioned where hoarding is erected, dust emission height is increased to 3 meters above ground. The relevant arrangement is further discussed in **Section 3.10.3** and **Section 3.10.9**.

3.6.6 For the ASRs which would still be adversely affected by the construction works under Tier 2 assessment, a Tier 3 assessment would be conducted with further limitation of size of the active work area of the work contract closest to the affected ASR to one third of monthly average work area of the work contract. The limitation of the active work area in Tier 2 and Tier 3 assessment is workable as confirmed by the Engineer, and would be implementation by means of contract management. Dust emission height is increased to 3 meters above ground when hoarding is erected at the construction site boundary adjacent to the non-construction areas as discussed in **Section 3.6.5**.

3.6.7 In terms of the annual average prediction, the highest percentage active work area is less than 10% during any short period of time based on the engineers' estimation. It is assumed that 10% of hourly/daily assessment emission rates are adopted for the long-term annual predictions for RSP/FSP in both assessment scenarios.

3.6.8 The impact of fugitive dust sources on air quality depends upon the quantity as well as the drift potential of the dust particles emitted into the atmosphere. Large dust particles (i.e. over 100 μm in diameter) would settle out near the source and particles that are between 30 and 100 μm in diameter are likely to undergo impeded settling. The major dust impacts are likely to arise from particles less than 30 μm in diameter, which have a greater potential to disperse over greater distances.

3.6.9 According to United States Environmental Protection Agency (USEPA) AP-42[13], construction dust particles may be grouped into five particle size classes. The particle size distribution as referenced to AP-42, Section 13.2.4 (Aggregate Handling Emission), is normalised to obtain the particle size distribution in PM_{30} (TSP). Details are shown in **Table 3.7**. The mass of RSP, which is defined as suspended particulates in air with aerodynamic size of 10 μm or smaller, is estimated to be 47.3% of PM_{30}, i.e. RSP emission = 47.3% × TSP emission. Similarly, the mass of FSP is estimated to be 7.2% of PM_{30}.

[13] USEPA, AP-42 Compilation of Air Pollution Emission Factors (AP-42), Section 13.2.4.3, 1st Table, November 2006

Table 3.7 Particle Size Distribution

Size Range (μm)	Average size(μm)	Distribution	Normalized Distribution
15 – 30	22.5	26%	35%
10 – 15	12.5	13%	18%
5 – 10	7.5	15%	20%
2.5 – 5	3.75	15%	20%
< 2.5	1.25	5%	7%

3.6.10 According to the preliminary engineering design information, dust control measures have been incorporated into the design, as presented in **Table 3.8**. These dust control measures have also been taken into account in the assessment.

Table 3.8 Dust Emission Design Control Measures

Process	Description	Dust Emission Design Control Measures
Haul road within construction sites facilities	Transportation of spoil	All road surfaces within the construction sites would be paved and watering frequently along the haul road during the working hours.
Unloading of materials	Unloading of spoil materials	The unloading process would be undertaken within a 3-sided screen with top tipping hall. Water spraying and flexible dust curtains would be provided at the discharge point for dust suppression.
Trucks	Vehicles leaving the loading/unloading facility	Vehicles would be required to pass through the wheel washing facilities provided at the site exit.

3.6.11 Details of emission factors of dusty construction activities are given in **Table 3.9**. The detail calculation of dust emission rates is presented in **Appendix 3.1**. The development phasing plan is attached in **Appendix 3.13**.

Table 3.9 Emission Factors for Dusty Construction Activities

Source	Activity	Emission Factor	Remarks
Site formation work or Carriageway construction	Heavy Construction Activities	E(TSP)=2.69 Mg/hectare /month of activity E(RSP) = 2.69 x 47.3% Mg/hectare/month [1] E(FSP) = 2.69 x 7.2% Mg/hectare/month [1]	100% area actively operating for hourly and daily concentration prediction 10% area actively operating for annual concentration prediction AP-42, Section 13.2.3, 1/95 ed.
	Wind Erosion	E(TSP)=0.85 Mg/hectare /year E(RSP) = 0.85 x 47.3% Mg/hectare/month [1] E(FSP) = 0.85 x 7.2% Mg/hectare/month [1]	100% area actively operating for hourly and daily concentration prediction 10% area actively operating for annual concentration prediction AP-42, 5th ed., Table 11.9.4

Source	Activity	Emission Factor	Remarks
Work Area at Park, Open Space, Community Farm and Amenity Area (For Tier 2 and 3 Assessment only)	Spoil Handling Activity	E(TSP) = 0.00076 kg/Mg E(RSP) = 0.00076 x 47.3% kg/Mg E(FSP) = 0.00076 x 7.2% kg/Mg	100% area actively operating for hourly and daily concentration prediction AP42, Section 13.2.4
	Wind Erosion	E(TSP)=0.85 Mg/hectare /year E(RSP) = 0.85 x 47.3% Mg/hectare/month [1] E(FSP) = 0.85 x 7.2% Mg/hectare/month [1]	100% area actively operating for hourly and daily concentration prediction AP-42, 5th ed., Table 11.9.4

Remarks:
[1] USEPA AP-42, 5th ed. 11/06 ed. S13.2.3

3.6.12 For the prediction of the highest hourly average TSP, the 10th highest daily average and annual average RSP and FSP concentrations, 12-hour (07:00-19:00) per day, 7 days a week is assumed for the construction period in the assessment. Only wind erosion is assumed for other non-working hours (19:00 to 07:00 of the following day).

Secondary Contributions

3.6.13 Secondary contributions are the air pollutant emissions in the immediate neighborhood of the Project Site contributing further to the local air quality impact. Any particulates emission sources within 500 m of the Project boundary with notable impact shall be identified and considered in the construction phase impact assessment. The secondary contribution in particulates comes from the vehicular emission and industrial emissions within and in the vicinity of the Project area. The details of the emission inventories of the vehicular traffic and industrial sources refer to **Section 3.7**. Based on the available information of the concurrent projects as presented in **Section 2**.**12**, the work sites of the concurrent projects with confirmed programme would not be located within 500 m assessment area during construction phase of the Project, therefore no cumulative dust impact from these concurrent projects would be anticipated. Regarding the projects in the vicinity of the Project but without the confirmed construction programme (e.g. TMWB), its construction dust impact would not be considered in the assessment as the relevant engineering information is unavailable.

Background Contributions

3.6.14 As suggested by "Guidelines on Assessing the 'TOTAL' Air Quality Impacts", an integrated modelling system, Pollutants in the Atmosphere and their Transport over Hong Kong model (PATH-2016) which is developed and maintained by EPD is applied to estimate the background pollutant concentrations. The details of the background concentrations for the assessment area refer to **Section 3.7**.

Determination of the Assessment Year

3.6.15 Referring to the construction programme as presented in **Appendix 2.4**, the construction work is targeted to commence in Year 2019 and be completed by Year 2038. For these 19-years construction period, the construction works in the Project area would be generally divided into five stages work contracts:

- Advance Works (Year 2019 – 2029)
- Stage 1 Works (Year 2022 – 2025)
- Stage 2 Works (Year 2026 – 2031)

- Stage 3 Works (Year 2031 – 2035)
- Stage 4 Works (Year 2031 – 2038)

3.6.16 A review is conducted on the tentative construction and implementation programme, as presented in **Appendix 3.13**. Advance Works, Stage 1 Works and Stage 2 Works contracts would be commenced in Year 2019, Year 2022 and Year 2026, respectively. These three stages work contracts would be overlapped and all dusty construction activities would be completed by Year 2030 (while the completion date of Stage 2 Works is in Year 2031, all the site formation works, which are dust generating activities, would be completed by Year 2030. Hence, for construction dust assessment purpose, Stage 2 Works are included in Scenario 1 in **Table 3.10** below only). Stage 3 Works and Stage 4 Works contracts would be commenced in Year 2031 and would be completed by Year 2035 and Year 2038, respectively. While the completion date of Stage 4 Works is in Year 2038, all the site formation works, which are dust generating activities, would be completed by Year 2036. For construction dust assessment purpose, Scenario 2 is up to Year 2036. Therefore, the construction dust impact assessment is divided into two assessment scenarios, "Year 2019 – Year 2030" Scenario for Advance Works, Stage 1 Works and Stage 2 Works contracts and "Year 2031 – Year 2036" Scenario for Stage 3 Works and Stage 4 Works contracts. Both assessment scenarios are based on the conservative assumptions as mentioned in **Table 3.10** to determine the worst construction dust impact on the representative ASRs over the whole construction period.

Table 3.10 Assumption for Two Construction Dust Assessment Scenarios

Scenario	Justification	Work Site Area & ASRs Considered in the Assessment
Scenario 1: Year 2019 – Year 2030	A few development sites under Advance Works, Stage 1 Works and Stage 2 Works would commence the site formation works during Year 2019 – 2025, but a number of development sites would commence the site formation works in Year 2026. All works under Advance Works, Stage 1 Works and Stage 2 Works would be completed by Year 2031. As a conservative approach, assume all development sites under Advance Works, Stage 1 Works and Stage 2 Works undertaking the site formation works concurrently in the assessment except Sites 2-18, 2-19, 3-6, 3-7, 3-8, 4-20, part of 5-24 and 5-26. As the site formation of these sites would be completed in Year 2021 – Year 2026, these sites are considered as ASRs in the assessment. The intake year of Site 5-32 would be by Year 2030, therefore, this site is considered as construction dust source and the dust impact on this site is addressed in Scenario 2.	All development sites (work sites under Advance Works, Stage 1 Works and Stage 2 Works) undertaking site formation works during Year 2019 –2030 are assumed to be conducted concurrently. All representative existing ASRs and planned ASRs occupied between Year 2019 – 2030 are considered in the assessment, except the ASRs which would intake at end Year 2030 would be considered in Scenario 2: Year 2031 – 2036. The locations of the ASRs and the work site area considered in the model are shown in **Figure 3. 2**.

Scenario	Justification	Work Site Area & ASRs Considered in the Assessment
Scenario 2: Year 2031 - 2036	A number of development sites under Stage 3 Works and Stage 4 Works would commence the site formation works since Year 2031 and all these site formation works would be completed by Year 2036. As a conservative approach, assume all development sites under Stage 3 Works and Stage 4 Works undertaking the site formation works concurrently in the assessment.	All development sites (work sites under Stage 3 Works and Stage 4 Works) undertaking site formation works during Year 2031 – 2036 are assumed to be conducted concurrently. All representative existing ASRs and planned ASRs occupied before Year 2036 within the Project area are considered in the assessment. The locations of the ASRs and the work site area considered in the model are shown in **Figure 3. 3**.

3.6.17 The cumulative emission sources considered in these two scenarios are listed in **Table 3.11** and **Table 3.12**, respectively. As mentioned in **Section 3.5.1**, the construction programme for TMWB is not available at the time of the assessment, its construction dust impact would not be considered in the assessment, however, its operation phase vehicular emission impact would be considered in the assessment since Year 2031.

Table 3.11 Cumulative Dust Emission Sources in Scenario 1: Year 2019-2030

Emission	Cumulative Dust Sources
Construction Dust	Dust sources from Advance Works, Stage 1 Works and Stage 2 Works
Open Road Traffic	• With a section of Road P1 (DP1), a section of Road D1 (DP2), junction of D8/P1 (DP5) connecting "PBU+SWU", and all existing roads within 500 m assessment area. The locations of the roads considered in the model refer to **Appendix 3.4**. • Traffic data: Year 2026 • Vehicular emission factor: Year 2026 (There is no new road opening under the Project until Year 2024, therefore traffic before Year 2024 is irrelevant for study. Referring to the emission burden calculation as shown in **Table 3.20**, Year 2026 is considered the worst emission burden year between Year 2024 and Year 2030. Thus traffic flow and emission factor of Year 2026 are used.)
Industrial Emissions	• Existing chimneys and SPs within 500 m assessment area • 3 new concrete batching plants (CBPs) and 2 relocated chimneys within the Project area • Non-road mobile source from "PBU+SWU" at Site 3-6, 3-7 & 3-8

Table 3.12 Cumulative Dust Emission Sources in Scenario 2: Year 2031-2036

Emission	Cumulative Dust Sources
Construction Dust	Dust sources from Stage 3 Works and Stage 4 Works
Open Road Traffic	• All existing roads within 500 m assessment area and planned roads including all DP roads within the Project area. The locations of the roads considered in the model refer to **Appendix 3.4**. • Traffic data: Year 2036 (which has considered the traffic flow of all DPs roads) • Vehicular emission factor: Year 2031 (Year 2031 is the highest emission factor but it covers parts of new roads within the Project area only. In Year 2036, all new roads within the Project area including all DP roads would be open, therefore, the NOx emission burden is calculated based on Year 2036 traffic flow and Year 2031 emission factor. The NOx, RSP and FSP emission burden based on Year 2036 traffic flow and Year 2031 emission factor would be 233.2, 8.728 and 8.114 tonnes per annum, higher than the emission burden in any year from Year 2031 to 2039 as shown in **Table 3.20**)
Industrial Emissions	• Existing chimneys and SPs within 500 m assessment area • 3 new CBPs and 2 relocated chimneys within the Project area • Non-road mobile source from "PBU+SWU" at Site 3-1 to 3-14 • Boiler from the proposed hospital
Portal Emission and Ventilation Building	TMWB

Dispersion Modelling & Concentration Calculation

3.6.18 AERMOD, Gaussian dispersion model recommended by HKEPD, is adopted to assess the potential primary dust impact from the construction works and secondary ones due to industrial/chimney emissions. Assessment heights up to the roof level of the respective ASRs are adopted for the construction dust impact assessment.

3.6.19 CALINE4, the EPD approved line source air dispersion model developed by the California Department of Transport is used to assess the secondary contribution due to vehicular emissions from the roads within 500 m assessment area.

3.6.20 Under the current EPD guideline, the hourly meteorological data including wind speed, wind direction, and air temperature from the relevant grids from the WRF Meteorological data (same basis for PATH-2016 model), are employed for the model run.

3.6.21 The future prevailing RSP background concentrations for both assessment scenarios are extracted from the PATH-2016 model run for Year 2020 without vehicular emission for the corresponding grids. The traffic emission within the assessment area in the PATH model has been removed to avoid the double counting. The FSP concentration is calculated in accordance with the EPD's "Guidelines on the Estimation of $PM_{2.5}$ for Air Quality Assessment In Hong Kong". (That is Annual FSP background = 0.71 * Annual RSP from PATH model and Daily FSP background = 0.75 * Daily RSP from PATH-2016 model.). The hourly TSP concentration are assumed to be the same as that for RSP, as the best estimation.

3.7 Assessment Approach and Methodology (Operation Phase Air Quality Impact)

General Approach

3.7.1 Similar to the construction dust impact assessment, in accordance with EPD's "Guidelines on Assessing the 'TOTAL' Air Quality Impacts", three components of contribution as follows are considered in evaluating the air quality impact due to the Project upon ASRs.

- Primary contributions: project induced;
- Secondary contributions: pollutant-emitting activities in the immediate neighbourhood. Any emission sources in an area within 500 m radius of the Project area with notable impacts are identified and included in an air quality assessment to cover the short-range contributions using EPD's approved local-scale Gaussian type model; and
- Background contributions: Year 2020 background concentrations from PATH-2016 (Pollutants in the Atmosphere and their Transport over Hong Kong) model is adopted to estimate future concentrations for Year 2024 and afterwards. Vehicular emissions within the assessment area in the PATH model is removed to avoid double counting.

3.7.2 The following approach has been adopted for the operation phase cumulative air quality impact assessment:

- Use Gaussian dispersion models i.e. CALINE4 for line sources and AERMOD for point, area and volume sources to quantify the air quality impacts from sources including open road emission, TMWB portal emission, emission from TMWB ventilation building, container back-up area and industrial emissions in close proximity of the identified ASRs.
- The PATH-2016 model results are added to the sum of the CALINE4 and AERMOD model results sequentially on an hour-to-hour basis to derive the short-term and long-term cumulative impacts at the ASRs. The maximum hourly, daily and annual average results have been then calculated in accordance with the Title 40, Code of Federal Regulations, US Environmental Protection Agency (USEPA 40 CFR) Part 51 "Revision to the Guideline on Air Quality Models, Version 2005". The pollutant concentration predicted at an ASR amongst the 8760 hours (a year) have been ranked/ averaged to assess the cumulative impact. The number of exceedances for each ASR have been counted and compared with the acceptance values in the prevailing AQO criteria.

Emission from Open Road Traffic

3.7.3 The principal pollutants associated with traffic emission are NO_2, RSP and FSP. Major roads in the proximity of the assessment area include Yuen Long Highway, KSWH, Castle Peak Road, Ping Ha Road and Hung Tin Road. Local access roads, namely Tan Kwan Tsuen Road, Ha Tsuen Road, and Tin Ha Road are also potential sources of air pollution due to the heavy container traffic accessing the nearby "PBU+SWU".

3.7.4 The projected 24-hour traffic flows and vehicle compositions which have taken into account the development of concurrent projects including Yuen Long South, Kam Tin South and Tuen Mun Central are adopted in this air quality assessment. Owing to the release of the 2011-based Territorial Population and Employment Data Matrices (TPEDM) population and employment data, a new Technical Note illustrating the methodology to produce the traffic data (include 24-hr traffic flow, trips, daily vehicle-kilometer-travelled (VKT) & speed fraction) has been issued to the Transport Department (TD) for comment. TD has no comment on the Technical Note and the correspondence from TD is attached in **Appendix 4.7.1**. The traffic data is presented in **Appendix 3.3**.

3.7.5 EMFAC-HK v3.1.1 model is adopted to estimate the vehicle emission rates of nitrogen oxides and particulate matters, both RSP and FSP. The "vehicle fleet" refers to all motor vehicles operating on roads within this assessment area. The modelled fleet is broken down into 16 vehicle classes based on the information in the Transport Monthly Digest and vehicle population provided by EPD.

3.7.6 The input parameters and model assumptions made in EMFAC-HK model are summarised as follows.

- Referring to the EPD's Guideline on Modelling Vehicle Emissions, "Burden mode" is used for calculating area-specific emission inventories. It is selected for this Project, since it can provide hourly vehicular emissions, taking into account of ambient conditions and speeds combined with vehicle activity, i.e. the number of vehicles, the kilometers driven per day and the number of daily trips.

- Each vehicle class has diverse technological factors in different years. According to the underlying assumption in the EMFAC-HK model, each vehicle class can be modelled by the individual behaviour of the unique technology groups. Each technology group represents the vehicles from the same class but has distinct emission control technologies, has similar in-use deterioration rates and respond the same to repair. It means that the vehicles from the same class have the same emission standards or specific equipment installed in them (e.g. multi-port fuel injection, three-way catalyst, adaptive fuel controls, etc.) which make them have the same performance.

- The "2013 Licensed Vehicle by Age and Technology Group Fractions" provided in EPD's website, is adopted in this assessment. Since the provided exhaust technology fractions are only up to Year 2013 at the time of the assessment, those after Year 2013 are projected in accordance with EPD's Guideline on Modelling Vehicle Emissions – Appendix 3 "Implementation Schedule of Vehicle Emission Standards in Hong Kong (Updated as at 29 December 2015)" and Appendix 4 "EMFAC-HK Technology Group Indexes (Released in January 2016)".

- According to EPD's Guideline on Modelling Vehicle Emissions - Appendix 2, the implementation schedules of Euro V and Euro VI standards are in the middle of a year for some vehicle classes or fuel types. Since the detailed fraction data is not available after Year 2010, as a conservative approach, the exhaust technology fractions of these vehicle classes or fuel types are assumed to be kept as the previous standards fully for the scheduled year, while upgraded to the higher standards fully at the following year. Evaporative technology fraction in the model is based on the default value.

- As recommended in the EPD's Guideline on Modelling Vehicle Emissions, default vehicle populations forecast in EMFAC-HK is used.

- Vehicle-kilometer-travelled (VKT) represents the total distance travelled on a weekday. The VKT is calculated by multiplying the number of vehicles, which based on the highest predicted hourly traffic flow, and the length of road travelled in the

assessment area. The diurnal variation of VKT in the assessment area is provided by the traffic consultant, and the input in the model is by vehicle/fuel/hour.

- The default accrual rates in EMFAC-HK are estimated from the local mileage data adjusted to reflect the total VKT for each vehicle class. The default value has been used.

- For those roads with cold starts as advised by the traffic consultant, the diurnal variation of daily trips in the assessment area of the Project applied in the EMFAC-HK model is provided by the traffic consultant.

- Speed fraction represents the percentage in different speed ranges of each vehicle type weighted by VKT. The speed limits of existing road were made reference to the Traffic AIDs (plan marked the road marking, traffic sign and speed limits) from TD, while the speed limits of the proposed road are provided by traffic consultant.

- In accordance with the Road Traffic Ordinance, for any road with design speed limit of 70 kph or above, the speed limit for medium goods vehicles, heavy goods vehicles, buses and buses would be limited to not more than 70 kph. Thus, the speeds of medium goods vehicles, heavy goods vehicles and buses from the flow speed or 70 kph, whichever is lower, are adopted. For the public light buses, the speed limit should be limited to post speed of the carriageway or 80 kph, whichever is lower, are adopted.

3.7.7 According to "Guidelines on Choice of Models and Model Parameters", NO_2/NOx ratios of each vehicle type varies from type to type. NO_2/NOx ratio of each vehicle is extracted from the EMFAC-HK. With these extracted ratios, the emission rate of NOx estimated by EMFAC-HK can be further divided into emission rates of NO and NO_2. The NO_2/NOx ratios are presented in **Appendix 3.4**.

3.7.8 The hourly emissions of NO, NO_2, RSP and FSP are divided by the number of vehicles and the distance travelled to obtain the emission factors in gram per miles per vehicle. The calculated 24-hour emission factors of 16 vehicle classes for each road type are adopted in this air quality impact assessment. The detailed calculation of vehicular emission source is presented in **Appendix 3.4**.

3.7.9 The United States Environmental Protection Agency (USEPA) approved CALINE4 dispersion model has been used to assess traffic emissions impact from existing and planned road network. Owing to various phases of development of the Project, surface roughness coefficient of each modelling grid adopted in the CALINE4 model varies with time, ranging from 100 cm to 370 cm for mid-rise and high-rise development areas respectively. The surface roughness coefficient of each grid in each assessment year is summarised in **Table 3.13**. The grid-specific WRF meteorological data have also been adopted to calculate the hourly impact. PCRAMMET has been applied to generate Pasquill-Gifford stability class for the meteorological input to CALINE4 model based on the WRF meteorological data.

Table 3.13 Surface Roughness adopted in CALINE4 of Modelling Grid

Grid	Year 2024 – Year 2030	Year 2031 – Year 2039	
		Construction Phase	Operation Phase
20, 47	370 cm	370 cm	370 cm
20, 48	370 cm	370 cm	370 cm
20, 49	100 cm	100 cm	100 cm
21, 44	100 cm	100 cm	100 cm
21, 45	100 cm	370 cm	370 cm
21, 46	100 cm	370 cm	370 cm
21, 47	100 cm	100 cm	370 cm

Grid	Year 2024 – Year 2030	Year 2031 – Year 2039	
		Construction Phase	Operation Phase
21, 48	370 cm	370 cm	370 cm
21, 49	100 cm	370 cm	370 cm
21, 50	100 cm	100 cm	100 cm
22, 44	100 cm	100 cm	100 cm
22, 45	100 cm	370 cm	370 cm
22, 46	370 cm	370 cm	370 cm
22, 47	100 cm	100 cm	370 cm
22, 48	100 cm	100 cm	370 cm
22, 49	370 cm	370 cm	370 cm
23, 46	100 cm	100 cm	100 cm
23, 47	370 cm	370 cm	370 cm
23, 48	370 cm	370 cm	370 cm

3.7.10 Owing to the height difference between at-grade roads and bridges, road emission has been grouped into two scenarios; namely “Bridge” and “At-grade”. The “Bridge” scenario features the KSWH as a bridge of 20 meters above ground. The height of the Highway ranges from 35 mPD to 40 mPD with ground surface at 15 mPD – 20 mPD. As a conservative consideration, the height is considered as 20 m above ground only. The discrepancy of receptor heights between “Bridge” scenario and other two source scenarios are due to the relative frame of Bridge scenario to the other two. Owing to the limitation of Caline4 model, the highway is set to 10 meters above ground in the model and the frame of modeling domain has a relative height of 10 meters above the actual ground. Receptors with actual heights equal or lower than 10 mAG are assumed to have the same prediction as lowest height in the model. In order to map the receptors correctly with other two source scenarios, the correction of receptor heights has been done in the input file before the summation calculation of cumulative results. Receptors with actual heights equal or lower than 10 mAG are set to the lowest assessment level in the modeling frame, i.e. 1.5 m. Also, receptors with actual heights higher than 10 mAG are decreased by 10 m. **Table 3.14** illustrates the mapping of the receptor results in the calculation.

Table 3.14 “Adjusted” Modelling Height in Caline4 Results

Actual receivers height above Local Ground	Modelling height for the bridge (10 mAG as the reference zero)	Modelling height for at-grade roads (0 mAG as the reference zero)	Cumulative results on all roads (bridge + at-grade contributions)
1.5 m	1.5 m (below reference zero modeled as the lowest height)	1.5 m	1.5 m + 1.5 m
5 m	1.5 m (below reference zero modeled as the lowest height)	5 m	1.5 m + 5 m
10 m	1.5 m (below reference zero modeled as the lowest height)	10 m	1.5 m + 10 m
20 m	10 m	20 m	10 m + 20 m

3.7.11 For the calculation of NO_2 concentrations, the predicted NO is converted to NO_2 based on the Ozone Limiting Method with the background ozone concentration in accordance with the EPD’s Guidelines on Choice of Models and Model Parameters. The predicted NO_2 is directly adopted for the determination of Total NO_2 concentration.

3.7.12 Secondary air quality impact arising from the implementation of roadside noise barriers has been incorporated into the air quality model. Extent of the proposed noise mitigation measures are presented in the Noise Section (**Section 4**) and are summarised in **Table 4.28**.

Emission from Portal and Ventilation Building of TMWB

3.7.13 The portal and the ventilation building of the northern section of TMWB are assumed to be located at the west of the Project. The design information of TMWB is not available at the time of the assessment. In reference to approved EIA Report for Tuen Mun - Chek Lap Kok Link, it is assumed that the emission split would be 10% from the northbound tube of the portal and 90% from the ventilation building. The traffic flow of road links relevant to the tunnel is adopted to estimate the portal emission, together with the composite emission rates derived from the EMFAC-HK results. The detailed calculation of portal emission rates is presented in **Appendix 3.5**.

3.7.14 Portal emissions from TMWB are modelled based on the procedures in Section III of the Permanent International Association of Road Congress Report (PIARC), 1991. It is assumed that emissions would emerge as an air jet along the axis of the road so that only the well-diluted parts gradually shear off. The pollutants are assumed to be ejected as a volume source in the model, with two-thirds of the total emissions dispersed in the first 50 m, and one third of the total emissions dispersed in the second 50 m. The portal emissions are modelled as volume sources in American Meteorological Society (AMS) and U.S. Environmental Protection Agency (EPA) Regulatory Model (AERMOD).

Emission from Industrial Source

3.7.15 Based on the best available information obtained under “Agreement No. CE 66/96, Planning and Development Study on North West New Territories”, there are some discrete chimneys located within and in vicinity of the Project area. Chimney validation surveys have been conducted in June and July 2012 to update the operation status of chimneys. A further validation check was conducted in early and end 2015. In total, 21 nos. of chimney existing within 500m assessment area are found active and considered in the air quality assessment. The details of these chimneys (labelling as CH) including emission inventory and its chimney parameters, and its locations are presented in **Appendix 3.2**.

3.7.16 A review has also been conducted for the SP licences within the assessment area, and the concerned plants are listed in **Table 3.15**. In total, there are seven concrete batching plants, one asphalt plant and one stone crushing plant found within 500 m assessment area. The concrete batching plant, asphalt plant and the stone crushing plant at Lam Tei Quarry would be terminated before Year 2024. There are four concrete batching plants would be affected by the Project development and are required to be removed. Therefore, the impacts from these five concrete batching plants, a stone crushing plant and an asphalt plant would not be considered in the assessment. The remaining two existing concrete batching plants in the vicinity of the Project area are assumed to be still in operation in future and considered in the assessment. Referring to the Revised RODP, the planned industrial zone, Site 3-47 in the Project area is assumed for locating three new concrete batching plants. For the purpose of the air quality impact assessment in the EIA Study, the capacity of each new concrete batching plant at Site 3-47 is assumed to have the highest production capacity among the four existing concrete batching plants. For these concrete batching plants to be operated in future, the particulates concentration limit follow the Guidance Notes on Best Practicable Means (BPM) for Specified Process – Cement Plant (Concrete Batching Plant), BPM 32/16, issued in February 2016. The detailed emission rate calculation for concrete batching plant is presented in **Appendix 3.2**.

Table 3.15 Existing SP Activities within the Assessment Area

Model ID	Company Name	Industry	Location	SP Licence No.	Remark
-	Glorious Concrete (H.K.) Limited	Concrete Batching	Lot 1824, DD 130 Yick Yuen	L-3-043(4)	To be affected by the Project development and would be removed
CBPa_1 – CBPa_16, CBPb_1 – CPBb_16, CBPc_1 – CBPc_16	Multi-Way Industrial Limited	Concrete Batching	Lot 1825, DD 130, Tuen Mun	L-3-128(4)	To be affected by the Project development and would be removed Highest production capacity among the affected CBPs and the capacity is taken as that of the new CBP in the industrial area.
-	Anderson Concrete Limited	Concrete Batching	Lot 362, DD 127, Kiu Tau Wai, Yuen Long	L-3-057(6)	To be affected by the Project development and would be removed
-	Heng Tat Concrete Limited	Concrete Batching	DD 124, Lot 793 & DD 127 Lot 70-77, 215RP & 216 Hung Uk Tsuen, Yuen Long	L-3-230(1)	To be affected by the Project development and would be removed
GO1 – GO11	Golik Concrete Limited	Concrete Batching	Lot No. 1842 in DD121, 89 Tong Yan San Tsuen, Yuen Long, N.T.	L-3-229(2)	Existing CBP in the vicinity of the Project area. For the purpose of assessment in this EIA study, it assumes that the plant still be in operation in future
RED1, RED3 - RED24	Redland Concrete Limited	Concrete Batching	Lots Nos. 1207RP, 1208A, 1263RP(part), 1265RP, 1842RP & Adjoining G.L. in DD121, Tong Yan San Tsuen Road, Tong Yan San Tsuen, Yuen Long, N.T.	L-3-199(4)	Existing CBP in the vicinity of the Project For the purpose of assessment in this EIA study, it assumes that the plant still be in operation in future
-	Faith Oriental Investment Limited	Stone Crushing	Lam Tei Quarry	L-11-52(1)	Terminate before 2024
-		Concrete Batching [1]	Lam Tei Quarry	-	Terminate before 2024
-		Asphalt [1]	Lam Tei Quarry	-	Terminate before 2024

Note: [1] The SP Licences for concrete batching plant and asphalt plant at Lam Tei Quarry are under review.

3.7.17 Emissions (NOx, SO_2, RSP and FSP) from chimneys identified within the assessment area have been modelled as point sources while emissions (TSP, RSP and FSP) from the above mentioned SPs are modelled as point sources and area sources. The emission rates of the chimneys are presented in **Appendix 3.2**. For the chimneys newly identified during the site survey, an average fuel usage of 200 Liters/hour is assumed for the calculation of emissions, while the heights of the chimneys are based on observation. Use of distillate oil is assumed for the chimneys. Based on the size of the boiler, the corresponding emission factors with reference to Tables 1.3-1 and 1.3-7 of AP-42, USEPA have been adopted in the calculation. Sulphur diesel content of 0.005% is adopted for the chimneys, in accordance with the requirement under the APCO (Fuel Restriction) (Amendment) Regulation 2008.

3.7.18 The methodology as recommended in the EPDs' Guidelines on "Estimating Height Restriction and Position of Fresh Air Intake Using Gaussian Plume Models" have been adopted. For those stacks with no available data on stack parameter, emission temperature of 373K, exit velocity of 6 m/s and stack diameter of 1 m are assumed. For those concrete batching plants (CBPs), their emission is assumed to release at ambient temperature.

3.7.19 The use of fuel for the future stack emission sources within the areas zoned "Government" is still undetermined at this stage. Since any installation of chimneys for commercial/hospitals/medical centre would require separate approval/application from EPD in order to make sure that there is no adverse impact to the adjoining environment, for the purpose of this assessment, distillate oil and town gas are assumed to be used as power source for steam boilers and emergency generator. With such application, it is understood that the total chimney emission from a hospital should be proportional to the total number of patient accommodated in the hospital (i.e. number of bed). According to the North East New Territories (NENT) EIA report, it was assumed that there would be a total of 860 beds available in the proposed hospital in NENT, while the number of beds in the proposed hospital is estimated to be about 1000 with reference to the design assumption of the Revised RODP. Therefore, in order to estimate the emission from the proposed hospital, a correction factor of 1.16 has been multiplied to the chimney emission from NENT proposed hospital. The emission rate calculation for the chimney of planned hospital (labelling as "HP") is presented in **Appendix 3.2.**

Emission from Port Backup and Logistic Facilities

3.7.20 For the purpose of this assessment, the number of off-road vehicles and plants has been roughly assumed based on the similar operation of Container Terminal 1 – 9 in the Tsing Yi area. Since the actual layout and the configuration of these facilities have not been formulated, it is assumed that the emissions (NOx & RSP) from off-road vehicles and plants at open area would be area source at ground level at Revised RODP stage. The emission rates from off-road equipment and back-up facilities have been estimated with reference to the US Tier 3 Emission Standard (130≤kW<225) following the Air Pollution Control (Non-road Mobile Machinery) (Emission) Regulation. The post speed of these off-road vehicles within the Port Backup and Logistic Facilities is assumed as 10KPH in the assessment. Idling emission is based on "Road Tunnels: Vehicle Emissions and Air Demand for Ventilation" published by Permanent International Association of Road Congress (PIARC, 2012). Emission standard of Euro V is assumed for non-road vehicles according to the requirements as stipulated under the Air Pollution Control (Vehicle Design Standards) (Emission) Regulations (Cap. 311J). The 3-minutes idling of motor vehicles as stipulated in Motor Vehicle Idling (Fixed Penalty) Ordinance is considered for general case. The details of port backup sources (labelling with prefix P) such as emission rates in detail are presented in **Appendix 3.2**.

Dispersion Modelling & Concentration Calculation

3.7.21 American Meteorological Society (AMS) and U.S. Environmental Protection Agency (EPA) Regulatory Model (AERMOD), the EPD approved air dispersion model, has been employed to predict the air quality impacts due to industrial emission, portal emission from TMWB and emission from Port Backup/Logistic Facilities. The air pollutants concentrations are predicted at each identified ASRs at various assessment height, ranging from 1.5 mAG up to the roof level of the respective ASRs.

3.7.22 Hourly meteorological conditions including wind data, temperature, relative humidity, pressure cloud cover and mixing height of Year 2010 are extracted from the WRF meteorological data adopted in the PATH-2016 system. The minimum wind speed is capped at 1 metre per second. The height of the input data is assumed to be 9 and 26 meters above ground for the first and second layers of the WRF data as input respectively.

3.7.23 Surface characteristic parameters such as albedo, Bowen ratio and surface roughness are required in the AERMET (the meteorological pre-processor of AERMOD). The land use characteristics of the surrounding are classified into sectors. The parameters of each sector are then suggested by AERMET by default according to its land use characteristics. The detailed assumptions are discussed in **Appendix 3.17**.

3.7.24 The spot heights and contour heights on the basemap are extracted to be used as the terrain input data of the AERMOD's terrain pre-processor, AERMAP.

3.7.25 Gradual plume rise option has been employed for all modelled pollutants. Dry deposition and depletion have been employed for particulate matter. Particle size distribution adopted for chimney, CBP and off-road vehicle and plant are summarised in **Table 3.16**, **Table 3.17** and **Table 3.18**. Since the size distribution (less than 10 µm but larger than 2.5 µm) are known for various sources, Method 1 of particle dry deposition has been adopted for modelling using AERMOD.

Table 3.16 Particle Size Distribution of Chimney

Particle Size, µm	Average Particle Size (µm)	Percentage %
0.625	0.31	58.3
1	0.81	3.3
1.25	1.13	1.7
2.5	1.88	6.7
6	4.25	11.7
10	8.00	10.0
15	12.50	8.3
30	22.50	40.0

Reference: Table 1.3-5, Section 1.3, AP-42.

Table 3.17 Particle Size Distribution of Concrete Batching Plant

Particle Size, µm	Average Particle Size (µm)	Percentage %
General Activity(i)		
1.0	0.50	4.0
2.0	1.50	7.0
2.5	2.25	4.0
3.0	2.75	3.0
4.0	3.50	7.0
5.0	4.50	5.0
6.0	5.50	4.0
10.0	8.00	17.0

Particle Size, µm	Average Particle Size (µm)	Percentage %
30.0	20.00	49.0
Haul Road[ii]		
2.5	1.25	4.6
10	6.25	14.6
15	12.5	4.6
30	22.5	76.2

Reference:
i) Reference: Table B.2.2 - Category: 3, Appendix B2, AP42.
ii) Table 13.2.1-1,Section 13.2.1, AP42.

Table 3.18 Particle Size Distribution for Off-Road Vehicles and Plants

Particle Size, µm	Average Particle Size (µm)	Percentage %
1.0	0.50	82.0
2.0	1.50	6.0
2.5	2.25	2.0
3.0	2.75	0.0
4.0	3.50	2.0
5.0	4.50	1.0
6.0	5.5	0.0
10.0	8	3.0
30.0	20	4.0

Reference: Table B.2.2 - Category: 1, Appendix B2, AP-42.

3.7.26 For the calculation of NO_2 concentrations from Port Backup/Logistic Facilities plus Industrial Sources, the conversion from NOx to NO_2 is based on the Ozone Limiting Method and the background ozone concentration, assuming the NO_2 to be 28% of NOx for Port Backup/Logistic source while 10% for Industrial Sources.

3.7.27 Conversion factor for 10-minute average of SO_2 concentration refers to Guidelines on Estimation of 10-minute Average SO_2 Concentration for Air Quality Assessment in Hong Kong, HKEPD. The P-G stability classes and corresponding conversion factors are summarised in **Table 3.19**.

Table 3.19 Conversion Factor for 10-min Average SO_2 Concentration

P-G Stability Class	Conversion Factor
A	2.45
B	2.45
C	1.82
D	1.43
E	1.35
F	1.35

Determination of Assessment Year

3.7.28 Referring to the construction programme as presented in **Appendix 2.4**, the first population intake of the planned residential development under this Project would be Site 5-26 in Year 2024. During Year 2025 – 2030, there are some port back-up and logistic facilities sites to be occupied. More planned sites to be available for intake since Year 2031 and the development sites are expected to be completed and are occupied by Year 2036 - 2039. For determination of the worst case year for the assessment, emission burden of open roads for Year 2024 (first year intake of proposed Project), Year 2026 to 2028, Year 2031 (first major population intake year), Year 2036 to 2039 (15 years after commissioning of proposed Project) are calculated by EMFAC-HK model and the results are presented in **Table 3.20**. The detailed calculation is attached in **Appendix 3.4**. Results indicate that the maximum emission burden among the 15 years is Year 2026.

Despite the vehicular emission factors in Year 2024 are higher than that in Year 2026, Year 2026 traffic flow has taken into account the major development of concurrent projects including Yuen Long South, Kam Tin South and Tuen Mun Central. The emission burden in Year 2026 is therefore higher than Year 2024. The assessment scenario in Year 2026 can cover the impact in Year 2024, model run for Year 2024 is considered unnecessary.

Table 3.20 Emission Burden of Open Roads for Various Years

Year	NO_X (tonnes per annum)	RSP (tonnes per annum)	FSP (tonnes per annum)
Year 2024	298.9	12.2	11.2
Year 2026	307.8	12.6	11.6
Year 2027	300.2	12.4	11.4
Year 2028	291.5	12.1	11.1
Year 2031	189.7	7.1	6.6
Year 2036	186.6	4.9	4.6
Year 2037	186.4	5.0	4.6
Year 2038	186.9	5.1	4.7
Year 2039	188.0	5.4	5.0

3.7.29 Based on the population intake years, the following two assessment scenarios are studied for the evaluation of the operational air quality impact. The cumulative emission sources considered in these two scenarios are listed in **Table 3.21** and **Table 3.22**, respectively. As mentioned in **Section 3.5.1**, the construction programme for TMWB is not available at the time of the assessment, its construction dust impact would not be considered in the assessment, however, its operation phase vehicular emission impact would be considered in the assessment since Year 2031.

Table 3.21 Operation Phase Scenario Justification

Scenario	Justification	ASRs Considered in the Assessment
Scenario 1: Year 2024 – Year 2030	First population intake in Project area is Year 2024. Referring to the emission burden calculation as shown in **Table 3.20**, Year 2026 is considered the worst emission burden year between Year 2024 and Year 2030. Thus traffic flow and emission factor of Year 2026 are used.	All representative existing ASRs and planned ASRs occupied during Year 2024 - 2030. The locations of the ASRs considered in the model are shown in **Figure 3.2.1-3.2.6**.
Scenario 2: Year 2031 – Year 2039	First major population intake in Project area is Year 2031. Among Year 2031 – 2039, Year 2031 is the highest emission burden rate but it cover parts of new roads within the Project area only. In Year 2036, all new roads within the Project area including all DP roads would be open, therefore, the NOx emission burden is calculated based on Year 2036 traffic flow and Year 2031 emission factor. The calculated NOx, RSP and FSP emission burden would be 233.2, 8.728 and 8.114 tonnes per annum respectively, higher than the emission burden in any year from Year 2031 to 2039 as shown in **Table 3.20**.	All representative existing ASRs and planned ASRs within 500m assessment area of the Project are considered in the assessment. The locations of the ASRs considered in the model refer to **Figure 3.3.1-3.3.6**.

Table 3.22 Cumulative Emission Sources in Scenario 1: Year 2024-2030 (Operation Phase)

Source	Remarks
Open Road Traffic	• With a section of Road P1 (DP1), a section of Road D1(DP2) and junction of D8/P1 (DP5) connecting "PBU+SWU", and all existing roads within 500 m assessment area • Traffic data: Year 2026 • Emission Factor: Year 2026
Industrial source	• Existing Chimneys and SPs • 3 new CBPs and 2 relocated chimneys • "PBU+SWU" at Site 3-6, 3-7 & 3-8

Table 3.23 Cumulative Emission Sources in Scenario 2: Year 2031-2039 (Operation Phase)

Source	Remarks
Open Road Traffic	• All existing roads within 500 m assessment area and planned roads including all DP roads within the Project area • Traffic data: Year 2036 (which has considered the traffic flow of all DPs roads) • Emission Factor: Year 2031
Industrial source	• Existing Chimneys and SPs • 3 new CBPs and 2 relocated chimneys • "PBU+SWU" at Site 3-1 to 3-14
Portal Emission and Ventilation Building	TMWB
Boiler at planned hospital (commencement in Year 2036)	Pro-rata emission with designed capacity (1000 beds) with reference to the one of NENT EIA.

3.7.30 It is anticipated that NO_2, RSP and FSP are the key pollutants to be considered for vehicular emission and chimneys emission, and SO_2 for chimneys emission only during operation phase.

3.8 Assessment Approach and Methodology (Operational Phase Odour Impact)

Odour Emission Inventory of the Existing Chicken Farm

3.8.1 Site visits were made in end 2014 and July 2015, and a brief interview and site walk have been made with the owner of the chicken farm. Each chicken house could house about 5,000 – 10,000 numbers of chicken depending on its size. As a common practice in the trade that once the chicken are sold, the whole chicken house would then be cleansed and sterilised, and leave it vacant for about 10 days in order to prevent infection of disease.

3.8.2 Natural ventilation with the aid of fans is generally adopted in chicken houses to ensure the fresh air supply. As a common practice, medium and large chickens would be caged in the form of "A-shaped" on top of a purposely built channel to collect the excrements from chickens. In general, accumulated excrements along the channels will gather from the end of channel to the temporary storage tank manually once per day (at around 2:00pm). The excrements will then transfer into enclosed rubbish bins, which will be collected by licensed waste collector few times a week.

3.8.3 In terms of small chickens, they are kept separately from other mature chickens to avoid cross-infection. Excrements would be retained on papers underneath for ultimate disposal into rubbish bins.

3.8.4 There are also on-site covered septic tanks for treatment of domestic effluent. Based on the site observation, there is no "noticeable smell" identified. During the cleansing/maintenance period, a very short-term transient smell might be detected.

3.8.5 The number of chickens kept within the chicken farm would be varied, and such variation is very much affected by demand and supply from the market. In general, several high-demand periods would occur right before Chinese Festivals, such as Lunar New Year, Dragon Boat Festival, Mid-Autumn Festival and Winter Solstice. A maximum of about 102,000 chickens can be kept under the licence.

3.8.6 Based on the current information from EPD, there is no complaint on odour received from the existing/adjoining villages in the past 10 years. With the proper operation conditions/ maintenance procedures required under AFCD licence, adverse odour impact is not expected in the past.

3.8.7 The layout of the existing chicken farm and the photo records during the site visit in July 2015 are attached in **Appendix 3.6**. Based on the observation during the site visit, potential odour emission sources within the chicken farm are identified as below.

- Chicken Houses for small, medium or large sized chicken; and
- Garbage Collection Area.

3.8.8 For establishment of the odour emission inventory for odour impact assessment, odour sampling was conducted at the potential odour emission sources within the chicken farm on 18, 19 and 24 August 2015 of a three non-rainy days with daytime's temperature greater than 30°C (during sampling period) at site. Odour patrol/spot-check survey was also conducted at these three days at different time slots to understand the actual odour condition at the existing ASRs in the vicinity of the chicken farm. Specifically, the operation status of all the odour emitting sources within the chicken farm should also be determined to check whether or not the odour emitting process is underway during the odour patrol. As such, the odour patrol/spot-check was conducted at the same time frame as the in-situ odour sampling within the chicken farm as far as possible ("in order to synchronise the data at the same time/day").

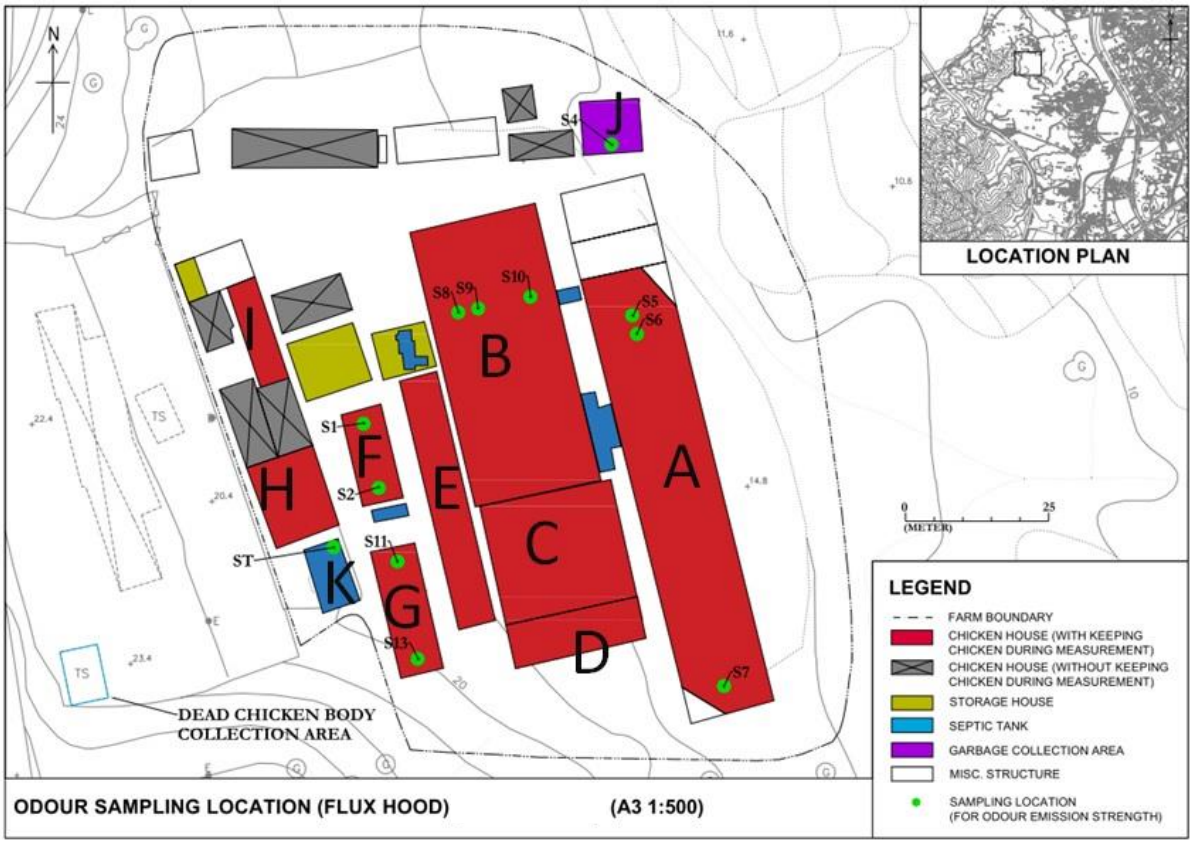
3.8.9 The details of the odour sampling and olfactometric analysis, and odour patrol/spot-check survey are presented in **Appendix 3.7**.

3.8.10 A specific odour emission rate (SOER) of each sample taken in chicken house, garage collection area and septic tank is determined in the odour survey. With reference to the SOER determined by the odour sampling, the odour emission rate of each chicken house is determined according to the total area of the excretion collection channels inside each chicken house or the size of the exposed area. The details for deriving the odour emission rates from the measurement results are presented in **Appendix 3.6**. The derived odour emission rate of each structure in the chicken farm is summarised in below **Table 3.24**.

Table 3.24 Effective Odour Emission Rate of Odour Source in Chicken Farm

ID	Description	SOER (OU/m^2·s)	Total Exposed Area (m^2)	Odour Emission Rate by Measurement (OU/s)
A [1]	Large-sized Chicken House	211.2 [3]	45	9504.0
B	Medium-sized Chicken House	2.19	640	1401.6
C	Medium-sized Chicken House	2.19	200	438.0
D	Medium-sized Chicken House	2.19	120	262.8
E [1]	Small-sized Chicken House (Active Vent)	48.4 [4]	4.00	193.6
F	Small-sized Chicken House	0.39	60	23.4
G	Small-to-Medium Chicken House	0.43	120	52.0
H	Small-sized Chicken House	0.39	130	50.6
I	Small-sized Chicken House	0.39	35	13.6
J [2]	Garbage Collection Area	0.48	N/A	N/A
K [2]	Septic Tank	4.55	N/A	N/A

Remark:
[1] The ventilation fans were turned on during sampling, thus the emission rate is determined by the concentration measured downstream of the ventilation fan.
[2] Direct adoption of SOER determined by odour sampling for area sources.
[3] Determined by odour concentration of 64 OU/m^3 and flow rate of 3.3 m/s, details refer to Appendix 3.6.
[4] Determined by odour concentration of 22 OU/m^3 and flow rate of 2.2 m/s, details refer to Appendix 3.6.



3.8.11 The odour emission rate in each hour would not achieve as high as the figures estimated above in **Table 3.24** because the excretion collection channels are cleaned regularly at around 2:00pm everyday as described in **Section 3.8.2**. However, as a conservative assessment approach, the odour emission rate is considered as same in every hour in the assessment.

Odour Emission Inventory for Upgraded San Wai STW, New HSK STW and Planned Refuse Transfer Station

3.8.12 The upgraded San Wai STW, new HSK STW and planned Refuse Transfer Station (RTS) would be located within 500 m area, therefore, cumulative odour impact from these three facilities are assessed.

Upgraded San Wai STW

3.8.13 An EIA Study for "Upgrading and Expansion of San Wai STW and the Expansion of Ha Tsuen PS" has been conducted in Year 2002 and was approved in Year 2003. The study has proposed that all potential odour emission sources of the STW would be enclosed and the vented air would be treated in the deodourising unit with 96% of H_2S removal efficiency. As there is no further change in the design of this STW, the mitigated odour emission rate as presented in the EIA Report is adopted in this assessment. The emission rate of the upgraded San Wai STW is presented in **Appendix 3.8**.

New HSK STW

3.8.14 The design capacity of the new Sewage Treatment Works (STW) would be 85,500 m^3 per day and the sewage will be treated by tertiary treatment and secondary plus treatment (with UV disinfection and 75% nitrogen removal). A separate EIA study would be conducted for this new STW in future. For the purpose of the assessment in this EIA Study, the design of STW is made reference to the design of Shek Wu Hui STW as (i) it is Tertiary Sewage Treatment Plant and (ii) the influent of both STWs are municipal sewage.

3.8.15 The NENT EIA study has conducted odour impact assessment on the Shek Wu Hui STW based on the ultimate capacity of 190,000 m^3 per day and the odour source emission areas of the STW would be enclosed and the vented air would be treated in the deodorising unit. The design of the planned STW would make reference to the treatment design of the Shek Wu Hui STW in the NENT EIA Report. The capacity of Shek Wu Hui is about two times of that of the new HSK STW, it is assumed that the emission source area of the new HSK STW would be about half of that of Shek Wu Hui STW. As the odour emission rates depends on the exposed area of the sewage treatment facilities such as inlet channel, primary sedimentation tanks, final sedimentation tanks, the odour emission rates of new HSK STW would be assumed as half of total odour emission rates of all enclosed odour emission sources areas of Shek Wu Hui STW. The details of the emission rate and deodourising unit of the new HSK STW are presented in **Appendix 3.8**.

Planned Refuse Transfer Station

3.8.16 The design capacity of the planned RTS would be 1,100 tonnes per day and the design would be similar to the existing RTS in Hong Kong. A separate EIA study for this planned new RTS would be conducted in future. For the purpose of the assessment in this EIA Study, the design of STW (i.e. number of stacks, height and diameter of stacks, etc.) is made reference to similar capacity of existing RTS, North Lantau Transfer Station which is 1,200 tonnes per day of the design capacity.

3.8.17 There is no available odour emission data for existing RTS in Hong Kong. Therefore, the odour emission rate of planned RTS considered in the assessment adopts the average measured SOER at tipping face (for municipal solid waste) of NENT Landfill extracted from the approved NENT Landfill EXT EIA Report. The details of the emission rate and the design assumption for new RTS are presented in **Appendix 3.8**.

Modelling Approach and Parameter

3.8.18 The existing chicken farm is located at northwest of the Project area (about 200 m from south-west of Fung Kong Tsuen) while San Wai STW, new HSK STW and planned RTS are located at the west of the Project area (next to KSWH and Ha Tsuen Road). The separation distance between existing chicken farm and the STWs/RTS is at least 850 m. There is no common ASR from these two groups of odour sources (chicken farm and the STWs/RTS), therefore, no cumulative odour impact would be anticipated. The evaluation of the odour impact from these two groups of odour sources is conducted separately using AERMOD model. The hourly meteorological data including wind data, temperature, relative humidity, pressure, cloud cover and mixing height of Year 2010 are extracted from the WRF meteorological data adopted in the PATH-2016 system. The minimum wind speed is capped at 1 metre per second. The height of the anemometer is assumed to be 9 meters above ground for the first layer of the WRF data as input.

3.8.19 Surface characteristic parameters such as albedo, Bowen ratio and surface roughness are required in the AERMET (the meteorological pre-processor of AERMOD). The land use characteristics of the surrounding are classified into sectors. The parameters of each sector are then suggested by AERMET by default according to its land use characteristics. The detailed assumptions are discussed in **Appendix 3.17**.

3.8.20 More recent research indicated that the peak-to-mean ratio of odour dispersion would depend upon the type of source, atmospheric stability and distance downwind. For the purpose of this assessment to produce more reasonable predictions for odour dispersion from wake-affected point sources and area/volume sources, reference is made to the peak-to-mean ratio stipulated in "Approved Methods for Modelling and Assessment of Air Pollutants in New South Wales" published by the Department of Environment and Conservation, New South Wales, Australia (NSW Approved Method).

3.8.21 The dispersion modelling techniques employed for this assessment follows those described in EPD's "Guidelines on Choice of Models and Model Parameters" except the use of alternative peak-to-mean ratios for wake-affected point sources and volume sources. However, it should be noted that the peak-to-mean ratios stated in the NSW Approved Method are derived based on experimental and theoretical analyses and assuming a 0.1% exceedance level (Ref.: Statistical Elements of Predicting the Impact of a Variety of Odour Sources, Peter R. Best, Karen E. Lunney and Christine A. Killip, Water Science and Technology, Australia, 44: 9 pp 157-164 2001). In other words, there would be a 0.1% probability that the actual peak concentration would be higher than those derived with the peak-to-mean ratios stated in the NSW Approved Method.

3.8.22 The chimneys of the deodourising units of expanded and upgraded San Wai STW, new HSK STW and planned RTS are considered wake-affected point sources while the emission sources of the chicken farm are assumed to be volume sources for input into model. **Table 3.25** shows the conversion factors to be used for converting the maximum modelled 1-hour average concentrations to corresponding maximum 1-second average concentrations that could occur during that hour.

3.8.23 Regarding the odour emission sources of chicken farm, the size of each volume source is determined by the averaged wind flux (m/s) within each chicken house multiplied by the cross-sectional area of openings (or house). With the general mixing by internal fans or natural wind, the dilution/diffusion factor between the excrement channel concentration and the room ambient odour concentration is identified. The odour spot-check results at the downwind location are used to verify the modelling setting in general.

Table 3.25 Conversion Factors for Volume Sources and Wake-affected Point Sources

Pasquill Stability Class	Conversion Factor (1 hour to 5 seconds)	
	Wake-affected Point Source	Volume Source
A	2.3	2.3
B	2.3	2.3
C	2.3	2.3
D	2.3	2.3
E	2.3	2.3
F	2.3	2.3

3.9 Prediction and Evaluation of Environmental Impacts

Construction Phase

3.9.1 The cumulative air quality impact due to emissions from construction activities, open roads, ventilation building and tunnel portal of TMWB (for Scenario 2 only), industrial and port backup / logistics facilities at the Project sites and representative existing ASRs in Scenario 1: Year 2019-2030 and Scenario 2: Year 2031 to 2036 has been evaluated. In the Scenario 1, the predicted unmitigated cumulative maximum hourly average TSP, the 10th highest daily average and annual average RSP, the 10th highest daily average and annual average FSP concentrations at the Project sites and existing representative ASRs would be ranged from 173 – 25849 µg/m^3, 91 – 1005 µg/m^3, 40 – 77 µg/m^3, 68 – 176 µg/m^3, and 29 – 34 µg/m^3, respectively. In Scenario 2, the predicted unmitigated cumulative maximum hourly average TSP, the 10th highest daily average and annual average RSP, the 10th highest daily average and annual average FSP concentrations at the Project sites and existing representative ASRs would be ranged from 198 – 18540 µg/m^3, 86 – 616 µg/m^3, 38 – 63 µg/m^3, 69 – 149 µg/m^3, and 29 – 33 µg/m^3, respectively. It is noted that exceedance would be predicted at the ASRs which are located in proximity to the work sites. Mitigation measures are required for these ASRs. The detailed prediction unmitigated results in Scenarios 1 and 2 are presented in **Appendix 3.9**.

Operation Phase Air Quality

Year 2024 – 2030 Scenario

3.9.2 The cumulative air quality impact due to emissions from open roads, industrial and port backup / logistics facilities at the Project sites and representative existing ASRs in this scenario has been evaluated. The predicted unmitigated cumulative 19th highest hourly and annual average NO_2, 4th highest 10-min and 4th highest daily average SO_2, 10th highest daily and annual average RSP and FSP concentrations at the Project sites and representative existing ASRs are summarised in **Table 3.26**, **Table 3.27, Table 3.28** and **Table 3.29** respectively. The detailed results are presented in **Appendix 3.10**.

3.9.3 According to these result tables, the predicted NO_2, RSP, FSP and SO_2 concentrations at the representative planned ASRs within the Project sites and existing ASRs would comply with AQOs.

Table 3.26 Predicted Operational Air Quality Impact on Sites with Air Sensitive Use in Year 2024 – Year 2030

Site	19th Highest Hourly NO_2 Conc. (µg/m^3)	Annual NO_2 Conc. (µg/m^3)	4th Highest 10-min SO_2 Conc. (µg/m^3)	4th Highest Daily SO_2 Conc. (µg/m^3)
2-18	94 - 96	20 - 20	132 - 153	30 - 30
2-19	96 - 96	20 - 20	132 - 132	30 - 30
3-43	96 - 124	21 - 33	134 - 134	29 - 29
3-44	96 - 110	21 - 27	134 - 134	29 - 29
3-45	96 - 126	21 - 34	134 - 134	29 - 29
3-50	96 - 137	21 - 35	127 - 134	28 - 29
3-51	96 - 130	22 - 32	127 - 127	28 - 28
3-6	90 - 99	18 - 21	132 - 132	30 - 30
3-7	90 - 99	18 - 23	132 - 170	30 - 32
3-8	90 - 104	18 - 23	131 - 170	30 - 33
4-20	110 - 120	25 - 28	127 - 127	28 - 28
4-22	91 - 119	19 - 30	127 - 127	28 - 28
4-28	86 - 110	18 - 30	111 - 127	26 - 28
5-16	92 - 122	18 - 36	110 - 110	28 - 28
5-17	91 - 113	18 - 32	110 - 110	28 - 28

Site	19th Highest Hourly NO_2 Conc. ($\mu g/m^3$)	Annual NO_2 Conc. ($\mu g/m^3$)	4th Highest 10-min SO_2 Conc. ($\mu g/m^3$)	4th Highest Daily SO_2 Conc. ($\mu g/m^3$)
5-18a	91 - 106	18 - 26	110 - 110	28 - 28
5-18b	91 - 114	18 - 31	110 - 110	28 - 28
5-2	94 - 107	21 - 27	111 - 111	27 - 27
5-21	95 - 109	21 - 27	110 - 110	28 - 28
5-22	91 - 99	21 - 27	117 - 117	26 - 26
5-23	91 - 101	21 - 25	117 - 117	26 - 26
5-24	88 - 101	18 - 26	117 - 117	26 - 26
5-26	88 - 102	19 - 26	117 - 117	26 - 26
5-27	95 - 104	22 - 27	117 - 117	26 - 26
5-28	95 - 105	22 - 27	117 - 117	26 - 26
5-32	88 - 119	18 - 33	117 - 117	26 - 26
5-33	92 - 103	21 - 26	117 - 117	26 - 26
5-3a	91 - 112	18 - 26	110 - 111	27 - 28
5-3b	92 - 104	20 - 25	111 - 111	27 - 27
5-6	102 - 104	23 - 25	111 - 111	27 - 27

Note: Bolded and underlined figure represents exceedance of relevant AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.27 Predicted Operational Air Quality Impact on Representative Existing ASRs in Year 2024 – Year 2030

ASR ID	19th Highest Hourly NO_2 Conc. ($\mu g/m^3$)	Annual NO_2 Conc. ($\mu g/m^3$)	4th Highest 10-min SO_2 Conc. ($\mu g/m^3$)	4th Highest Daily SO_2 Conc. ($\mu g/m^3$)
A1001	97 - 97	21 - 21	132 - 132	30 - 30
A1002	96 - 96	20 - 20	132 - 132	30 - 30
A1003	96 - 96	20 - 20	132 - 132	30 - 30
A1004	94 - 94	20 - 20	132 - 132	30 - 30
A1005	95 - 96	20 - 20	132 - 132	30 - 30
A102	96 - 114	21 - 31	114 - 114	26 - 26
A103	99 - 107	23 - 29	114 - 114	26 - 26
A104	106 - 107	27 - 28	114 - 114	26 - 26
A105	106 - 120	26 - 32	114 - 114	26 - 26
A106	111 - 114	28 - 30	114 - 114	26 - 26
A107	109 - 110	28 - 28	114 - 114	26 - 26
A108	107 - 107	27 - 27	114 - 114	26 - 26
A109	102 - 102	25 - 25	114 - 114	26 - 26
A110	113 - 113	28 - 28	114 - 114	26 - 26
A1101	99 - 102	24 - 25	131 - 131	29 - 29
A1102	98 - 100	23 - 23	131 - 131	29 - 29
A1103	92 - 104	18 - 23	131 - 131	29 - 29
A1104	92 - 104	18 - 22	131 - 131	29 - 29
A1105	92 - 101	18 - 23	131 - 131	29 - 29
A1106	92 - 101	18 - 22	131 - 131	29 - 29
A1107	92 - 102	18 - 24	131 - 131	29 - 29
A1108	94 - 101	20 - 22	131 - 131	29 - 29
A1109	92 - 111	18 - 24	131 - 131	29 - 29
A111	108 - 108	26 - 26	114 - 114	26 - 26
A112	110 - 110	26 - 26	114 - 114	26 - 26
A1201	99 - 113	20 - 29	110 - 110	27 - 27
A1202	99 - 108	20 - 26	110 - 110	27 - 27
A1203	99 - 118	20 - 28	110 - 110	27 - 27
A1300	100 - 101	20 - 20	256 - 256	36 - 36
A1301	92 - 93	19 - 19	153 - 153	30 - 30
A1302	93 - 95	20 - 21	153 - 153	30 - 30

ASR ID	19th Highest Hourly NO_2 Conc. ($\mu g/m^3$)	Annual NO_2 Conc. ($\mu g/m^3$)	4th Highest 10-min SO_2 Conc. ($\mu g/m^3$)	4th Highest Daily SO_2 Conc. ($\mu g/m^3$)
A1303	100 - 104	22 - 24	153 - 153	30 - 30
A1304	98 - 107	21 - 24	153 - 153	30 - 30
A1305	99 - 110	20 - 24	153 - 153	30 - 30
A1306	100 - 105	20 - 22	153 - 153	30 - 30
A1307	93 - 95	20 - 20	153 - 153	30 - 30
A1308	96 - 102	19 - 20	153 - 153	30 - 30
A1309	93 - 95	19 - 19	153 - 153	30 - 30
A1401	94 - 95	20 - 20	126 - 126	29 - 29
A1402	91 - 102	18 - 23	126 - 126	29 - 29
A1403	91 - 100	18 - 22	126 - 126	29 - 29
A1404	91 - 102	17 - 21	126 - 126	29 - 29
A1405	91 - 101	17 - 21	126 - 126	29 - 29
A1501	98 - 98	22 - 22	131 - 131	33 - 33
A201	105 - 105	25 - 25	127 - 127	28 - 28
A202	111 - 112	26 - 27	127 - 127	28 - 28
A203	112 - 114	26 - 27	127 - 127	28 - 28
A204	110 - 116	26 - 28	127 - 127	28 - 28
A205	110 - 113	26 - 27	127 - 127	28 - 28
A206	110 - 111	30 - 31	127 - 127	28 - 28
A207	100 - 102	25 - 25	127 - 127	28 - 28
A208	111 - 111	29 - 29	127 - 127	28 - 28
A209	110 - 113	26 - 29	127 - 127	28 - 28
A301	98 - 100	24 - 24	111 - 111	26 - 26
A302	100 - 100	24 - 25	111 - 111	26 - 26
A303	99 - 101	24 - 24	111 - 111	26 - 26
A304	99 - 100	24 - 25	111 - 111	26 - 26
A305	102 - 103	25 - 25	111 - 111	26 - 26
A306	97 - 99	24 - 25	111 - 111	26 - 26
A307	101 - 107	24 - 25	111 - 111	26 - 26
A308	100 - 109	25 - 25	111 - 111	26 - 26
A309	98 - 103	24 - 24	111 - 111	26 - 26
A310	108 - 108	30 - 30	111 - 111	26 - 26
A311	97 - 103	23 - 26	111 - 111	26 - 26
A312	101 - 103	25 - 26	111 - 111	26 - 26
A313	98 - 107	23 - 28	111 - 111	26 - 26
A314	96 - 108	23 - 28	111 - 111	26 - 26
A401	96 - 99	23 - 24	117 - 117	26 - 26
A402	98 - 99	24 - 27	117 - 117	26 - 26
A403	94 - 102	21 - 27	117 - 117	26 - 26
A404	98 - 105	24 - 26	117 - 117	26 - 26
A405	93 - 103	22 - 27	117 - 117	26 - 26
A406	98 - 105	24 - 28	117 - 117	26 - 26
A407	98 - 101	24 - 27	117 - 117	26 - 26
A408	97 - 98	23 - 23	117 - 117	26 - 26
A409	93 - 108	21 - 30	117 - 117	26 - 26
A410	96 - 98	23 - 24	117 - 117	26 - 26
A411	99 - 100	25 - 25	117 - 117	26 - 26
A412	96 - 98	23 - 24	117 - 117	26 - 26
A413	106 - 110	27 - 29	117 - 117	26 - 26
A414	98 - 99	24 - 24	117 - 117	26 - 26
A415	96 - 96	23 - 23	117 - 117	26 - 26
A416	94 - 103	21 - 26	117 - 117	26 - 26
A502	92 - 116	21 - 31	118 - 118	26 - 26

ASR ID	19th Highest Hourly NO_2 Conc. ($\mu g/m^3$)	Annual NO_2 Conc. ($\mu g/m^3$)	4th Highest 10-min SO_2 Conc. ($\mu g/m^3$)	4th Highest Daily SO_2 Conc. ($\mu g/m^3$)
A503	105 - 114	27 - 30	118 - 118	26 - 26
A504	106 - 109	25 - 26	118 - 118	26 - 26
A505	105 - 106	23 - 24	118 - 118	26 - 26
A506	101 - 101	24 - 24	118 - 118	26 - 26
A507	99 - 105	22 - 26	118 - 118	26 - 26
A508	101 - 106	22 - 26	118 - 118	26 - 26
A601	99 - 99	21 - 21	119 - 119	30 - 30
A602	100 - 100	25 - 25	119 - 119	30 - 30
A603	101 - 101	24 24	119 - 119	30 - 30
A701	97 - 102	23 26	110 - 110	28 - 28
A702	100 - 109	24 26	110 - 110	28 - 28
A703	102 - 103	25 26	110 - 110	28 - 28
A704	101 - 102	25 26	110 - 110	28 - 28
A705	107 - 111	25 26	110 - 110	28 - 28
A706	103 - 104	23 24	110 - 110	28 - 28
A707	96 - 109	20 24	110 - 110	28 - 28
A708	97 - 97	22 23	110 - 110	28 - 28
A801	102 - 102	23 24	111 - 111	27 - 27
A802	102 - 102	23 23	111 - 111	27 - 27
A803	98 - 102	22 23	111 - 111	27 - 27
A804	101 - 102	22 23	111 - 111	27 - 27
A805	101 - 105	22 25	111 - 111	27 - 27
A806	101 - 101	21 22	111 - 111	27 - 27
A807	101 - 102	22 22	111 - 111	27 - 27
A808	98 - 116	22 30	111 - 111	27 - 27
A809	97 - 105	21 24	111 - 111	27 - 27
A810	97 - 105	21 24	111 - 111	27 - 27
A811	96 - 111	21 27	111 - 111	27 - 27
A812	92 - 106	20 26	111 - 111	27 - 27
A813	92 - 111	20 26	111 - 111	27 - 27
A901	97 - 97	22 22	170 - 170	32 - 32
A902	96 - 97	20 20	170 - 170	32 - 32
A903	97 - 97	20 20	170 - 170	32 - 32

Note: Bolded and underlined figure represents exceedance of relevant AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.28 Predicted Operational Air Quality Impact (Particulates) on Sites with Air Sensitive Use in Year 2024 – Year 2030

Site	10th Highest Daily RSP Conc. ($\mu g/m^3$)	Annual RSP Conc. ($\mu g/m^3$)	10th Highest Daily FSP Conc. ($\mu g/m^3$)	Annual FSP Conc. ($\mu g/m^3$)
2-18	92 - 94	41 - 42	69 - 71	29 - 30
2-19	94 - 94	42 - 42	71 - 71	30 - 30
3-43	94 - 95	42 - 43	70 - 71	30 - 30
3-44	94 - 99	42 - 44	70 - 71	30 - 30
3-45	94 - 98	42 - 43	70 - 71	30 - 31
3-50	93 - 95	42 - 43	70 - 71	30 - 31
3-51	93 - 93	42 - 42	70 - 70	30 - 30
3-6	94 - 94	42 - 42	71 - 71	30 - 30
3-7	94 - 95	42 - 42	71 - 72	30 - 30
3-8	94 - 97	42 - 43	71 - 73	30 - 30
4-20	93 - 93	42 - 42	70 - 70	30 - 30
4-22	93 - 93	42 - 42	70 - 70	30 - 30
4-28	93 - 93	42 - 43	70 - 70	30 - 30
5-16	92 - 93	41 - 41	69 - 70	29 - 30

Site	10th Highest Daily RSP Conc. (µg/m³)	Annual RSP Conc. (µg/m³)	10th Highest Daily FSP Conc. (µg/m³)	Annual FSP Conc. (µg/m³)
5-17	92 - 92	41 - 41	69 - 69	29 - 29
5-18a	92 - 92	41 - 41	69 - 69	29 - 29
5-18b	92 - 92	41 - 41	69 - 69	29 - 29
5-2	92 - 92	41 - 41	69 - 69	29 - 29
5-21	92 - 92	41 - 41	69 - 69	29 - 29
5-22	92 - 92	41 - 41	69 - 69	29 - 29
5-23	92 - 92	41 - 41	69 - 69	29 - 29
5-24	92 - 92	41 - 41	69 - 69	29 - 29
5-26	92 - 92	41 - 41	69 - 69	29 - 29
5-27	92 - 92	41 - 41	69 - 69	29 - 29
5-28	92 - 92	41 - 41	69 - 69	29 - 29
5-32	92 - 93	41 - 42	69 - 70	29 - 30
5-33	92 - 92	41 - 41	69 - 69	29 - 29
5-3a	92 - 92	41 - 41	69 - 69	29 - 29
5-3b	92 - 92	41 - 41	69 - 69	29 - 29
5-6	92 - 92	41 - 41	69 - 69	29 - 29

Note: Bolded and underlined figure represents exceedance of relevant AQO. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.29 Predicted Operational Air Quality Impact (Particulates) on Representative Existing ASRs in Year 2024 – Year 2030

ASR ID	10th Highest Daily RSP Conc. (µg/m³)	Annual RSP Conc. (µg/m³)	10th Highest Daily FSP Conc. (µg/m³)	Annual FSP Conc. (µg/m³)
A1001	94 - 94	42 - 42	71 - 71	30 - 30
A1002	94 - 94	42 - 42	71 - 71	30 - 30
A1003	94 - 94	42 - 42	71 - 71	30 - 30
A1004	94 - 94	42 - 42	71 - 71	30 - 30
A1005	94 - 94	42 - 42	71 - 71	30 - 30
A102	93 - 93	42 - 43	70 - 70	30 - 30
A103	93 - 93	42 - 42	70 - 70	30 - 30
A104	93 - 93	42 - 42	70 - 70	30 - 30
A105	93 - 93	42 - 43	70 - 70	30 - 30
A106	93 - 93	42 - 43	70 - 70	30 - 30
A107	93 - 93	42 - 42	70 - 70	30 - 30
A108	93 - 93	42 - 42	70 - 70	30 - 30
A109	93 - 93	42 - 42	70 - 70	30 - 30
A110	93 - 93	42 - 42	70 - 70	30 - 30
A1101	92 - 92	41 - 41	69 - 69	29 - 29
A1102	92 - 92	41 - 41	69 - 69	29 - 29
A1103	92 - 92	41 - 41	69 - 69	29 - 29
A1104	92 - 92	41 - 41	69 - 69	29 - 29
A1105	92 - 92	41 - 41	69 - 69	29 - 29
A1106	92 - 92	41 - 41	69 - 69	29 - 29
A1107	92 - 92	41 - 41	69 - 69	29 - 29
A1108	92 - 92	41 - 41	69 - 69	29 - 29
A1109	92 - 92	41 - 41	69 - 69	29 - 29
A111	93 - 93	42 - 42	70 - 70	30 - 30
A112	93 - 93	42 - 42	70 - 70	30 - 30
A1201	92 - 92	41 - 41	69 - 69	29 - 29
A1202	92 - 92	41 - 41	69 - 69	29 - 29
A1203	92 - 92	41 - 41	69 - 69	29 - 29
A1300	93 - 93	41 - 41	70 - 70	29 - 29
A1301	92 - 92	41 - 41	69 - 69	29 - 29
A1302	92 - 92	41 - 41	69 - 69	29 - 29
A1303	92 - 92	41 - 41	69 - 69	29 - 29

ASR ID	10th Highest Daily RSP Conc. (µg/m³)	Annual RSP Conc. (µg/m³)	10th Highest Daily FSP Conc. (µg/m³)	Annual FSP Conc. (µg/m³)
A1304	92 - 92	41 - 41	69 - 69	29 - 29
A1305	92 - 92	41 - 41	69 - 69	29 - 29
A1306	92 - 92	41 - 41	69 - 69	29 - 29
A1307	92 - 92	41 - 41	69 - 69	29 - 29
A1308	92 - 92	41 - 41	69 - 69	29 - 29
A1309	92 - 92	41 - 41	69 - 69	29 - 29
A1401	91 - 91	41 - 41	69 - 69	29 - 29
A1402	91 - 92	40 - 41	68 - 69	29 - 29
A1403	91 - 91	40 - 41	68 - 69	29 - 29
A1404	91 - 91	40 - 41	68 - 69	29 - 29
A1405	91 - 92	40 - 41	68 - 69	29 - 29
A1501	97 - 97	43 - 43	73 - 73	30 - 30
A201	93 - 93	42 - 42	70 - 70	30 - 30
A202	93 - 93	42 - 42	70 - 70	30 - 30
A203	93 - 93	42 - 42	70 - 70	30 - 30
A204	93 - 93	42 - 42	70 - 70	30 - 30
A205	93 - 93	42 - 42	70 - 70	30 - 30
A206	94 - 94	42 - 42	70 - 70	30 - 30
A207	93 - 93	42 - 42	70 - 70	30 - 30
A208	93 - 93	42 - 42	70 - 70	30 - 30
A209	93 - 93	42 - 42	70 - 70	30 - 30
A301	93 - 93	42 - 42	70 - 70	30 - 30
A302	93 - 93	42 - 42	70 - 70	30 - 30
A303	93 - 93	42 - 42	70 - 70	30 - 30
A304	93 - 93	42 - 42	70 - 70	30 - 30
A305	93 - 93	42 - 42	70 - 70	30 - 30
A306	93 - 93	42 - 42	70 - 70	30 - 30
A307	93 - 93	42 - 42	70 - 70	30 - 30
A308	93 - 93	42 - 42	70 - 70	30 - 30
A309	93 - 93	42 - 42	70 - 70	30 - 30
A310	93 - 93	43 - 43	70 - 70	30 - 30
A311	93 - 93	42 - 42	70 - 70	30 - 30
A312	93 - 93	42 - 42	70 - 70	30 - 30
A313	93 - 93	42 - 42	70 - 70	30 - 30
A314	93 - 93	42 - 42	70 - 70	30 - 30
A401	92 - 92	41 - 41	69 - 69	29 - 29
A402	92 - 92	41 - 41	69 - 69	29 - 29
A403	92 - 92	41 - 41	69 - 69	29 - 29
A404	92 - 92	41 - 41	69 - 69	29 - 29
A405	92 - 92	41 - 41	69 - 69	29 - 29
A406	92 - 92	41 - 41	69 - 69	29 - 29
A407	92 - 92	41 - 41	69 - 69	29 - 29
A408	92 - 92	41 - 41	69 - 69	29 - 29
A409	92 - 92	41 - 41	69 - 69	29 - 30
A410	92 - 92	41 - 41	69 - 69	29 - 29
A411	92 - 92	41 - 41	69 - 69	29 - 29
A412	92 - 92	41 - 41	69 - 69	29 - 29
A413	92 - 92	41 - 41	69 - 69	29 - 30
A414	92 - 92	41 - 41	69 - 69	29 - 29
A415	92 - 92	41 - 41	69 - 69	29 - 29
A416	92 - 92	41 - 41	69 - 69	29 - 29
A502	92 - 93	41 - 42	69 - 70	29 - 30
A503	93 - 93	42 - 42	70 - 70	30 - 30
A504	92 - 92	42 - 42	69 - 69	30 - 30

ASR ID	10th Highest Daily RSP Conc. (µg/m3)	Annual RSP Conc. (µg/m3)	10th Highest Daily FSP Conc. (µg/m3)	Annual FSP Conc. (µg/m3)
A505	92 - 92	42 - 42	69 - 69	29 - 29
A506	92 - 92	42 - 42	69 - 69	29 - 29
A507	92 - 92	41 - 42	69 - 69	29 - 29
A508	92 - 92	41 - 42	69 - 69	29 - 29
A601	94 - 94	41 - 41	70 - 70	29 - 29
A602	94 - 94	42 - 42	71 - 71	30 - 30
A603	94 - 94	42 42	71 - 71	30 - 30
A701	92 - 92	41 41	69 - 69	29 - 29
A702	92 - 92	41 41	69 - 69	29 - 29
A703	92 - 92	41 41	69 - 69	29 - 29
A704	92 - 92	41 41	69 - 69	29 - 29
A705	92 - 92	41 41	69 - 69	29 - 29
A706	92 - 92	41 41	69 - 69	29 - 29
A707	92 - 92	41 41	69 - 69	29 - 29
A708	92 - 92	41 41	69 - 69	29 - 29
A801	92 - 92	41 41	69 - 69	29 - 29
A802	92 - 92	41 41	69 - 69	29 - 29
A803	92 - 92	41 41	69 - 69	29 - 29
A804	92 - 92	41 41	69 - 69	29 - 29
A805	92 - 92	41 41	69 - 69	29 - 29
A806	92 - 92	41 41	69 - 69	29 - 29
A807	92 - 92	41 41	69 - 69	29 - 29
A808	92 - 92	41 41	69 - 69	29 - 29
A809	92 - 92	41 41	69 - 69	29 - 29
A810	92 - 92	41 41	69 - 69	29 - 29
A811	92 - 92	41 41	69 - 69	29 - 29
A812	92 - 92	41 41	69 - 69	29 - 29
A813	92 - 92	41 41	69 - 69	29 - 29
A901	95 - 95	42 42	72 - 72	30 - 30
A902	95 - 95	42 42	71 - 71	30 - 30
A903	95 - 95	42 42	71 - 71	30 - 30

Note: Bolded and underlined figure represents exceedance of relevant AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

3.9.4 According to the assessment results, the worst hit level generally appears at 1.5 meters above ground (mAG) of the receptors. As NO_2 and particulates are considered the potential key air pollutant associated with the operation of the Project, the contour plots for NO_2, RSP and FSP are presented. The contour plot of 19th highest hourly average and annual average NO_2 concentrations at 1.5 mAG are illustrated in **Figure 3.4** and **Figure 3.5**. The contour plot of 10th highest daily average and annual average RSP concentrations at 1.5 mAG are illustrated in **Figure 3.8** and **Figure 3.9**. The contour plot of 10th highest daily average and annual average FSP concentrations at 1.5 mAG are illustrated in **Figure 3.10** and **Figure 3.11**. Referring to the contour plots, no existing ASRs and planned ASRs occupied during Year 2024 - 2030 are found exceedance of AQOs.

Year 2031 – 2039 Scenario

3.9.5 The cumulative air quality impact due to emissions from open roads, ventilation building and tunnel portal of TMWB, planned hospital, industrial and port backup / logistics facilities at the Project sites and representative existing ASRs in this scenario has been evaluated. The predicted unmitigated cumulative 19th highest hourly and annual average NO_2, 4th highest 10-min and 4th highest daily average SO_2, 10th highest daily and annual average RSP and FSP concentrations at the Project sites and representative existing ASRs are summarised in **Table 3.30**, **Table 3.31, Table 3.32** and **Table 3.33** respectively. The detailed results are presented in **Appendix 3.10**. Results indicate that all representative existing and planned ASRs would comply with the AQOs.

Table 3.30 Predicted Operational Air Quality Impact on Sites with Air Sensitive Use in Year 2031 – Year 2039

Site	19th Highest Hourly NO_2 Conc. ($\mu g/m^3$)	Annual NO_2 Conc. ($\mu g/m^3$)	4th Highest 10-min SO_2 Conc. ($\mu g/m^3$)	4th Highest Daily SO_2 Conc. ($\mu g/m^3$)
1-1	91 - 99	18 - 21	126 - 131	29 - 29
1-12	92 - 98	18 - 21	131 - 131	29 - 29
1-13	91 - 97	18 - 21	110 - 131	28 - 29
1-15	94 - 99	19 - 21	110 - 110	28 - 28
1-16	91 - 98	18 - 21	110 - 110	28 - 28
1-17	91 - 99	18 - 21	110 - 110	28 - 28
1-18	95 - 99	20 - 21	110 - 110	28 - 28
1-19	91 - 101	19 - 22	110 - 110	28 - 28
1-2	91 - 97	18 - 21	126 - 126	29 - 29
1-21	91 - 100	18 - 22	110 - 110	28 - 28
1-22	96 - 104	20 - 25	110 - 110	28 - 28
1-29	94 - 100	21 - 24	117 - 117	26 - 26
1-3	92 - 98	19 - 20	131 - 131	29 - 29
1-30	91 - 98	21 - 23	117 - 117	26 - 26
1-4	92 - 97	18 - 20	131 - 131	29 - 29
1-5	92 - 100	18 - 21	131 - 131	29 - 29
1-6	92 - 100	18 - 21	131 - 131	29 - 29
1-8	92 - 99	18 - 21	131 - 131	29 - 29
1-9	92 - 101	18 - 21	131 - 131	29 - 29
2-1	91 - 93	18 - 19	153 - 153	30 - 30
2-10	91 - 94	18 - 20	153 - 153	30 - 30
2-11	91 - 97	18 - 22	153 - 153	30 - 30
2-14	91 - 94	18 - 20	153 - 153	30 - 30
2-15	92 - 93	18 - 19	153 - 153	30 - 30
2-16	91 - 92	18 - 19	153 - 153	30 - 30
2-17	92 - 93	18 - 20	153 - 153	30 - 30
2-18	93 - 97	19 - 20	132 - 153	30 - 30
2-19	95 - 96	20 - 20	132 - 132	30 - 30
2-2	91 - 93	18 - 19	153 - 153	30 - 30
2-20	94 - 95	19 - 21	132 - 132	30 - 30
2-21	96 - 100	19 - 21	132 - 132	30 - 30
2-22	96 - 98	19 - 21	132 - 132	30 - 30
2-24	92 - 98	19 - 22	131 - 132	29 - 30
2-25	96 - 98	19 - 20	132 - 132	30 - 30
2-26	96 - 99	19 - 20	132 - 132	30 - 30
2-28	97 - 100	19 - 20	132 - 132	30 - 30
2-29	96 - 101	19 - 21	132 - 132	30 - 30
2-3	91 - 92	18 - 19	153 - 153	30 - 30
2-30	92 - 102	18 - 21	132 - 132	30 - 30
2-31	92 - 98	18 - 22	131 - 132	29 - 30

Site	19th Highest Hourly NO_2 Conc. ($\mu g/m^3$)	Annual NO_2 Conc. ($\mu g/m^3$)	4th Highest 10-min SO_2 Conc. ($\mu g/m^3$)	4th Highest Daily SO_2 Conc. ($\mu g/m^3$)
2-33	97 - 107	19 - 23	132 - 132	30 - 30
2-7	91 - 93	18 - 19	153 - 153	30 - 30
2-9	91 - 93	18 - 20	153 - 153	30 - 30
3-1	92 - 98	18 - 23	132 - 132	30 - 30
3-11	92 - 104	19 - 24	119 - 131	30 - 33
3-13	91 - 101	18 - 23	119 - 132	30 - 30
3-14	92 - 103	19 - 23	119 - 119	30 - 30
3-15	91 - 101	18 - 25	132 - 132	30 - 30
3-16	93 - 106	19 - 26	119 - 119	30 - 30
3-18	92 - 111	19 - 26	119 - 119	30 - 30
3-20	93 - 109	19 - 23	119 - 119	30 - 30
3-24	93 - 119	19 - 25	119 - 119	30 - 30
3-25	91 - 106	19 - 22	110 - 119	28 - 30
3-27	93 - 112	19 - 25	119 - 119	30 - 30
3-28	93 - 109	19 - 24	119 - 119	30 - 30
3-29	93 - 107	19 - 23	119 - 134	29 - 30
3-32	94 - 108	19 - 24	119 - 134	29 - 30
3-33	95 - 105	20 - 23	134 - 134	29 - 29
3-34	95 - 100	21 - 22	117 - 134	26 - 29
3-39	95 - 109	20 - 24	134 - 134	29 - 29
3-4	91 - 102	18 - 24	132 - 132	30 - 30
3-40	95 - 105	20 - 23	134 - 134	29 - 29
3-42	89 - 103	19 - 24	117 - 134	26 - 29
3-43	97 - 125	21 - 36	134 - 134	29 - 29
3-44	97 - 111	21 - 28	134 - 134	29 - 29
3-45	98 - 125	21 - 36	134 - 134	29 - 29
3-5	91 - 98	18 - 23	132 - 132	30 - 30
3-50	96 - 134	21 - 36	127 - 134	28 - 29
3-51	96 - 140	21 - 36	127 - 127	28 - 28
3-52	94 - 145	21 - 36	127 - 127	28 - 28
3-6	90 - 100	18 - 24	132 - 132	30 - 30
3-7	90 - 98	18 - 23	132 - 170	30 - 32
3-8	90 - 103	18 - 25	131 - 170	30 - 33
4-1	94 - 115	19 - 26	134 - 134	29 - 29
4-10	94 - 101	19 - 22	134 - 134	29 - 29
4-12a	91 - 127	19 - 30	127 - 127	28 - 28
4-12b	91 - 118	19 - 27	127 - 127	28 - 28
4-12c	94 - 107	19 - 24	134 - 134	29 - 29
4-12d	91 - 106	19 - 26	127 - 127	28 - 28
4-13a	91 - 109	19 - 24	127 - 127	28 - 28
4-13b	91 - 103	19 - 22	127 - 134	28 - 29
4-14	100 - 103	22 - 23	127 - 134	28 - 29
4-15	91 - 114	20 - 29	127 - 127	28 - 28
4-16	91 - 104	19 - 23	127 - 127	28 - 28
4-17	94 - 129	21 - 31	127 - 127	28 - 28
4-2	99 - 116	21 - 26	134 - 134	29 - 29
4-20	105 - 115	24 - 27	127 - 127	28 - 28
4-21	92 - 126	21 - 30	127 - 127	28 - 28
4-22	91 - 110	19 - 27	127 - 127	28 - 28
4-24	87 - 99	18 - 24	111 - 111	26 - 26
4-25a	87 - 101	18 - 23	111 - 127	26 - 28
4-25b	87 - 98	18 - 24	111 - 117	26 - 26
4-25c	91 - 101	19 - 23	127 - 134	28 - 29

Site	19th Highest Hourly NO_2 Conc. ($\mu g/m^3$)	Annual NO_2 Conc. ($\mu g/m^3$)	4th Highest 10-min SO_2 Conc. ($\mu g/m^3$)	4th Highest Daily SO_2 Conc. ($\mu g/m^3$)
4-26	87 - 99	18 - 23	111 - 111	26 - 26
4-28	87 - 108	18 - 25	111 - 127	26 - 28
4-29	91 - 105	19 - 24	127 - 127	28 - 28
4-3	94 - 103	19 - 24	134 - 134	29 - 29
4-31	91 - 107	19 - 25	127 - 127	28 - 28
4-32	91 - 107	19 - 26	127 - 127	28 - 28
4-33	96 - 112	21 - 27	127 - 127	28 - 28
4-36	96 - 105	21 - 25	127 - 127	28 - 28
4-4	94 - 111	19 - 25	134 - 134	29 - 29
4-5	94 - 110	19 - 24	134 - 134	29 - 29
4-6	94 - 101	19 - 23	134 - 134	29 - 29
4-8	97 - 101	21 - 22	134 - 134	29 - 29
4-9	94 - 101	19 - 22	134 - 134	29 - 29
5-1	92 - 103	20 - 24	111 - 111	27 - 27
5-13	90 - 108	19 - 25	110 - 118	26 - 28
5-14	92 - 104	21 - 24	118 - 118	26 - 26
5-16	91 - 102	18 - 26	110 - 110	28 - 28
5-17	91 - 98	18 - 24	110 - 110	28 - 28
5-18a	91 - 98	18 - 22	110 - 110	28 - 28
5-18b	91 - 99	18 - 24	110 - 110	28 - 28
5-2	94 - 103	21 - 25	111 - 111	27 - 27
5-21	95 - 98	20 - 23	110 - 110	28 - 28
5-22	91 - 99	20 - 25	117 - 117	26 - 26
5-23	91 - 98	20 - 23	117 - 117	26 - 26
5-24	88 - 98	18 - 23	117 - 117	26 - 26
5-26	89 - 99	19 - 24	117 - 117	26 - 26
5-27	93 - 99	21 - 24	117 - 117	26 - 26
5-28	93 - 99	21 - 24	117 - 117	26 - 26
5-32	88 - 105	18 - 28	117 - 117	26 - 26
5-33	91 - 99	20 - 24	117 - 117	26 - 26
5-34	91 - 96	20 - 24	117 - 117	26 - 26
5-37	92 - 99	20 - 24	117 - 117	26 - 26
5-38	94 - 104	21 - 25	117 - 117	26 - 26
5-3a	91 - 102	18 - 23	110 - 111	27 - 28
5-3b	92 - 101	20 - 23	111 - 111	27 - 27
5-4	92 - 102	20 - 25	111 - 111	27 - 27
5-42	98 - 107	22 - 24	117 - 117	26 - 26
5-6	100 - 103	22 - 24	111 - 111	27 - 27
5-7a	92 - 102	20 - 24	111 - 111	27 - 27
5-7b	92 - 101	20 - 25	111 - 111	27 - 27
5-8	92 - 102	20 - 25	111 - 111	27 - 27
5-9	92 - 102	20 - 23	111 - 111	27 - 27

Note: Bolded and underlined figure represents exceedance of relevant AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.31 Predicted Operational Air Quality Impact on Representative Existing ASRs in Year 2031 – Year 2039

ASR ID	19th Highest Hourly NO_2 Conc. ($\mu g/m^3$)	Annual NO_2 Conc. ($\mu g/m^3$)	4th Highest 10-min SO_2 Conc. ($\mu g/m^3$)	4th Highest Daily SO_2 Conc. ($\mu g/m^3$)
A1001	97 - 97	20 - 20	132 - 132	30 - 30
A1002	97 - 97	21 - 21	132 - 132	30 - 30
A1003	95 - 95	20 - 20	132 - 132	30 - 30
A1004	95 - 95	19 - 19	132 - 132	30 - 30
A1005	95 - 95	19 - 20	132 - 132	30 - 30
A102	95 - 111	21 - 30	114 - 114	26 - 26
A103	98 - 104	23 - 27	114 - 114	26 - 26
A104	102 - 103	25 - 26	114 - 114	26 - 26
A105	101 - 112	25 - 28	114 - 114	26 - 26
A106	103 - 105	26 - 27	114 - 114	26 - 26
A107	104 - 104	26 - 26	114 - 114	26 - 26
A108	102 - 102	25 - 26	114 - 114	26 - 26
A109	98 - 99	24 - 24	114 - 114	26 - 26
A110	111 - 112	26 - 26	114 - 114	26 - 26
A1101	97 - 97	21 - 21	131 - 131	29 - 29
A1102	96 - 96	20 - 20	131 - 131	29 - 29
A1103	92 - 97	18 - 20	131 - 131	29 - 29
A1104	92 - 97	18 - 20	131 - 131	29 - 29
A1105	92 - 97	18 - 21	131 - 131	29 - 29
A1106	92 - 97	18 - 20	131 - 131	29 - 29
A1107	92 - 97	18 - 22	131 - 131	29 - 29
A1108	93 - 96	19 - 20	131 - 131	29 - 29
A1109	92 - 104	18 - 22	131 - 131	29 - 29
A111	105 - 106	24 - 24	114 - 114	26 - 26
A112	108 - 108	25 - 25	114 - 114	26 - 26
A1201	99 - 108	20 - 26	110 - 110	27 - 27
A1202	99 - 105	20 - 24	110 - 110	27 - 27
A1203	99 - 115	20 - 26	110 - 110	27 - 27
A1300	100 - 100	20 - 20	256 - 256	36 - 36
A1301	91 - 91	19 - 19	153 - 153	30 - 30
A1302	92 - 92	19 - 19	153 - 153	30 - 30
A1303	95 - 96	20 - 21	153 - 153	30 - 30
A1304	94 - 97	19 - 20	153 - 153	30 - 30
A1305	94 - 98	19 - 20	153 - 153	30 - 30
A1306	93 - 95	19 - 19	153 - 153	30 - 30
A1307	91 - 92	19 - 19	153 - 153	30 - 30
A1308	92 - 93	19 - 19	153 - 153	30 - 30
A1309	91 - 92	18 - 19	153 - 153	30 - 30
A1401	92 - 92	19 - 19	126 - 126	29 - 29
A1402	91 - 95	18 - 21	126 - 126	29 - 29
A1403	91 - 95	18 - 21	126 - 126	29 - 29
A1404	91 - 97	17 - 20	126 - 126	29 - 29
A1405	91 - 97	17 - 20	126 - 126	29 - 29
A1501	97 - 97	22 - 22	131 - 131	33 - 33
A201	103 - 104	23 - 23	127 - 127	28 - 28
A202	106 - 106	24 - 24	127 - 127	28 - 28
A203	107 - 107	24 - 25	127 - 127	28 - 28
A204	105 - 109	24 - 25	127 - 127	28 - 28
A205	104 - 108	23 - 24	127 - 127	28 - 28
A206	108 - 110	30 - 31	127 - 127	28 - 28
A207	97 - 97	24 - 24	127 - 127	28 - 28

ASR ID	19th Highest Hourly NO_2 Conc. ($\mu g/m^3$)	Annual NO_2 Conc. ($\mu g/m^3$)	4th Highest 10-min SO_2 Conc. ($\mu g/m^3$)	4th Highest Daily SO_2 Conc. ($\mu g/m^3$)
A208	107 - 109	28 - 29	127 - 127	28 - 28
A209	104 - 109	23 - 26	127 - 127	28 - 28
A301	96 - 96	22 - 22	111 - 111	26 - 26
A302	95 - 96	22 - 22	111 - 111	26 - 26
A303	95 - 97	22 - 22	111 - 111	26 - 26
A304	96 - 98	22 - 22	111 - 111	26 - 26
A305	97 - 97	22 - 22	111 - 111	26 - 26
A306	95 - 97	22 - 22	111 - 111	26 - 26
A307	95 - 96	22 - 22	111 - 111	26 - 26
A308	96 - 97	22 - 22	111 - 111	26 - 26
A309	96 - 97	22 - 22	111 - 111	26 - 26
A310	100 - 100	25 - 25	111 - 111	26 - 26
A311	92 - 99	21 - 23	111 - 111	26 - 26
A312	98 - 99	22 - 23	111 - 111	26 - 26
A313	93 - 103	21 - 25	111 - 111	26 - 26
A314	93 - 101	21 - 25	111 - 111	26 - 26
A401	95 - 98	21 - 22	117 - 117	26 - 26
A402	97 - 98	22 - 24	117 - 117	26 - 26
A403	93 - 99	20 - 24	117 - 117	26 - 26
A404	97 - 102	22 - 24	117 - 117	26 - 26
A405	92 - 100	21 - 24	117 - 117	26 - 26
A406	97 - 99	22 - 25	117 - 117	26 - 26
A407	95 - 97	22 - 24	117 - 117	26 - 26
A408	96 - 98	22 - 22	117 - 117	26 - 26
A409	92 - 100	20 - 26	117 - 117	26 - 26
A410	97 - 98	22 - 24	117 - 117	26 - 26
A411	97 - 98	22 - 23	117 - 117	26 - 26
A412	95 - 96	22 - 24	117 - 117	26 - 26
A413	98 - 98	23 - 24	117 - 117	26 - 26
A414	96 - 96	22 - 22	117 - 117	26 - 26
A415	94 - 95	22 - 22	117 - 117	26 - 26
A416	92 - 99	20 - 24	117 - 117	26 - 26
A502	90 - 103	21 - 26	118 - 118	26 - 26
A503	98 - 104	24 - 26	118 - 118	26 - 26
A504	101 - 101	23 - 24	118 - 118	26 - 26
A505	99 - 99	22 - 23	118 - 118	26 - 26
A506	98 - 99	23 - 23	118 - 118	26 - 26
A507	98 - 101	21 - 24	118 - 118	26 - 26
A508	98 - 102	22 - 25	118 - 118	26 - 26
A601	99 - 100	21 - 21	119 - 119	30 - 30
A602	99 - 99	24 - 24	119 - 119	30 - 30
A603	99 - 99	23 - 23	119 - 119	30 - 30
A701	97 - 98	22 - 24	110 - 110	28 - 28
A702	98 - 105	22 - 23	110 - 110	28 - 28
A703	99 - 100	22 - 22	110 - 110	28 - 28
A704	97 - 98	22 - 22	110 - 110	28 - 28
A705	99 - 99	22 - 22	110 - 110	28 - 28
A706	97 - 98	21 - 21	110 - 110	28 - 28
A707	95 - 97	19 - 21	110 - 110	28 - 28
A708	97 - 98	21 - 21	110 - 110	28 - 28
A801	100 - 100	22 - 23	111 - 111	27 - 27
A802	99 - 100	22 - 22	111 - 111	27 - 27
A803	96 - 99	22 - 22	111 - 111	27 - 27

ASR ID	19th Highest Hourly NO_2 Conc. ($\mu g/m^3$)	Annual NO_2 Conc. ($\mu g/m^3$)	4th Highest 10-min SO_2 Conc. ($\mu g/m^3$)	4th Highest Daily SO_2 Conc. ($\mu g/m^3$)
A804	98 - 99	22 - 23	111 - 111	27 - 27
A805	98 - 102	22 - 24	111 - 111	27 - 27
A806	97 - 98	21 - 21	111 - 111	27 - 27
A807	99 - 100	21 - 21	111 - 111	27 - 27
A808	95 - 108	21 - 26	111 - 111	27 - 27
A809	95 - 102	21 - 23	111 - 111	27 - 27
A810	95 - 101	21 - 22	111 - 111	27 - 27
A811	94 - 104	21 - 25	111 - 111	27 - 27
A812	92 - 102	20 - 24	111 - 111	27 - 27
A813	92 - 101	20 - 23	111 - 111	27 - 27
A901	96 - 96	21 - 21	170 - 170	32 - 32
A902	96 - 96	20 - 20	170 - 170	32 - 32
A903	96 - 96	20 - 20	170 - 170	32 - 32

Note: Bolded and underlined figure represents exceedance of relevant AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.32 Predicted Operational Air Quality Impact (Particulates) on Sites with Air Sensitive Use in Year 2031 – Year 2039

Site	10th Highest Daily RSP Conc. ($\mu g/m^3$)	Annual RSP Conc. ($\mu g/m^3$)	10th Highest Daily FSP Conc. ($\mu g/m^3$)	Annual FSP Conc. ($\mu g/m^3$)
1-1	91 - 92	40 - 41	69 - 69	29 - 29
1-12	92 - 92	41 - 41	69 - 69	29 - 29
1-13	92 - 92	41 - 41	69 - 69	29 - 29
1-15	92 - 92	41 - 41	69 - 69	29 - 29
1-16	92 - 92	41 - 41	69 - 69	29 - 29
1-17	92 - 92	41 - 41	69 - 69	29 - 29
1-18	92 - 92	41 - 41	69 - 69	29 - 29
1-19	92 - 92	41 - 41	69 - 69	29 - 29
1-2	91 - 91	40 - 41	68 - 69	29 - 29
1-21	92 - 92	41 - 41	69 - 69	29 - 29
1-22	92 - 92	41 - 41	69 - 69	29 - 29
1-29	92 - 92	41 - 41	69 - 69	29 - 29
1-3	92 - 92	41 - 41	69 - 69	29 - 29
1-30	92 - 92	41 - 41	69 - 69	29 - 29
1-4	92 - 92	41 - 41	69 - 69	29 - 29
1-5	92 - 92	41 - 41	69 - 69	29 - 29
1-6	92 - 92	41 - 41	69 - 69	29 - 29
1-8	92 - 92	41 - 41	69 - 69	29 - 29
1-9	92 - 92	41 - 41	69 - 69	29 - 29
2-1	92 - 92	41 - 41	69 - 69	29 - 29
2-10	92 - 92	41 - 41	69 - 69	29 - 29
2-11	92 - 92	41 - 41	69 - 69	29 - 29
2-14	92 - 92	41 - 41	69 - 69	29 - 29
2-15	92 - 92	41 - 41	69 - 69	29 - 29
2-16	92 - 92	41 - 41	69 - 69	29 - 29
2-17	92 - 92	41 - 41	69 - 69	29 - 29
2-18	92 - 94	41 - 42	69 - 71	29 - 30
2-19	94 - 94	42 - 42	71 - 71	30 - 30
2-2	92 - 92	41 - 41	69 - 69	29 - 29
2-20	94 - 94	42 - 42	71 - 71	30 - 30
2-21	94 - 94	42 - 42	71 - 71	30 - 30
2-22	94 - 94	42 - 42	71 - 71	30 - 30
2-24	92 - 94	41 - 42	69 - 71	29 - 30
2-25	94 - 94	42 - 42	71 - 71	30 - 30

Site	10th Highest Daily RSP Conc. (µg/m³)	Annual RSP Conc. (µg/m³)	10th Highest Daily FSP Conc. (µg/m³)	Annual FSP Conc. (µg/m³)
2-26	94 - 94	42 - 42	71 - 71	30 - 30
2-28	94 - 94	42 - 42	71 - 71	30 - 30
2-29	94 - 94	42 - 42	71 - 71	30 - 30
2-3	92 - 92	41 - 41	69 - 69	29 - 29
2-30	94 - 94	42 - 42	71 - 71	29 - 30
2-31	92 - 94	41 - 42	69 - 71	29 - 30
2-33	94 - 94	42 - 42	71 - 71	30 - 30
2-7	92 - 92	41 - 41	69 - 69	29 - 29
2-9	92 - 92	41 - 41	69 - 69	29 - 29
3-1	94 - 94	42 - 42	71 - 71	29 - 30
3-11	94 - 97	41 - 43	70 - 73	29 - 30
3-13	94 - 94	41 - 42	70 - 71	29 - 30
3-14	94 - 94	41 - 42	70 - 70	29 - 30
3-15	94 - 94	42 - 42	71 - 71	29 - 30
3-16	94 - 94	41 - 42	70 - 71	29 - 30
3-18	94 - 94	41 - 42	70 - 70	29 - 30
3-20	94 - 94	41 - 41	70 - 70	29 - 29
3-24	94 - 94	41 - 42	70 - 70	29 - 30
3-25	92 - 94	41 - 41	69 - 70	29 - 29
3-27	94 - 94	41 - 42	70 - 70	29 - 30
3-28	94 - 94	41 - 41	70 - 70	29 - 29
3-29	94 - 94	41 - 42	70 - 70	29 - 30
3-32	94 - 94	41 - 42	70 - 70	29 - 30
3-33	94 - 94	42 - 42	70 - 70	30 - 30
3-34	92 - 94	41 - 42	69 - 70	29 - 30
3-39	94 - 94	42 - 42	70 - 70	30 - 30
3-4	94 - 94	42 - 42	71 - 71	29 - 30
3-40	94 - 94	42 - 42	70 - 70	30 - 30
3-42	92 - 94	41 - 42	69 - 70	29 - 30
3-43	94 - 95	42 - 43	70 - 71	30 - 30
3-44	94 - 98	42 - 43	70 - 71	30 - 30
3-45	94 - 97	42 - 43	70 - 71	30 - 30
3-5	94 - 94	42 - 42	71 - 71	29 - 30
3-50	93 - 95	42 - 43	70 - 71	30 - 30
3-51	93 - 93	42 - 42	70 - 70	30 - 30
3-52	93 - 94	42 - 43	70 - 71	30 - 31
3-6	94 - 94	42 - 42	71 - 71	29 - 30
3-7	94 - 95	42 - 42	71 - 72	29 - 30
3-8	94 - 97	42 - 43	71 - 73	29 - 30
4-1	94 - 94	42 - 42	70 - 70	30 - 30
4-10	94 - 94	42 - 42	70 - 70	30 - 30
4-12a	93 - 93	42 - 42	70 - 70	30 - 30
4-12b	93 - 93	42 - 42	70 - 70	30 - 30
4-12c	94 - 94	42 - 42	70 - 70	30 - 30
4-12d	93 - 93	42 - 42	70 - 70	30 - 30
4-13a	93 - 93	42 - 42	70 - 70	30 - 30
4-13b	93 - 94	42 - 42	70 - 70	30 - 30
4-14	93 - 94	42 - 42	70 - 70	30 - 30
4-15	93 - 93	42 - 42	70 - 70	30 - 30
4-16	93 - 93	42 - 42	70 - 70	30 - 30
4-17	93 - 93	42 - 42	70 - 70	30 - 30
4-2	94 - 95	42 - 42	70 - 70	30 - 30
4-20	93 - 93	42 - 42	70 - 70	30 - 30
4-21	93 - 93	42 - 42	70 - 70	30 - 30

Site	10th Highest Daily RSP Conc. (µg/m3)	Annual RSP Conc. (µg/m3)	10th Highest Daily FSP Conc. (µg/m3)	Annual FSP Conc. (µg/m3)
4-22	93 - 93	42 - 42	70 - 70	30 - 30
4-24	93 - 93	42 - 42	70 - 70	30 - 30
4-25a	93 - 93	42 - 42	70 - 70	30 - 30
4-25b	92 - 93	41 - 42	69 - 70	29 - 30
4-25c	93 - 94	42 - 42	70 - 70	30 - 30
4-26	93 - 93	42 - 42	70 - 70	30 - 30
4-28	93 - 93	42 - 42	70 - 70	30 - 30
4-29	93 - 93	42 - 42	70 - 70	30 - 30
4-3	94 - 94	42 - 42	70 - 70	30 - 30
4-31	93 - 93	42 - 42	70 - 70	30 - 30
4-32	93 - 93	42 - 42	70 - 70	30 - 30
4-33	93 - 93	42 - 42	70 - 70	30 - 30
4-36	93 - 93	42 - 42	70 - 70	30 - 30
4-4	94 - 94	42 - 42	70 - 70	30 - 30
4-5	94 - 94	42 - 42	70 - 70	30 - 30
4-6	94 - 94	42 - 42	70 - 70	30 - 30
4-8	94 - 94	42 - 42	70 - 70	30 - 30
4-9	94 - 94	42 - 42	70 - 70	30 - 30
5-1	92 - 92	41 - 41	69 - 69	29 - 29
5-13	92 - 92	41 - 41	69 - 69	29 - 29
5-14	92 - 93	41 - 41	69 - 69	29 - 29
5-16	92 - 92	41 - 41	69 - 69	29 - 29
5-17	92 - 92	41 - 41	69 - 69	29 - 29
5-18a	92 - 92	41 - 41	69 - 69	29 - 29
5-18b	92 - 92	41 - 41	69 - 69	29 - 29
5-2	92 - 92	41 - 41	69 - 69	29 - 29
5-21	92 - 92	41 - 41	69 - 69	29 - 29
5-22	92 - 92	41 - 41	69 - 69	29 - 29
5-23	92 - 92	41 - 41	69 - 69	29 - 29
5-24	92 - 92	41 - 41	69 - 69	29 - 29
5-26	92 - 92	41 - 41	69 - 69	29 - 29
5-27	92 - 92	41 - 41	69 - 69	29 - 29
5-28	92 - 92	41 - 41	69 - 69	29 - 29
5-32	92 - 92	41 - 41	69 - 69	29 - 29
5-33	92 - 92	41 - 41	69 - 69	29 - 29
5-34	92 - 92	41 - 41	69 - 69	29 - 29
5-37	92 - 92	41 - 41	69 - 69	29 - 29
5-38	92 - 92	41 - 41	69 - 69	29 - 29
5-3a	92 - 92	41 - 41	69 - 69	29 - 29
5-3b	92 - 92	41 - 41	69 - 69	29 - 29
5-4	92 - 92	41 - 41	69 - 69	29 - 29
5-42	92 - 92	41 - 41	69 - 69	29 - 29
5-6	92 - 92	41 - 41	69 - 69	29 - 29
5-7a	92 - 92	41 - 41	69 - 69	29 - 29
5-7b	92 - 92	41 - 41	69 - 69	29 - 29
5-8	92 - 92	41 - 41	69 - 69	29 - 29
5-9	92 - 92	41 - 41	69 - 69	29 - 29

Note: Bolded and underlined figure represents exceedance of relevant AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.33 Predicted Operational Air Quality Impact (Particulates) on Representative Existing ASRs in Year 2031 – Year 2039

ASR ID	10th Highest Daily RSP Conc. (µg/m³)	Annual RSP Conc. (µg/m³)	10th Highest Daily FSP Conc. (µg/m³)	Annual FSP Conc. (µg/m³)
A1001	94 - 94	42 - 42	71 - 71	30 - 30
A1002	94 - 94	42 - 42	71 - 71	30 - 30
A1003	94 - 94	42 - 42	71 - 71	30 - 30
A1004	94 - 94	42 - 42	71 - 71	30 - 30
A1005	94 - 94	42 - 42	71 - 71	30 - 30
A102	93 - 93	42 - 43	70 - 70	30 - 30
A103	93 - 93	42 - 42	70 - 70	30 - 30
A104	93 - 93	42 - 42	70 - 70	30 - 30
A105	93 - 93	42 - 42	70 - 70	30 - 30
A106	93 - 93	42 - 42	70 - 70	30 - 30
A107	93 - 93	42 - 42	70 - 70	30 - 30
A108	93 - 93	42 - 42	70 - 70	30 - 30
A109	93 - 93	42 - 42	70 - 70	30 - 30
A110	93 - 93	42 - 42	70 - 70	30 - 30
A1101	92 - 92	41 - 41	69 - 69	29 - 29
A1102	92 - 92	41 - 41	69 - 69	29 - 29
A1103	92 - 92	41 - 41	69 - 69	29 - 29
A1104	92 - 92	41 - 41	69 - 69	29 - 29
A1105	92 - 92	41 - 41	69 - 69	29 - 29
A1106	92 - 92	41 - 41	69 - 69	29 - 29
A1107	92 - 92	41 - 41	69 - 69	29 - 29
A1108	92 - 92	41 - 41	69 - 69	29 - 29
A1109	92 - 92	41 - 41	69 - 69	29 - 29
A111	93 - 93	42 - 42	70 - 70	30 - 30
A112	93 - 93	42 - 42	70 - 70	30 - 30
A1201	92 - 92	40 - 41	69 - 69	29 - 29
A1202	92 - 92	40 - 41	69 - 69	29 - 29
A1203	92 - 92	40 - 41	69 - 69	29 - 29
A1300	93 - 93	41 - 41	70 - 70	29 - 29
A1301	92 - 92	41 - 41	69 - 69	29 - 29
A1302	92 - 92	41 - 41	69 - 69	29 - 29
A1303	92 - 92	41 - 41	69 - 69	29 - 29
A1304	92 - 92	41 - 41	69 - 69	29 - 29
A1305	92 - 92	41 - 41	69 - 69	29 - 29
A1306	92 - 92	41 - 41	69 - 69	29 - 29
A1307	92 - 92	41 - 41	69 - 69	29 - 29
A1308	92 - 92	41 - 41	69 - 69	29 - 29
A1309	92 - 92	41 - 41	69 - 69	29 - 29
A1401	91 - 91	40 - 40	69 - 69	29 - 29
A1402	91 - 92	40 - 41	68 - 69	29 - 29
A1403	91 - 91	40 - 41	68 - 69	29 - 29
A1404	91 - 91	40 - 41	68 - 69	29 - 29
A1405	91 - 91	40 - 41	68 - 69	29 - 29
A1501	97 - 97	43 - 43	73 - 73	30 - 30
A201	93 - 93	42 - 42	70 - 70	30 - 30
A202	93 - 93	42 - 42	70 - 70	30 - 30
A203	93 - 93	42 - 42	70 - 70	30 - 30
A204	93 - 93	42 - 42	70 - 70	30 - 30
A205	93 - 93	42 - 42	70 - 70	30 - 30
A206	93 - 93	42 - 42	70 - 70	30 - 30
A207	93 - 93	42 - 42	70 - 70	30 - 30
A208	93 - 93	42 - 42	70 - 70	30 - 30

ASR ID	10th Highest Daily RSP Conc. (µg/m3)	Annual RSP Conc. (µg/m3)	10th Highest Daily FSP Conc. (µg/m3)	Annual FSP Conc. (µg/m3)
A209	93 - 93	42 - 42	70 - 70	30 - 30
A301	93 - 93	42 - 42	70 - 70	30 - 30
A302	93 - 93	42 - 42	70 - 70	30 - 30
A303	93 - 93	42 - 42	70 - 70	30 - 30
A304	93 - 93	42 - 42	70 - 70	30 - 30
A305	93 - 93	42 - 42	70 - 70	30 - 30
A306	93 - 93	42 - 42	70 - 70	30 - 30
A307	93 - 93	42 - 42	70 - 70	30 - 30
A308	93 - 93	42 - 42	70 - 70	30 - 30
A309	93 - 93	42 - 42	70 - 70	30 - 30
A310	93 - 93	42 - 42	70 - 70	30 - 30
A311	93 - 93	42 - 42	70 - 70	30 - 30
A312	93 - 93	42 - 42	70 - 70	30 - 30
A313	93 - 93	42 - 42	70 - 70	30 - 30
A314	93 - 93	42 - 42	70 - 70	30 - 30
A401	92 - 92	41 - 41	69 - 69	29 - 29
A402	92 - 92	41 - 41	69 - 69	29 - 29
A403	92 - 92	41 - 41	69 - 69	29 - 29
A404	92 - 92	41 - 41	69 - 69	29 - 29
A405	92 - 92	41 - 41	69 - 69	29 - 29
A406	92 - 92	41 - 41	69 - 69	29 - 29
A407	92 - 92	41 - 41	69 - 69	29 - 29
A408	92 - 92	41 - 41	69 - 69	29 - 29
A409	92 - 92	41 - 41	69 - 69	29 - 29
A410	92 - 92	41 - 41	69 - 69	29 - 29
A411	92 - 92	41 - 41	69 - 69	29 - 29
A412	92 - 92	41 - 41	69 - 69	29 - 29
A413	92 - 92	41 - 41	69 - 69	29 - 29
A414	92 - 92	41 - 41	69 - 69	29 - 29
A415	92 - 92	41 - 41	69 - 69	29 - 29
A416	92 - 92	41 - 41	69 - 69	29 - 29
A502	92 - 93	41 - 41	69 - 69	29 - 29
A503	93 - 93	41 - 42	69 - 69	29 - 29
A504	92 - 92	42 - 42	69 - 69	29 - 29
A505	92 - 92	41 - 41	69 - 69	29 - 29
A506	92 - 92	41 - 41	69 - 69	29 - 29
A507	92 - 92	41 - 41	69 - 69	29 - 29
A508	92 - 92	41 - 41	69 - 69	29 - 29
A601	94 - 94	41 - 41	70 - 70	29 - 29
A602	94 - 94	42 - 42	71 - 71	30 - 30
A603	94 - 94	42 - 42	71 - 71	30 - 30
A701	92 - 92	41 - 41	69 - 69	29 - 29
A702	92 - 92	41 - 41	69 - 69	29 - 29
A703	92 - 92	41 - 41	69 - 69	29 - 29
A704	92 - 92	41 - 41	69 - 69	29 - 29
A705	92 - 92	41 - 41	69 - 69	29 - 29
A706	92 - 92	41 - 41	69 - 69	29 - 29
A707	92 - 92	41 - 41	69 - 69	29 - 29
A708	92 - 92	41 - 41	69 - 69	29 - 29
A801	92 - 92	41 - 41	69 - 69	29 - 29
A802	92 - 92	41 - 41	69 - 69	29 - 29
A803	92 - 92	41 - 41	69 - 69	29 - 29
A804	92 - 92	41 - 41	69 - 69	29 - 29
A805	92 - 92	41 - 41	69 - 69	29 - 29

ASR ID	10th Highest Daily RSP Conc. (µg/m3)	Annual RSP Conc. (µg/m3)	10th Highest Daily FSP Conc. (µg/m3)	Annual FSP Conc. (µg/m3)
A806	92 - 92	41 - 41	69 - 69	29 - 29
A807	92 - 92	41 - 41	69 - 69	29 - 29
A808	92 - 92	41 - 41	69 - 69	29 - 29
A809	92 - 92	41 - 41	69 - 69	29 - 29
A810	92 - 92	41 - 41	69 - 69	29 - 29
A811	92 - 92	41 - 41	69 - 69	29 - 29
A812	92 - 92	41 - 41	69 - 69	29 - 29
A813	92 - 92	41 - 41	69 - 69	29 - 29
A901	95 - 95	42 - 42	71 - 72	30 - 30
A902	95 - 95	42 - 42	71 - 71	30 - 30
A903	95 - 95	42 - 42	71 - 71	30 - 30

Note: Bolded and underlined figure represents exceedance of relevant AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

3.9.6 According to the assessment results, the worst-hit generally appears at 1.5 meters above ground (mAG) of the receptors. As NO_2 and particulates are considered the potential key air pollutant associated with the operation of the Project, the contour plots for NO_2, RSP and FSP are presented. The contour plot of 19th highest hourly average and annual average NO_2 concentrations at 1.5 mAG are illustrated in **Figure 3.12** and **Figure 3.13**. The contour plot of 10th highest daily average and annual average RSP concentrations at 1.5 mAG are illustrated in **Figure 3.16** and **Figure 3.17**. The contour plot of 10th highest daily average and annual average FSP concentrations at 1.5 mAG are illustrated in **Figure 3.18** and **Figure 3.19**.

Predicted Odour Impact by Existing Chicken Farm

3.9.7 The odour concentrations due to the subject chicken farm at the nearest planned development sites are predicted and presented in **Table 3.34**. A contour plot of odour concentration covering 500 m area from chicken farm boundary at 1.5 mAG is presented in **Figure 3.20**. According to the spatial plot of odour concentration, it is noted that exceedance of 5 odour units would be predicted at portion of Site 3-1 (OU, PBU + SWU) at 1.5 mAG and 5 mAG. It is proposed that air sensitive uses at Site 3-1 should not be located at this exceedance zone or the fresh air intake of the building located at 6 mAG or above. The detailed assessment results are presented in **Appendix 3.15**.

Table 3.34 Predicted Cumulative Odour Concentration (OU/m3) due to Existing Chicken Farm

Site / ASR ID	Maximum 5-second Odour Concentration (OU/m3)
2-18	0.3 - 0.7
2-19	0.3 - 0.7
2-20	0.2 - 1.3
2-21	0.0 - 0.9
2-22	0.1 - 0.6
2-24	0.0 - 0.5
2-25	0.1 - 0.6
2-26	0.0 - 0.9
2-28	0.0 - 1.1
2-29	0.0 - 1.5
2-30	0.0 - 0.9
2-31	0.0 - 0.6
2-33	0.0 - 1.4
3-1	0.0 - **7.8**
3-13	0.0 - 0.5
3-15	0.0 - 1.0
3-4	0.0 - 2.0

Site / ASR ID	Maximum 5-second Odour Concentration (OU/m^3)
3-5	0.1 - 1.3
3-6	0.0 - 0.7
3-7	0.1 - 0.6
3-8	0.0 - 0.4
A1001	0.1 - 0.5
A1002	0.6 - 1.0
A1003	0.4 - 0.8
A1004	0.3 - 0.6
A1005	0.3 - 0.7

Note: The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights. For Site 3-1, exceedance of 5 odour units would be predicted at 1.5 mAG and 5 mAG, no exceedance is predicted at 6 mAG, detailed results refer to Appendix 3.15.

Predicted Cumulative Odour Impact from Upgraded San Wai STW, New HSK STW and planned RTS

3.9.8 The predicted cumulative odour concentrations due to the operation of expanded and upgraded San Wai STW, new HSK STW and planned RTS at the nearest ASRs are summarised in **Table 3.35**. Results indicate that no exceedance would be predicted at the nearest ASRs. Contour plots of odour concentration covering 500 m area from these facilities boundary at 1.5 mAG are presented in **Figure 3.21**. A 50 x 50 m grid is used to generate the pollution contours in order to investigate the pollutant dispersion. The detailed assessment results are presented in **Appendix 3.16**.

Table 3.35 Predicted Cumulative Odour Concentration (OU/m^3) due to STWs and RTS

Site / ASR ID	Maximum 5-second Odour Concentration (OU/m^3)
2-18	0.1 - 0.1
2-19	0.1 - 0.1
2-20	0.1 - 0.1
2-21	0.1 - 0.1
2-22	0.1 - 0.1
2-24	0.1 - 0.1
2-25	0.1 - 0.1
2-26	0.1 - 0.1
2-28	0.1 - 0.1
2-29	0.1 - 0.1
2-30	0.1 - 0.2
2-31	0.1 - 0.1
2-33	0.1 - 0.2
3-1	0.1 - 0.1
3-11	0.2 - 1.3
3-13	0.1 - 0.3
3-14	0.1 - 0.4
3-15	0.1 - 0.4
3-16	0.1 - 0.3
3-18	0.1 - 0.7
3-20	0.1 - 0.4
3-24	0.1 - 0.3
3-25	0.1 - 0.3
3-27	0.1 - 0.5
3-28	0.1 - 0.4
3-29	0.1 - 0.4
3-32	0.2 - 0.6
3-4	0.1 - 0.1
3-5	0.1 - 0.2
3-6	0.1 - 0.2
3-7	0.1 - 0.2

Site / ASR ID	Maximum 5-second Odour Concentration (OU/m^3)
3-8	0.1 - 1.2
A1001	0.1 - 0.1
A1002	0.1 - 0.1
A1003	0.1 - 0.1
A1004	0.1 - 0.1
A1005	0.1 - 0.1
A1501	0.9 - 0.9
A601	0.3 - 0.3
A602	1.5 - 1.5
A603	1.1 - 1.1
A901	0.4 - 0.4
A902	0.1 - 0.1
A903	0.1 - 0.1

Note: The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Potential Odour Impact from Sewage Pumping Stations

3.9.9 The four new SPS (DP9) would be constructed at Sites 2-34, 3-41, 3-48 and 4-35 to convey the sewage to San Wai STW and the new HSK STW. The design capacity of the SPS at Site 2-34 (SPS1), Site 3-41 (SPS2), Site 3-48 (SPS3) and Site 4-35 (SPS4) would be 27,000 m per day, 39,500 m^3 per day, 11,000 m^3 per day and 68,000 m^3 per day, respectively. The separation distance between the nearest ASRs and the SPSs would be 76 m for SPS1, 93 m for SPS2, 100 m for SPS3 and 10 m for SPS4.

3.9.10 Potential odour emission from wet well, inlet chamber and screen chamber are possible odour sources of the SPSs. All facilities and areas with potential odour emission such as wet wells, inlet chamber and screen chambers would be covered by reinforced concrete structure and the exhausted air would be conveyed to a deodouriser with odour removal efficiency of at least 90% (e.g. 90% of odour removal efficiency of deodouriser (granular activated carbon) for the SPS as stated in Approved EIA Report for Tuen Mun Area 54 Sewage Pumping Station) for treatment before discharge to the environment. The ventilation system would also maintain a slight negative pressure within the facilities. In addition, the exhaust outlet of the deodouriser would be located away from the nearby ASRs as far as practicable. Similar odour mitigation measures have also been implemented at other SPSs in urban area to successfully control the potential odour impacts. The screening wastes would be removed regularly (i.e. at least twice a week) from the SPSs, and would be properly packed and handled within the SPS structures and transported to designated landfill for disposal immediately after collection from the SPSs. With proper implementation of these odour mitigation measures, adverse odour impact from SPS would not be expected.

3.10 Mitigation of Adverse Environmental Impacts

Construction Phase

3.10.1 In order to minimise the construction dust impact, the following dust mitigation measures shall be implemented:

- Watering once per hour on active works areas, exposed areas and unpaved haul roads to reduce dust emission by 91.7%. Any potential dust impact and watering mitigation would be subject to the actual site condition. For example, a construction activity that produces inherently wet conditions or in cases under rainy weather, the above water application intensity may not be unreservedly applied. While the above watering frequency is to be followed, the extent of watering may vary depending on actual site conditions but should be sufficient to achieve the removal efficiency. The dust levels would be monitored and managed under an EM&A programme as specified in the EM&A Manual.

Year 2026 – Year 2030 Scenario

Tier 1 Assessment

3.10.2 With the implementation of the above measures, the predicted cumulative maximum hourly average TSP, 10th highest daily and annual average RSP and FSP concentrations at the representative ASRs are summarised in **Table 3.36** and **Table 3.37**. It is noted that the annual average RSP and FSP concentrations at the representative ASRs would comply with the AQOs. However, exceedance of hourly average TSP, 10th highest daily average RSP and FSP would still be predicted at some ASRs which are located in proximity to the work sites. Tier 2 assessment is required for these ASRs. The detailed prediction results are presented in **Appendix 3.11**. The contour plots of cumulative maximum hourly average TSP, 10th highest daily and annual average RSP and FSP concentrations at 1.5 mAG are presented in **Figure 3.22**, **Figure 3.23**, **Figure 3.24**, **Figure 3.25**, and **Figure 3.26** respectively.

Table 3.36 Predicted Construction Dust Impact on Project Sites with Air Sensitive Use in Year 2019 – Year 2030 (Mitigated, Tier 1)

Site	Maximum Hourly TSP Conc. (µg/m³)	10th Highest Daily RSP Conc. (µg/m³)	Annual RSP Conc. (µg/m³)	10th Highest Daily FSP Conc. (µg/m³)	Annual FSP Conc. (µg/m³)
2-18	348 - **544**	95 - 98	41 - 42	69 - 71	29 - 30
2-19	366 - 409	96 - 97	42 - 42	71 - 71	30 - 30
3-6	174 - **1523**	94 - **120**	42 - 43	71 - 72	30 - 30
3-7	174 - **729**	94 - **112**	42 - 43	71 - 72	30 - 30
3-8	174 - **863**	94 - **111**	42 - 43	71 - 75	30 - 30
4-20	**1153** - **2111**	**117** - **156**	43 - 45	75 - **81**	30 - 30
5-2	176 - **591**	92 - 98	41 - 41	69 - 70	29 - 29
5-22	173 - 476	92 - **101**	41 - 42	69 - 70	29 - 29
5-23	173 - 426	92 - 99	41 - 42	69 - 70	29 - 29
5-24	173 - **829**	92 - **113**	41 - 42	69 - 72	29 - 30
5-26	173 - **694**	92 - **112**	41 - 42	69 - 72	29 - 30
5-33	173 - 408	92 - 96	41 - 42	69 - 70	29 - 29
5-6	213 - 427	92 - 93	41 - 41	69 - 69	29 - 29

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.37 Predicted Construction Dust Impact on Representative Existing ASRs in Year 2019 – Year 2030 (Mitigated, Tier 1)

ASR ID	Maximum Hourly TSP Conc. (µg/m³)	19th Highest Daily RSP Conc. (µg/m³)	Annual RSP Conc. (µg/m³)	19th Highest Daily FSP Conc. (µg/m³)	Annual FSP Conc. (µg/m³)
A1001	314 - 319	**100** - **100**	42 - 42	71 - 71	30 - 30
A1002	**525** - **681**	**101** - **103**	42 - 42	71 - 71	30 - 30
A1003	381 - 393	97 - 97	42 - 42	71 - 71	30 - 30
A1004	403 - 422	100 - **100**	42 - 42	71 - 71	30 - 30
A1005	346 - 368	97 - 98	42 - 42	71 - 71	30 - 30
A102	180 - **726**	94 - 98	42 - 43	70 - 71	30 - 30
A103	180 - **634**	94 - 95	42 - 43	70 - 70	30 - 30
A104	**638** - **659**	96 - 96	43 - 43	70 - 70	30 - 30
A105	**600** - **655**	96 - 97	43 - 43	71 - 71	30 - 30
A106	**664** - **688**	98 - 98	43 - 43	71 - 71	30 - 30
A107	**806** - **841**	**104** - **105**	43 - 43	71 - 71	30 - 30
A108	**860** - **897**	**107** - **108**	43 - 43	72 - 72	30 - 30
A109	**809** - **826**	**109** - **109**	43 - 43	72 - 72	30 - 30
A110	497 - **593**	94 - 94	43 - 43	70 - 70	30 - 30
A1101	292 - 294	96 - 96	41 - 41	70 - 70	29 - 29

ASR ID	Maximum Hourly TSP Conc. (µg/m³)	19th Highest Daily RSP Conc. (µg/m³)	Annual RSP Conc. (µg/m³)	19th Highest Daily FSP Conc. (µg/m³)	Annual FSP Conc. (µg/m³)
A1102	280 - 286	95 - 96	41 - 41	70 - 70	29 - 29
A1103	172 - 246	92 - 93	41 - 41	69 - 69	29 - 29
A1104	172 - 222	92 - 93	41 - 41	69 - 69	29 - 29
A1105	172 - 228	92 - 93	41 - 41	69 - 69	29 - 29
A1106	172 - 256	92 - 94	41 - 41	69 - 70	29 - 29
A1107	172 - 226	92 - 94	41 - 41	69 - 69	29 - 29
A1108	218 - 241	93 - 94	41 - 41	69 - 70	29 - 29
A1109	172 - 195	92 - 93	41 - 41	69 - 69	29 - 29
A111	**710** - **834**	95 - 96	43 - 43	70 - 70	30 - 30
A112	**670** - **767**	97 - 98	43 - 43	70 - 70	30 - 30
A1201	173 - 173	92 - 92	41 - 41	69 - 69	29 - 29
A1202	173 - 173	92 - 92	41 - 41	69 - 69	29 - 29
A1203	173 - 173	92 - 92	41 - 41	69 - 69	29 - 29
A1300	**598** - **1144**	**103** - **117**	42 - 42	72 - **77**	29 - 30
A1301	284 - 288	94 - 94	41 - 41	69 - 69	29 - 29
A1302	433 - 446	94 - 94	41 - 41	69 - 69	29 - 29
A1303	280 - 288	94 - 94	41 - 41	69 - 69	29 - 29
A1304	268 - 276	94 - 94	41 - 41	69 - 69	29 - 29
A1305	309 - 328	94 - 94	41 - 41	69 - 69	29 - 29
A1306	363 - 369	94 - 94	41 - 41	69 - 69	29 - 29
A1307	392 - 415	93 - 93	41 - 41	69 - 69	29 - 29
A1308	384 - 392	93 - 93	41 - 41	69 - 69	29 - 29
A1309	319 - 376	93 - 93	41 - 41	69 - 69	29 - 29
A1401	170 - 170	92 - 92	41 - 41	69 - 69	29 - 29
A1402	170 - 170	91 - 92	40 - 41	68 - 69	29 - 29
A1403	170 - 170	91 - 92	40 - 41	68 - 69	29 - 29
A1404	170 - 170	91 - 92	40 - 41	68 - 69	29 - 29
A1405	170 - 170	91 - 92	40 - 41	68 - 69	29 - 29
A1501	**583** - **583**	**105** - **105**	43 - 43	73 - 73	30 - 30
A201	**816** - **955**	100 - **101**	42 - 42	72 - 72	30 - 30
A202	**988** - **1286**	**103** - **105**	42 - 43	72 - 72	30 - 30
A203	**997** - **1344**	**103** - **106**	42 - 43	72 - 73	30 - 30
A204	**954** - **1439**	**105** - **109**	43 - 43	73 - 73	30 - 30
A205	**863** - **1388**	**104** - **110**	43 - 43	73 - 73	30 - 30
A206	**1092** - **1338**	**122** - **124**	43 - 44	**77** - **77**	30 - 30
A207	**1067** - **1164**	**116** - **117**	43 - 43	**75** - **75**	30 - 30
A208	**1109** - **1611**	**122** - **133**	43 - 44	**76** - **78**	30 - 30
A209	**988** - **1739**	**112** - **138**	43 - 44	75 - **78**	30 - 30
A301	**631** - **783**	98 - 98	42 - 42	71 - 71	30 - 30
A302	**624** - **784**	98 - 98	42 - 43	71 - 71	30 - 30
A303	**661** - **830**	98 - 99	42 - 43	70 - 71	30 - 30
A304	**876** - **1282**	**105** - **114**	43 - 43	71 - 72	30 - 30
A305	**535** - **693**	96 - 98	42 - 42	70 - 70	30 - 30
A306	**771** - **1334**	**107** - **114**	43 - 43	72 - 73	30 - 30
A307	300 - **590**	96 - 98	42 - 42	70 - 71	30 - 30
A308	474 - **644**	97 - 98	42 - 43	71 - 71	30 - 30
A309	**565** - **724**	98 - 99	42 - 43	71 - 71	30 - 30
A310	**2126** **2126**	**152** **152**	45 45	**78** **78**	31 31
A311	411 - **1270**	98 - **108**	42 - 43	71 - 73	30 - 30
A312	**636** - **766**	100 - 100	43 - 43	71 - 72	30 - 30
A313	438 - **1040**	98 - **104**	42 - 43	71 - 73	30 - 30
A314	401 - **1566**	98 - **117**	42 - 43	71 - 73	30 - 30

ASR ID	Maximum Hourly TSP Conc. (μg/m³)	19th Highest Daily RSP Conc. (μg/m³)	Annual RSP Conc. (μg/m³)	19th Highest Daily FSP Conc. (μg/m³)	Annual FSP Conc. (μg/m³)
A401	351 - 459	94 - 95	41 - 41	69 - 69	29 - 29
A402	376 - 387	94 - 95	41 - 41	69 - 69	29 - 29
A403	174 - 343	93 - 93	41 - 41	69 - 69	29 - 29
A404	269 - 282	93 - 93	41 - 41	69 - 69	29 - 29
A405	173 - 253	93 - 94	41 - 41	69 - 69	29 - 29
A406	378 - 380	97 - 97	41 - 42	69 - 70	29 - 30
A407	343 - 368	96 - 96	41 - 42	70 - 70	29 - 29
A408	418 - 420	96 - 96	41 - 41	69 - 69	29 - 29
A409	173 - 272	92 - 94	41 - 42	69 - 70	29 - 30
A410	**570** - **571**	95 - 95	41 - 41	69 - 69	29 - 29
A411	275 - 275	93 - 94	41 - 41	69 - 69	29 - 29
A412	431 - 434	94 - 94	41 - 41	69 - 69	29 - 29
A413	241 - 246	93 - 93	41 - 42	69 - 69	29 - 30
A414	268 - 274	93 - 93	41 - 41	69 - 69	29 - 29
A415	**639** - **646**	**104** - **105**	42 - 42	71 - 71	29 - 29
A416	173 - **510**	93 - 97	41 - 42	69 - 70	29 - 29
A502	181 - **952**	93 - 100	41 - 42	69 - 71	29 - 30
A503	303 - **515**	94 - 98	42 - 42	70 - 71	30 - 30
A504	316 - 316	94 - 94	42 - 42	69 - 69	30 - 30
A505	283 - 287	93 - 93	42 - 42	69 - 69	29 - 29
A506	369 - 372	93 - 93	42 - 42	70 - 70	29 - 29
A507	269 - 290	92 - 93	41 - 42	69 - 69	29 - 30
A508	268 - 279	93 - 93	41 - 42	69 - 69	29 - 30
A601	375 - 385	**102** - **102**	42 - 42	71 - 71	30 - 30
A602	**1154** - **1154**	**120** - **120**	43 - 43	75 - 75	30 - 30
A603	**619** - **619**	**106** - **106**	42 - 42	72 - 72	30 - 30
A701	**591** - **615**	**102** - **102**	41 - 41	70 - 70	29 - 29
A702	**561** - **584**	**101** - **101**	41 - 41	70 - 70	29 - 29
A703	341 - 456	**103** - **104**	41 - 42	70 - 71	29 - 29
A704	378 - 382	**101** - **101**	41 - 41	70 - 70	29 - 29
A705	386 - 393	98 - 98	41 - 41	70 - 70	29 - 29
A706	387 - 398	97 - 97	41 - 41	70 - 70	29 - 29
A707	186 - 338	92 - 98	41 - 41	69 - 69	29 - 29
A708	473 - 488	98 - 99	41 - 41	70 - 70	29 - 29
A801	213 - 215	93 - 93	41 - 41	69 - 69	29 - 29
A802	255 - 283	92 - 92	41 - 41	69 - 69	29 - 29
A803	208 - 249	92 - 92	41 - 41	69 - 69	29 - 29
A804	203 - 219	92 - 92	41 - 41	69 - 69	29 - 29
A805	194 - 205	92 - 92	41 - 41	69 - 69	29 - 29
A806	176 - 176	92 - 92	41 - 41	69 - 69	29 - 29
A807	206 - 210	92 - 92	41 - 41	69 - 69	29 - 29
A808	176 - 327	92 - 93	41 - 41	69 - 69	29 - 29
A809	176 - 239	92 - 92	41 - 41	69 - 69	29 - 29
A810	176 - 226	92 - 92	41 - 41	69 - 69	29 - 29
A811	176 - 212	92 - 92	41 - 41	69 - 69	29 - 29
A812	176 - 247	92 - 92	41 - 41	69 - 69	29 - 29
A813	176 - 317	92 - 93	41 - 41	69 - 69	29 - 29
A901	208 - 214	97 - 97	42 - 42	72 - 72	30 - 30
A902	326 - 331	99 - 99	42 - 42	72 - 72	30 - 30
A903	**534** - **545**	99 - 99	42 - 42	72 - 72	30 - 30

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Tier 2 Assessment

3.10.3 Those ASRs with non-compliance in particulates levels under Tier 1 screening during the construction of the Project are selected to undergo Tier 2 assessment accordingly as discussed in **Section 3.6**. In Year 2019 – Year 2030 scenario, there are 69, 53 and 9 nos. of non-compliance ASRs required for TSP (hourly average), RSP (daily average) and FSP (daily average) Tier 2 assessment respectively. The active work areas of concurrent work contracts are positioned closest to the potentially worst affected ASRs. The individual affected ASR would be influenced by one active work area of each concurrent work contract. The predicted cumulative maximum hourly average TSP, 10th highest daily average RSP and FSP concentrations at the ASRs of Tier 2 scenarios are summarised in **Table 3.38** and **Table 3.39**. The detailed prediction results are presented in **Appendix 3.12**.

Table 3.38 Predicted Construction Dust Impact on Project Sites with Air Sensitive Use in Year 2019 – Year 2030 (Mitigated, Tier 2)

Site	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)
2-18	169 - 174	93 - 95	69 - 71
3-6	174 - 469	94 - **104**	71 - 72
3-7	174 - 197	94 - 97	71 - 72
3-8	174 - 374	94 - **102**	71 - 73
4-20	174 - **569**	95 - **102**	70 - 71
5-2	176 - 364	92 - 94	69 - 69
5-22	173 - 173	92 - 92	69 - 69
5-24	173 - 223	92 - 94	69 - 69
5-26	173 - 270	92 - 94	69 - 70

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.39 Predicted Construction Dust Impact on Representative Existing ASRs in Year 2019 – Year 2030 (Mitigated, Tier 2)

ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)
A1001	174 - 174	95 - 95	71 - 71
A1002	217 - 393	96 - 100	71 - 71
A1004	174 - 174	95 - 95	71 - 71
A102	180 - 180	93 - 94	70 - 70
A103	180 - 180	93 - 94	70 - 70
A104	180 - 180	94 - 94	70 - 70
A105	180 - 180	94 - 94	70 - 70
A106	180 - 180	94 - 94	70 - 70
A107	180 - 191	94 - 94	70 - 70
A108	180 - 180	95 - 95	70 - 70
A109	180 - 180	94 - 95	70 - 70
A110	180 - 180	93 - 93	70 - 70
A111	180 - 180	93 - 93	70 - 70
A112	181 - 204	93 - 94	70 - 70
A1300	185 - 259	94 - 96	70 - 71
A1501	248 - 248	98 - 98	73 - 73
A201	173 - 240	94 - 96	70 - 70
A202	173 - 313	94 - 97	70 - 70
A203	174 - 320	94 - 97	70 - 70
A204	174 - 363	95 - 98	70 - 70
A205	174 - 354	96 - 98	70 - 70
A206	178 - 279	97 - 98	71 - 71
A207	175 - 224	97 - 97	70 - 71
A208	187 - 440	99 - **102**	71 - 71
A209	174 - **814**	97 - **109**	70 - 72

ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)
A301	179 - 179	94 - 94	70 - 70
A302	179 - 179	93 - 94	70 - 70
A303	179 - 179	93 - 94	70 - 70
A304	179 - 329	94 - 96	70 - 70
A305	180 - 180	93 - 93	70 - 70
A306	179 - 310	94 - 96	70 - 70
A307	179 - 179	93 - 94	70 - 70
A308	179 - 179	93 - 93	70 - 70
A309	179 - 179	93 - 94	70 - 70
A310	**683** - **683**	**100** - **100**	71 - 71
A311	179 - 332	93 - 97	70 - 70
A312	179 - 179	94 - 94	70 - 70
A313	179 - 226	93 - 95	70 - 70
A314	179 - 359	93 - 97	70 - 70
A410	192 - 220	92 - 92	69 - 69
A415	193 - 341	94 - 95	69 - 69
A416	173 - 200	92 - 92	69 - 69
A502	181 - 197	92 - 93	69 - 70
A503	181 - 181	93 - 93	70 - 70
A601	175 - 175	94 - 94	70 - 70
A602	175 - 175	96 - 96	71 - 71
A603	176 - 176	95 - 95	71 - 71
A701	172 - 172	93 - 93	69 - 69
A702	172 - 172	93 - 93	69 - 69
A703	172 - 188	93 - 94	69 - 69
A704	172 - 172	93 - 93	69 - 69
A903	174 - 174	96 - 96	72 - 72

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

3.10.4 The contour plots of cumulative maximum hourly average TSP, 10th highest daily average RSP and FSP concentrations at 1.5 mAG of the above Tier 2 scenarios are presented in **Figure 3.32** – **Figure 3.34**.

3.10.5 According to the above Tier 2 assessments, ASR P240 of Site 4-20, A209 and A310 would still exceed the criterion of hourly average TSP levels stipulated in EIAO-TM. ASR P1032 of Site 3-6, P1501 of Site 3-8, P240 of Site 4-20, A208, A209 and A310 would still exceed the criterion of daily average RSP levels stipulated in the AQOs. Tier 3 assessments for these ASRs are then conducted with reduction in active works area nearby to one-third of monthly average work. As confirmed by the Engineer, one-third of monthly average work area during any short period of time is considered practical applying for these ASRs. The predicted cumulative maximum hourly average TSP and 10th highest daily average RSP concentrations at the ASRs of Tier 3 scenario are summarised in **Table 3.40**. The results show that construction dust impact on these ASRs would all comply with the criteria with implementation of reduced active works area nearby. The contour plots of cumulative maximum hourly average TSP and 10th highest daily average RSP at 1.5 mAG are illustrated in **Figure 3.38** and **Figure 3.39**.

Table 3.40 Predicted Construction Dust Impact on Affected ASRs in Year 2019 – Year 2030 (Mitigated, Tier 3)

ASR ID	Assessment Height (mAG)	Maximum Hourly TSP Conc. (µg/m³)	10th Highest Daily RSP Conc. (µg/m³)
P240	1.5	311	99
	5.0	177	96
	10.0	174	94
A209	1.5	421	100
	5.0	331	99
	10.0	174	95
A310	1.5	478	98
ASR ID	**Assessment Height (mAG)**	**10th Highest Daily RSP Conc. (µg/m³)**	
P1032	1.5	99	
	5.0	97	
	10.0	95	
	20.0	94	
	40.0	94	
	80.0	94	
P1501	1.5	98	
	5.0	97	
	10.0	97	
	20.0	97	
	40.0	97	
	80.0	96	
A208	1.5	98	
	5.0	98	
	10.0	97	

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Year 2031 – Year 2036 Scenario

Tier 1 Assessment

3.10.6 With the implementation of the above measures, the predicted cumulative maximum hourly average TSP, 10th highest daily and annual average RSP and FSP concentrations at the representative ASRs are summarised in **Table 3.41** and **Table 3.42**. The detailed assessment results are presented in **Appendix 3.11**.

Table 3.41 Predicted Construction Dust Impact on Project Sites with Air Sensitive Use in Year 2031 – Year 2036 (Mitigated, Tier 1)

Site / ASR ID	Maximum Hourly TSP Conc. (µg/m³)	10th Highest Daily RSP Conc. (µg/m³)	Annual RSP Conc. (µg/m³)	10th Highest Daily FSP Conc. (µg/m³)	Annual FSP Conc. (µg/m³)
2-18	**504** - **722**	**102** - **109**	41 - 42	71 - 72	29 - 30
2-19	425 - **506**	**100** - **101**	42 - 42	72 - 72	30 - 30
3-1	175 - **599**	95 - **103**	42 - 42	71 - 72	30 - 30
3-11	176 - **520**	95 - **106**	41 - 43	71 - 73	29 - 30
3-13	175 - **1199**	95 - **120**	41 - 43	71 - **76**	29 - 30
3-14	176 - **1127**	96 - **122**	41 - 43	71 - **75**	29 - 30
3-18	176 - **1372**	95 - **125**	41 - 43	71 - **75**	29 - 30
3-4	175 - **586**	95 - **102**	42 - 42	71 - 72	30 - 30
3-43	176 - **538**	95 - **102**	42 - 43	71 - 72	30 - 30
3-44	176 - 495	95 - 99	42 - 43	71 - 72	30 - 30
3-45	176 - 477	95 - 99	42 - 43	71 - 72	30 - 30

Site / ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	Annual RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)	Annual FSP Conc. (μg/m³)
3-5	175 - **537**	95 - 100	42 - 42	71 - 72	30 - 30
3-50	173 - 340	93 - 97	42 - 43	70 - 71	30 - 30
3-51	173 - 285	93 - 96	42 - 42	70 - 70	30 - 30
3-52	173 - 256	93 - 96	42 - 43	70 - 71	30 - 31
3-6	175 - **506**	95 - **100**	42 - 42	71 - 71	30 - 30
3-7	174 - 367	95 - 97	42 - 42	71 - 72	30 - 30
3-8	174 - 480	95 - **102**	42 - 43	71 - 73	30 - 30
4-1	176 - **650**	95 - **101**	42 - 43	70 - 73	30 - 30
4-10	176 - 314	94 - 98	42 - 42	70 - 71	30 - 30
4-12a	173 - 269	93 - 96	42 - 42	70 - 70	30 - 30
4-12b	173 - 280	93 - 96	42 - 42	70 - 70	30 - 30
4-12c	176 - 285	94 - 96	42 - 42	70 - 71	30 - 30
4-12d	173 - 250	93 - 95	42 - 42	70 - 70	30 - 30
4-13a	173 - 304	93 - 96	42 - 42	70 - 70	30 - 30
4-13b	173 - 339	93 - 96	42 - 42	70 - 71	30 - 30
4-14	227 - 382	95 - 97	42 - 42	70 - 71	30 - 30
4-15	173 - 259	93 - 96	42 - 42	70 - 70	30 - 30
4-16	173 - 237	93 - 95	42 - 42	70 - 70	30 - 30
4-17	173 - 240	93 - 95	42 - 42	70 - 70	30 - 30
4-2	176 - **516**	96 - 98	42 - 42	71 - 72	30 - 30
4-20	173 - 194	93 - 94	42 - 42	70 - 70	30 - 30
4-21	173 - 204	93 - 94	42 - 42	70 - 70	30 - 30
4-22	173 - 279	93 - 95	42 - 42	70 - 70	30 - 30
4-24	179 - 187	93 - 94	42 - 42	70 - 70	30 - 30
4-25a	173 - 386	93 - 95	42 - 42	70 - 70	30 - 30
4-25b	173 - 214	92 - 94	41 - 42	69 - 70	29 - 30
4-25c	173 - 424	93 - 96	42 - 42	70 - 71	30 - 30
4-26	179 - 179	93 - 94	42 - 42	70 - 70	30 - 30
4-28	173 - 345	93 - 95	42 - 42	70 - 70	30 - 30
4-29	173 - 400	93 - 95	42 - 42	70 - 70	30 - 30
4-3	176 - **591**	95 - **105**	42 - 42	70 - 73	30 - 30
4-31	173 - 367	93 - 95	42 - 42	70 - 70	30 - 30
4-32	173 - 352	93 - 95	42 - 42	70 - 70	30 - 30
4-33	208 - 319	93 - 95	42 - 42	70 - 70	30 - 30
4-36	188 - 284	93 - 95	42 - 42	70 - 70	30 - 30
4-4	176 - 422	94 - 98	42 - 42	70 - 72	30 - 30
4-5	176 - 339	94 - 97	42 - 42	70 - 71	30 - 30
4-6	176 - **769**	94 - **102**	42 - 42	70 - 72	30 - 30
4-8	176 - 493	95 - 99	42 - 42	71 - 72	30 - 30
4-9	176 - 334	94 - 97	42 - 42	70 - 71	30 - 30
5-1	176 - 453	92 - 97	41 - 41	69 - 70	29 - 29
5-16	172 - **1222**	92 - **110**	41 - 42	69 - 72	29 - 29
5-17	172 - **1137**	92 - **112**	41 - 42	69 - 72	29 - 29
5-18a	172 - **1215**	92 - **116**	41 - 42	69 - 73	29 - 29
5-18b	172 - **961**	92 - **115**	41 - 42	69 - 72	29 - 29
5-2	176 - 446	92 - 96	41 - 41	69 - 70	29 - 29
5-21	172 - **1609**	93 - **142**	41 - 44	69 - **76**	29 - 29
5-22	174 - **531**	93 - 96	41 - 41	69 - 69	29 - 29
5-23	174 - **514**	92 - 93	41 - 41	69 - 69	29 - 29
5-24	173 - **510**	92 - 94	41 - 41	69 - 69	29 - 29
5-26	174 - 461	92 - 92	41 - 41	69 - 69	29 - 29
5-27	199 - 428	92 - 93	41 - 41	69 - 69	29 - 29

Site / ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	Annual RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)	Annual FSP Conc. (μg/m³)
5-28	201 - 401	92 - 93	41 - 41	69 - 69	29 - 29
5-32	173 - 316	92 - 93	41 - 41	69 - 69	29 - 29
5-33	173 - 420	92 - 93	41 - 41	69 - 69	29 - 29
5-37	174 - 416	92 - 92	41 - 41	69 - 69	29 - 29
5-38	276 - 325	92 - 92	41 - 41	69 - 69	29 - 29
5-3a	172 - **647**	92 - **115**	41 - 42	69 - 72	29 - 29
5-3b	176 - **698**	92 - **108**	41 - 42	69 - 71	29 - 29
5-6	303 - **540**	94 - **101**	41 - 41	69 - 70	29 - 29

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.42 Predicted Construction Dust Impact on Representative Existing ASRs in Year 2031 – Year 2036 (Mitigated, Tier 1)

ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	Annual RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)	Annual FSP Conc. (μg/m³)
A1001	398 - 432	98 - 98	42 - 42	71 - 71	30 - 30
A1002	**556** - **649**	**103** - **103**	42 - 42	72 - 72	30 - 30
A1003	338 - 472	99 - **101**	42 - 42	71 - 72	30 - 30
A1004	332 - 363	98 - 99	42 - 42	71 - 71	30 - 30
A1005	287 - 391	99 - 100	42 - 42	71 - 71	30 - 30
A102	180 - 389	93 - 94	42 - 43	70 - 70	30 - 30
A103	180 - 369	93 - 94	42 - 42	70 - 70	30 - 30
A104	322 - 356	94 - 94	42 - 42	70 - 70	30 - 30
A105	293 - 327	94 - 94	42 - 42	70 - 70	30 - 30
A106	308 - 333	94 - 94	42 - 42	70 - 70	30 - 30
A107	308 - 346	94 - 94	42 - 42	70 - 70	30 - 30
A108	314 - 333	94 - 94	42 - 42	70 - 70	30 - 30
A109	320 - 327	94 - 95	42 - 42	70 - 70	30 - 30
A110	292 - 311	93 - 93	42 - 42	70 - 70	30 - 30
A1101	**786** - **1005**	**120** - **130**	42 - 43	74 - **75**	29 - 29
A1102	451 - **1336**	**111** - **126**	42 - 42	72 - 74	29 - 29
A1103	172 - **911**	92 - **110**	41 - 41	69 - 71	29 - 29
A1104	172 - **994**	92 - **111**	41 - 41	69 - 70	29 - 29
A1105	172 - **704**	92 - 99	41 - 41	69 - 70	29 - 29
A1106	172 - **1038**	92 - **110**	41 - 41	69 - 72	29 - 29
A1107	172 - **715**	92 - 100	41 - 41	69 - 70	29 - 29
A1108	172 - **1055**	93 - **111**	41 - 41	69 - 72	29 - 29
A1109	172 - **661**	92 - 99	41 - 41	69 - 70	29 - 29
A111	314 - 345	93 - 93	42 - 42	70 - 70	30 - 30
A112	311 - 330	93 - 93	42 - 42	70 - 70	30 - 30
A1201	173 - 313	92 - 93	41 - 41	69 - 69	29 - 29
A1202	173 - 288	92 - 92	41 - 41	69 - 69	29 - 29
A1203	173 - 305	92 - 92	41 - 41	69 - 69	29 - 29
A1300	484 - **1123**	**106** - **126**	42 - 43	73 - **78**	29 - 30
A1301	423 - 486	96 - 96	41 - 41	69 - 70	29 - 29
A1302	404 - **619**	97 - 99	41 - 41	70 - 70	29 - 29
A1303	**584** - **606**	99 - 100	41 - 41	70 - 70	29 - 29
A1304	**579** - **623**	99 - 100	41 - 41	70 - 70	29 - 29
A1305	**504** - **575**	98 - 98	41 - 41	70 - 70	29 - 29
A1306	446 - 457	96 - 97	41 - 41	70 - 70	29 - 29
A1307	446 - **687**	96 - 97	41 - 41	70 - 70	29 - 29

ASR ID	Maximum Hourly TSP Conc. (µg/m³)	10th Highest Daily RSP Conc. (µg/m³)	Annual RSP Conc. (µg/m³)	10th Highest Daily FSP Conc. (µg/m³)	Annual FSP Conc. (µg/m³)
A1308	308 - 320	94 - 96	41 - 41	69 - 69	29 - 29
A1309	248 - 272	93 - 93	41 - 41	69 - 69	29 - 29
A1401	433 - 446	93 - 93	41 - 41	69 - 69	29 - 29
A1402	170 - 340	92 - 94	40 - 41	68 - 69	29 - 29
A1403	170 - 319	91 - 93	40 - 41	68 - 69	29 - 29
A1404	170 - 225	91 - 93	40 - 41	68 - 69	29 - 29
A1405	170 - 261	91 - 93	40 - 41	68 - 69	29 - 29
A1501	294 - 294	100 - 100	43 - 43	73 - 73	30 - 30
A201	245 - 274	95 - 95	42 - 42	70 - 70	30 - 30
A202	219 - 238	95 - 95	42 - 42	70 - 70	30 - 30
A203	211 - 228	95 - 95	42 - 42	70 - 70	30 - 30
A204	235 - 259	95 - 95	42 - 42	70 - 70	30 - 30
A205	263 - 295	95 - 95	42 - 42	70 - 70	30 - 30
A206	173 - 183	93 - 93	42 - 42	70 - 70	30 - 30
A207	177 - 178	93 - 93	42 - 42	70 - 70	30 - 30
A208	189 - 193	94 - 94	42 - 42	70 - 70	30 - 30
A209	285 - 309	95 - 95	42 - 42	70 - 70	30 - 30
A301	179 - 179	93 - 93	42 - 42	70 - 70	30 - 30
A302	179 - 179	94 - 94	42 - 42	70 - 70	30 - 30
A303	179 - 179	94 - 94	42 - 42	70 - 70	30 - 30
A304	179 - 179	94 - 94	42 - 42	70 - 70	30 - 30
A305	179 - 179	94 - 94	42 - 42	70 - 70	30 - 30
A306	179 - 179	94 - 94	42 - 42	70 - 70	30 - 30
A307	179 - 179	94 - 94	42 - 42	70 - 70	30 - 30
A308	179 - 179	94 - 94	42 - 42	70 - 70	30 - 30
A309	179 - 179	94 - 94	42 - 42	70 - 70	30 - 30
A310	179 - 179	94 - 94	42 - 42	70 - 70	30 - 30
A311	179 - 179	94 - 94	42 - 42	70 - 70	30 - 30
A312	212 - 252	94 - 94	42 - 42	70 - 70	30 - 30
A313	179 - 187	94 - 94	42 - 42	70 - 70	30 - 30
A314	179 - 179	94 - 94	42 - 42	70 - 70	30 - 30
A401	265 - 281	92 - 92	41 - 41	69 - 69	29 - 29
A402	314 - 353	92 - 92	41 - 41	69 - 69	29 - 29
A403	173 - 362	92 - 92	41 - 41	69 - 69	29 - 29
A404	279 - 331	92 - 92	41 - 41	69 - 69	29 - 29
A405	173 - 285	92 - 92	41 - 41	69 - 69	29 - 29
A406	288 - 308	92 - 93	41 - 41	69 - 69	29 - 29
A407	**612** - **624**	93 - 94	41 - 41	69 - 69	29 - 29
A408	382 - 421	94 - 94	41 - 41	69 - 69	29 - 29
A409	174 - 366	93 - 94	41 - 41	69 - 70	29 - 29
A410	432 - 467	100 - **101**	42 - 42	70 - 70	29 - 29
A411	454 - 478	96 - 96	41 - 41	69 - 69	29 - 29
A412	443 - 482	99 - **100**	42 - 42	70 - 70	29 - 29
A413	358 - 376	96 - 97	41 - 41	69 - 70	29 - 29
A414	**524** - **544**	97 - 98	41 - 41	69 - 69	29 - 29
A415	264 - 299	93 - 93	41 - 41	69 - 69	29 - 29
A416	174 - 333	92 - 93	41 - 41	69 - 69	29 - 29
A502	181 - **612**	93 - 98	41 - 42	69 - 70	29 - 29
A503	272 - **579**	94 - 99	42 - 42	70 - 70	29 - 29
A504	**634** - **694**	96 - 97	42 - 42	70 - 70	29 - 29
A505	446 - **553**	95 - 96	42 - 42	69 - 70	29 - 29
A506	**652** - **909**	99 - 99	42 - 42	70 - 71	29 - 29

ASR ID	Maximum Hourly TSP Conc. (µg/m³)	10th Highest Daily RSP Conc. (µg/m³)	Annual RSP Conc. (µg/m³)	10th Highest Daily FSP Conc. (µg/m³)	Annual FSP Conc. (µg/m³)
A507	262 - **644**	93 - 97	41 - 42	69 - 70	29 - 29
A508	461 - **592**	96 - 97	41 - 42	69 - 70	29 - 29
A601	**741** - **1143**	**118** - **123**	42 - 43	74 - 75	30 - 30
A602	**694** - **694**	**110** - **110**	42 - 42	74 - 74	30 - 30
A603	**606** - **606**	**107** - **107**	42 - 42	73 - 73	30 - 30
A701	**737** - **1368**	**120** - **143**	42 - 43	74 - **76**	29 - 29
A702	**672** - **1013**	**121** - **133**	42 - 43	73 - **76**	29 - 29
A703	**598** - **1285**	**112** - **122**	42 - 42	72 - 73	29 - 29
A704	**547** - **1419**	**121** - **131**	42 - 43	74 - **75**	29 - 29
A705	**524** - **1422**	**120** - **146**	43 - 44	73 - **76**	29 - 29
A706	**594** - **793**	**117** - **125**	42 - 42	72 - 73	29 - 29
A707	172 - **1149**	93 - **109**	41 - 41	69 - 71	29 - 29
A708	**767** - **1392**	**125** - **134**	42 - 43	73 - 75	29 - 29
A801	**528** - **542**	96 - 96	41 - 41	70 - 70	29 - 29
A802	451 - 471	96 - 96	41 - 41	70 - 70	29 - 29
A803	330 - 408	93 - 94	41 - 41	69 - 70	29 - 29
A804	322 - 323	93 - 93	41 - 41	69 - 69	29 - 29
A805	311 - 332	93 - 93	41 - 41	69 - 69	29 - 29
A806	233 - 276	93 - 93	41 - 41	69 - 69	29 - 29
A807	243 - 250	93 - 93	41 - 41	69 - 69	29 - 29
A808	176 - 419	93 - 95	41 - 41	69 - 70	29 - 29
A809	176 - 265	92 - 93	41 - 41	69 - 69	29 - 29
A810	176 - 260	92 - 93	41 - 41	69 - 69	29 - 29
A811	176 - 271	92 - 93	41 - 41	69 - 69	29 - 29
A812	176 - 312	92 - 94	41 - 41	69 - 69	29 - 29
A813	176 - **523**	92 - 94	41 - 41	69 - 70	29 - 29
A901	187 - 203	96 - 96	42 - 42	72 - 72	30 - 30
A902	213 - 217	96 - 96	42 - 42	72 - 72	30 - 30
A903	277 - 287	96 - 96	42 - 42	72 - 72	30 - 30

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

3.10.7 The contour plots of cumulative maximum hourly average TSP, 10th highest daily and annual average RSP and FSP concentrations at 1.5 mAG (the worst hit level) are presented in **Figures 3.27 - 3.31**.

3.10.8 Based on the prediction of Tier 1 screening, the predicted hourly average TSP levels and daily average RSP and FSP levels would still exceed the criterion stipulated in EIAO-TM and AQOs while the predicted annual average RSP and FSP levels at all the Project sites and representative ASRs would comply with the criteria of AQOs.

3.10.9 Those ASRs with non-compliance in particulates levels under Tier 1 screening during the construction of the Project are selected to undergo Tier 2 assessment. As discussed in **Section 3.6**, Tier 2 assessment is undertaken whereby monthly average works areas of the concurrent work contracts according to the construction work programme are considered as active work areas in the assessment. The active work areas of concurrent work contracts are positioned closest to the potentially worst affected ASRs. The individual affected ASR would be influenced by one active work area of each concurrent work contract. In Year 2031 – 2036 scenario, 97, 84 and 12 nos. non-compliance ASRs are required for TSP (hourly average), RSP (daily average) and FSP (daily average) Tier 2 assessment respectively. As the non-compliance ASRs are scattered around the Project Site, Tier 2 has divided into 5 scenarios for modelling based on the locations of the affected ASRs:

- Tier 2 East – for the clusters of affected ASRs at East of the Project area
- Tier 2 South – for the clusters of affected ASRs at South of the Project area
- Tier 2 West – for the clusters of affected ASRs at West of the Project area
- Tier 2 North – for the clusters of affected ASRs at North of the Project area
- Tier 2 Central – for the clusters of affected ASRs in the Central of the Project area

3.10.10 The predicted cumulative maximum hourly average TSP, 10th highest daily average RSP and FSP concentrations at the ASRs of Tier 2 scenarios are summarised in **Table 3.43** to **Table 3.51**. The detailed prediction results are presented in **Appendix 3.12**.

Table 3.43 Predicted Construction Dust Impact on Representative Existing ASRs in Year 2031 – Year 2036 (Mitigated, Tier 2 Central)

ASR ID	Maximum Hourly TSP Conc. (µg/m³)	10th Highest Daily RSP Conc. (µg/m³)	10th Highest Daily FSP Conc. (µg/m³)
A1001	174 - 174	95 - 95	71 - 71
A1002	174 - 174	95 - 95	71 - 71
A601	175 - 204	95 - 96	71 - 71
A701	174 - 319	94 - 98	69 - 70
A702	172 - 430	93 - **101**	69 - 70
A703	173 - 312	94 - 95	69 - 69
A704	173 - 185	94 - 95	69 - 69
A705	173 - 224	94 - 98	69 - 70
A706	172 - 173	94 - 95	69 - 69
A708	172 - 303	94 - 95	69 - 69

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.44 Predicted Construction Dust Impact on Project Sites with Air Sensitive Use in Year 2031 – Year 2036 (Mitigated, Tier 2 East)

Site / ASR ID	Maximum Hourly TSP Conc. (µg/m³)	10th Highest Daily RSP Conc. (µg/m³)	10th Highest Daily FSP Conc. (µg/m³)
5-16	172 - 328	92 - 95	69 - 70
5-17	172 - 300	92 - 95	69 - 69
5-18a	172 - 336	92 - 95	69 - 69
5-18b	172 - 243	92 - 94	69 - 69
5-21	172 - 172	92 - 93	69 - 69
5-3a	172 - 267	92 - 94	69 - 69
5-3b	176 - 229	92 - 94	69 - 69
5-6	176 - 195	92 - 93	69 - 69

Note: Bolded and underlined figure represents exceedance of relevant AQO. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.45 Predicted Construction Dust Impact on Representative Existing ASRs in Year 2031 – Year 2036 (Mitigated, Tier 2 East)

ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)
A1103	172 - 172	92 - 93	69 - 69
A1104	172 - 203	92 - 93	69 - 69
A1105	172 - 172	92 - 93	69 - 69
A1106	172 - 232	92 - 93	69 - 69
A1107	172 - 172	92 - 92	69 - 69
A1108	172 - 225	92 - 93	69 - 69
A1109	172 - 172	92 - 92	69 - 69
A414	173 - 173	92 - 92	69 - 69
A707	172 - 172	92 - 93	69 - 69
A801	176 - 196	93 - 93	69 - 69
A813	176 - 176	92 - 92	69 - 69

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.46 Predicted Construction Dust Impact on Project Sites with Air Sensitive Use in Year 2031 – Year 2036 (Mitigated, Tier 2 North)

Site / ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)
2-18	169 - 295	93 - 95	69 - 71
2-19	174 - 182	95 - 95	71 - 71
3-1	174 - 191	94 - 95	71 - 71

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.47 Predicted Construction Dust Impact on Representative Existing ASRs in Year 2031 – Year 2036 (Mitigated, Tier 2 North)

ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)
A1002	174 - 183	95 - 95	71 - 71
A1003	174 - 175	95 - 95	71 - 71
A1300	169 - 216	95 - 98	70 - 71
A1302	169 - 169	93 - 93	69 - 69
A1303	169 - 169	92 - 93	69 - 69
A1304	169 - 193	92 - 93	69 - 69
A1305	169 - 171	92 - 93	69 - 69
A1307	169 - 169	92 - 93	69 - 69

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.48 Predicted Construction Dust Impact on Project Sites with Air Sensitive Use in Year 2031 – Year 2036 (Mitigated, Tier 2 West)

Site / ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)
3-11	175 - 180	94 - 97	70 - 73
3-13	174 - 245	94 - 96	70 - 71
3-14	175 - **592**	94 - **100**	70 - 71
3-18	175 - 381	94 - 96	70 - 71
3-4	174 - 181	94 - 95	71 - 71
3-5	174 - 174	94 - 95	71 - 71
3-6	174 - 174	94 - 95	71 - 71
3-8	174 - 180	94 - 97	71 - 73

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.49 Predicted Construction Dust Impact on Representative Existing ASRs in Year 2031 – Year 2036 (Mitigated, Tier 2 West)

ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)
A602	175 - 175	95 - 95	71 - 71
A603	189 - 189	96 - 96	71 - 71

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.50 Predicted Construction Dust Impact on Project Sites with Air Sensitive Use in Year 2031 – Year 2036 (Mitigated, Tier 2 South)

Site / ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)
3-43	175 - 204	94 - 95	70 - 71
4-1	175 - 178	94 - 96	70 - 71
4-2	176 - 176	94 - 95	70 - 71
4-3	175 - 258	94 - 96	70 - 71
4-6	175 - 197	94 - 96	70 - 71
5-22	173 - 259	92 - 93	69 - 69
5-23	173 - 173	92 - 92	69 - 69
5-24	173 - 252	92 - 93	69 - 69

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Table 3.51 Predicted Construction Dust Impact on Representative Existing ASRs in Year 2031 – Year 2036 (Mitigated, Tier 2 South)

ASR ID	Maximum Hourly TSP Conc. (μg/m³)	10th Highest Daily RSP Conc. (μg/m³)	10th Highest Daily FSP Conc. (μg/m³)
A407	174 - 178	92 - 93	69 - 69
A410	263 - 281	94 - 94	69 - 69
A412	173 - 173	92 - 92	69 - 69
A502	181 - 181	92 - 93	69 - 69
A503	181 - 181	93 - 93	69 - 69
A504	181 - 181	93 - 93	69 - 69
A505	181 - 181	92 - 92	69 - 69
A506	181 - 181	92 - 93	69 - 69
A507	181 - 181	92 - 92	69 - 69
A508	181 - 181	92 - 92	69 - 69

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

3.10.11 The contour plots of cumulative maximum hourly average TSP, 10th highest daily average RSP and FSP concentrations at 1.5 mAG (worst hit level) of the above Tier 2 scenarios are presented in **Figures 3.40 – 3.42** for Tier 2 Central, **Figures 3.46 – 3.48** for Tier 2 East, **Figures 3.49 – 3.51** for Tier 2 South, **Figures 3.52 – 3.54** for Tier 2 West, and **Figures 3.55 –3.57** for Tier 2 North.

3.10.12 According to the above Tier 2 assessment results, ASR P606 of Site 3-14 would still exceed the criteria of hourly average TSP and daily average RSP stipulated in the EIAO-TM and AQOs, while ASR A702 would exceed the criterion of daily average RSP stipulated in the AQOs. Tier 3 assessments for these two ASRs are then conducted with reduction in active works area nearby to one-third of monthly average work. As confirmed by the Engineer, one-third of monthly average work area during any short period of time is considered practical applying for these ASRs. The predicted cumulative maximum hourly average TSP and 10th highest daily average RSP concentrations at these ASRs of Tier 3 scenario are summarised in **Table 3.52** and the contour plots are

illustrated in **Figure 3.58 - 3.60** respectively. The results show that construction dust impact on P606 and A702 would comply with the criteria with implementation of reduced active works area nearby.

Table 3.52 Predicted Construction Dust Impact on Affected ASRs in Year 2031 – Year 2036 (Mitigated, Tier 3 Central & West)

ASR ID	Assessment Height (mAG)	Maximum Hourly TSP Conc. (μg/m^3)	10th Highest Daily RSP Conc. (μg/m^3)
P606	1.5	421	98
	5.0	267	96
	10.0	175	96
	20.0	175	94
	40.0	175	94
	80.0	175	94
ASR ID	**Assessment Height (mAG)**	**10th Highest Daily RSP Conc. (μg/m^3)**	
A702	1.5	98	
	5	97	
	10	93	

Note: Bolded and underlined figure represents exceedance of relevant limits stipulated in EIAO-TM and AQOs. The range of the results means the predicted concentrations at the ASRs within the site at all assessment heights.

Operation Phase Air Quality

Operational Air Quality

3.10.13 With the proper planning design, no unacceptable operational air quality impact would be anticipated. As mentioned in **Section 4.12.3**, the future planned development layout may be designed to avoid exposing excessive traffic noise levels so as to minimise the extent of the proposed noise barriers. To determine the implications of the proposed noise barrier on air quality impact, a sensitivity test is therefore conducted for the planned ASRs which are located behind the proposed noise barriers. The model setup and detailed results of the sensitivity test are presented in **Appendix 3.14**. Referring to the results as presented in **Appendix 3.14**, the differences between the results of with and without noise barrier scenarios are negligible.

Predicted Odour Impact by Existing Chicken Farm

3.10.14 With the proper planning design, i.e. avoid the location of air sensitive uses or fresh air intake in the exceedance zone, no unacceptable odour impact from existing chicken farm is anticipated.

Predicted Cumulative Odour Impact from Expanded and Upgraded San Wai STW, New HSK STW and Planned RTS

3.10.15 With the implementation of proposed mitigation measures, the odourous sources are all enclosed and installation of the deodorising units for treatment of the vented air from these enclosed odorous sources, no unacceptable odour impact from these two STWs and the RTS would be anticipated.

Odour Impact from Sewage Pumping Stations

3.10.16 The major odour sources (wet well, inlet chamber and screen chambers) of SPSs would be located within enclosed building structures. With proper enclosure and ventilation system to divert the odour emissions to deodouriser with at least 90% odour removal efficiency for treatment before discharge to environment, the ventilation system with a slight negative pressure within the facilities and locating the exhaust outlet of the deodouriser away from the nearby ASRs as far as practicable, it is expected that odour impact could be mitigated to an acceptable level.

3.11 Evaluation of Residual Environmental Impacts

Construction Phase

3.11.1 With the implementation measures specified in the Air Pollution Control (Construction Dust) Regulation together with recommended water frequency (dust removal efficiency of 91.7%) on active work site, exposed area and unpaved road, i.e. once per hour, and limitation of size of the active work area of the work contract closest to the affected ASR to one third of monthly average work area of the work contract in Tier 3 assessment, no adverse residual impact would be expected from the construction of the Project. When there are open excavation and spoil handling works, hoarding of 3 m high should be provided along the construction site boundary adjacent to the non-construction areas such as residential, educational institutes or recreation area in use so as to minimise the dust impact.

3.11.2 The following standard dust suppression measures are also proposed to be implemented to further minimise the dust impact. They should be incorporated by the Contractor as a General Specification to control the dust nuisance throughout the construction phase:

- Any excavated or stockpile of dusty material should be covered entirely by impervious sheeting or sprayed with water to maintain the entire surface wet and then removed or backfilled or reinstated where practicable within 24 hours of the excavation or unloading;
- Any dusty material remaining after a stockpile is removed should be wetted with water and cleared from the surface of roads;
- A stockpile of dusty material should not extend beyond the pedestrian barriers, fencing or traffic cones;
- The load of dusty materials on a vehicle leaving a construction site should be covered entirely by impervious sheeting to ensure that the dusty materials do not leak from the vehicle;
- Where practicable, vehicles washing facilities including a high pressure water jet should be provided at every discernible or designated vehicle exit point. The area where vehicle washing takes place and the road section between the washing facilities and the exit point should be paved with concrete, bituminous materials or hardcores;
- Provision of not less than 2.4 m high hoarding from ground level along site boundary where adjoins roads, streets or other accessible to the public except for a site entrance or exit. Good site practice shall also be adopted by the Contractor to ensure the conditions of the hoardings are properly maintained throughout the construction period;
- The portion of any road leading only to construction site that is within 30 m of a vehicle entrance or exit should be kept clear of dusty materials;

- Surfaces where any pneumatic or power-driven drilling, cutting, polishing or other mechanical breaking operation takes place should be sprayed with water or a dust suppression chemical continuously;
- Any area that involves demolition activities should be sprayed with water or a dust suppression chemical immediately prior to, during and immediately after the activities so as to maintain the entire surface wet;
- Where a scaffolding is erected around the perimeter of a building under construction, effective dust screens, sheeting or netting should be provided to enclose the scaffolding from the ground floor level of the building, or a canopy should be provided from the first floor level up to the highest level of the scaffolding;
- Any skip hoist for material transport should be totally enclosed by impervious sheeting;
- Every stock of more than 20 bags of cement or dry pulverised fuel ash (PFA) should be covered entirely by impervious sheeting or placed in an area sheltered on the top and the three sides;
- Cement or dry PFA delivered in bulk should be stored in a closed silo fitted with an audible high level alarm which is interlocked with the material filling line and no overfilling is allowed; and
- Exposed earth should be properly treated by compaction, turfing, hydroseeding, vegetation planting or sealing with latex, vinyl, bitumen, shortcrete or other suitable surface stabiliser within six months after the last construction activity on the construction site or part of the construction site where the exposed earth lies.

Operation Phase

3.11.3 During the operation phase of the Project, the predicted NO_2, RSP and FSP concentrations at all representative ASRs would comply with the AQOs. With the implementation of proposed mitigation measures, the predicted odour concentrations at the ASRs in the vicinity of the existing chicken farm, San Wai STW, new HSK STW and planned RTS would comply with the criterion stipulated in EIAO-TM. No adverse residual impact is expected.

3.12 Environmental Monitoring and Audit

Construction Phase

3.12.1 Environmental monitoring and audit for potential dust impacts should be conducted during the construction phase of the Project so as to check compliance with legislative requirements. Details of the monitoring and audit programme are contained in a stand-alone EM&A Manual.

Operation Phase

3.12.2 No adverse impact would be generated during the operation phase of this Project. Therefore the EM&A works related to air quality for the operational phase is considered not necessary. However, performance/compliance tests for deodourising units of the four new SPSs are recommended. The tests should be included as one of the requirements in the construction contract before the proposed SPSs in operation.

3.13 Environmental Acceptability of Schedule 2 Designated Projects

3.13.1 An application for an EP would be submitted under this EIA for DP1, DP2, DP5, DP6, DP9 and DP12 (refer to **Figure 1.3** and **Section 3.1.2**).

New Roads (DP1, DP2, DP5, DP6 and DP12)

3.13.2 With proper implementation of the recommended mitigation measures for construction activities (as detailed in **Sections 3.6**, **3.10.1**, **3.11.1** and **3.11.2** above), no unacceptable dust impact would be resulted from the proposed roads (DP1, DP2, DP5, DP6 and DP12). There is no operational air quality impact of these DP roads as mentioned in **Section 3.11.3**.

Sewage Pumping Stations (DP9)

3.13.3 With proper implementation of the recommended mitigation measures for construction activities (as detailed in **Sections 3.10.1**, **3.11.1** and **3.11.2** above), and the design of four SPSs follow the mitigation measures as described in **Section 3.9.10**, no unacceptable construction dust and odour impacts would be resulted from the proposed SPSs (DP9).

Other DPs

3.13.4 There will be separate EIA studies to assess the following Schedule 2 DPs (refer to **Figure 1.2**):

- DP3 - Construction of new West Rail HSK station (Site 4-34)
- DP4 - Potential construction of EFTS – subject to further review
- DP7 - Potential construction of a new container back-up and storage area (Site 3-1, 3-4, 3-5, 3-13 and 3-14) - subject to further review
- DP8 - Construction of new HSK STW (Site 3-26 and part of existing San Wai STW)
- DP10 – Construction of flushing water service reservoirs for reuse of reclaimed water of Tan Kwai Tsuen and Fung Kong Tsuen (Sites 3-3 and 5-40)
- DP11 - Construction of a RTS (Site 3-12)

3.13.5 The air quality impact of these Schedule 2 DPs during construction phase and operation phase will be further investigated in separate EIA studies under the EIAO. The EM&A requirements for these Schedule 2 DPs will also be provided under separate EIA studies.

3.14 Conclusion

Construction Phase

3.14.1 The potential air quality impact from the construction works of the Project would mainly relate to construction dust from excavation, material handling, spoil removal and wind erosion. With the implementation measures specified in the Air Pollution Control (Construction Dust) Regulation together with the recommended dust suppression measures, good site practices, and EM&A programme, the predicted dust impact at ASRs would comply with the hourly, daily and annual particulates criteria stipulated in the AQOs and EIAO-TM.

Operation Phase

3.14.2 Cumulative air quality impact arising from the vehicular emissions from the open roads, portal emission and emission from ventilation building of TMWB and chimney emissions within the 500 m assessment area has been assessed at the worst case years. The assessment results conclude that the predicted cumulative 1-hour and annual average NO_2, 10-min and daily average SO_2, daily and annual average RSP / FSP concentrations at representative existing and planned ASRs would comply with the AQOs.

3.14.3 The potential odour impact from nearby existing chicken farm has also been assessed. Exceedance of odour criterion would be expected at Site 3 – 1 (OU, PBU + SWU) at 1.5 mAG and 5 mAG. It is proposed that the air sensitive uses at Site 3-1 should not be located at this exceedance zone or with the fresh air intake of the building located at 6 mAG or above.

3.14.4 The potential cumulative odour impact from expanded and upgraded San Wai STW, new HSK STW and planned RTS are also assessed. No unacceptable odour impact is predicted from these facilities with the provision of deodourising units and enclosed all odorous treatment facilities/sources. The proposed four new SPSs are not expected to pose odour impact to the nearby ASRs with the implementation of proposed mitigation measures.